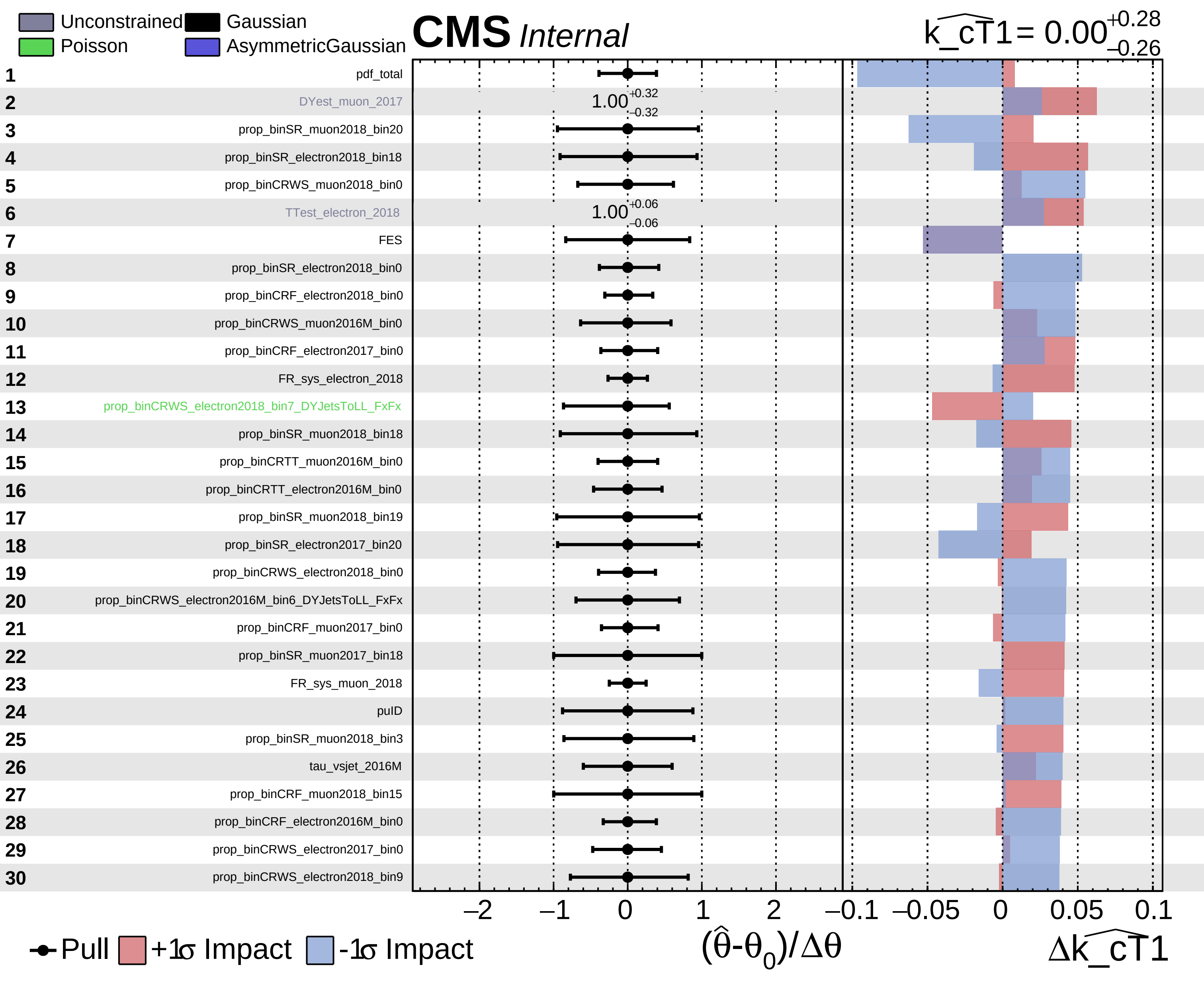
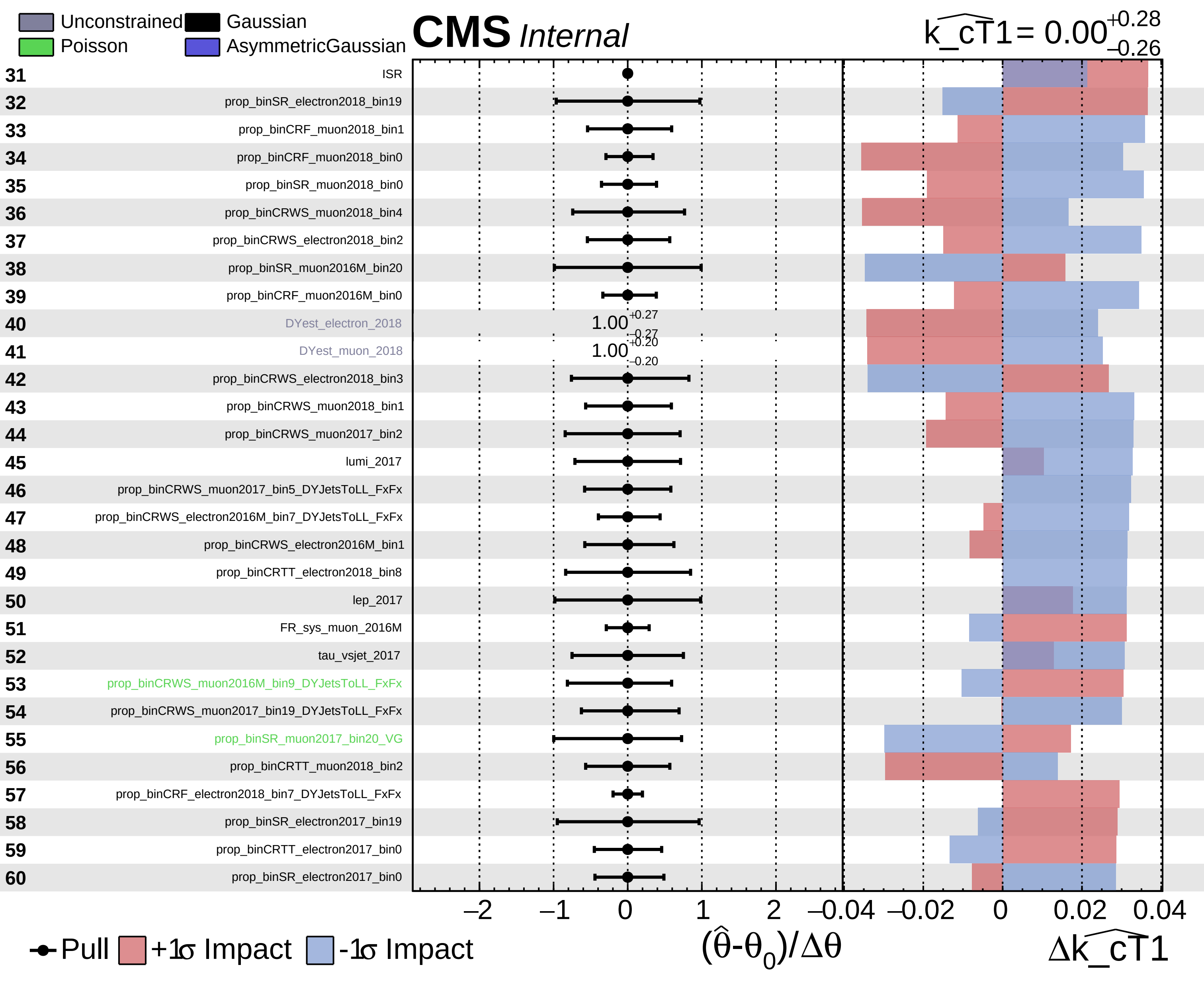
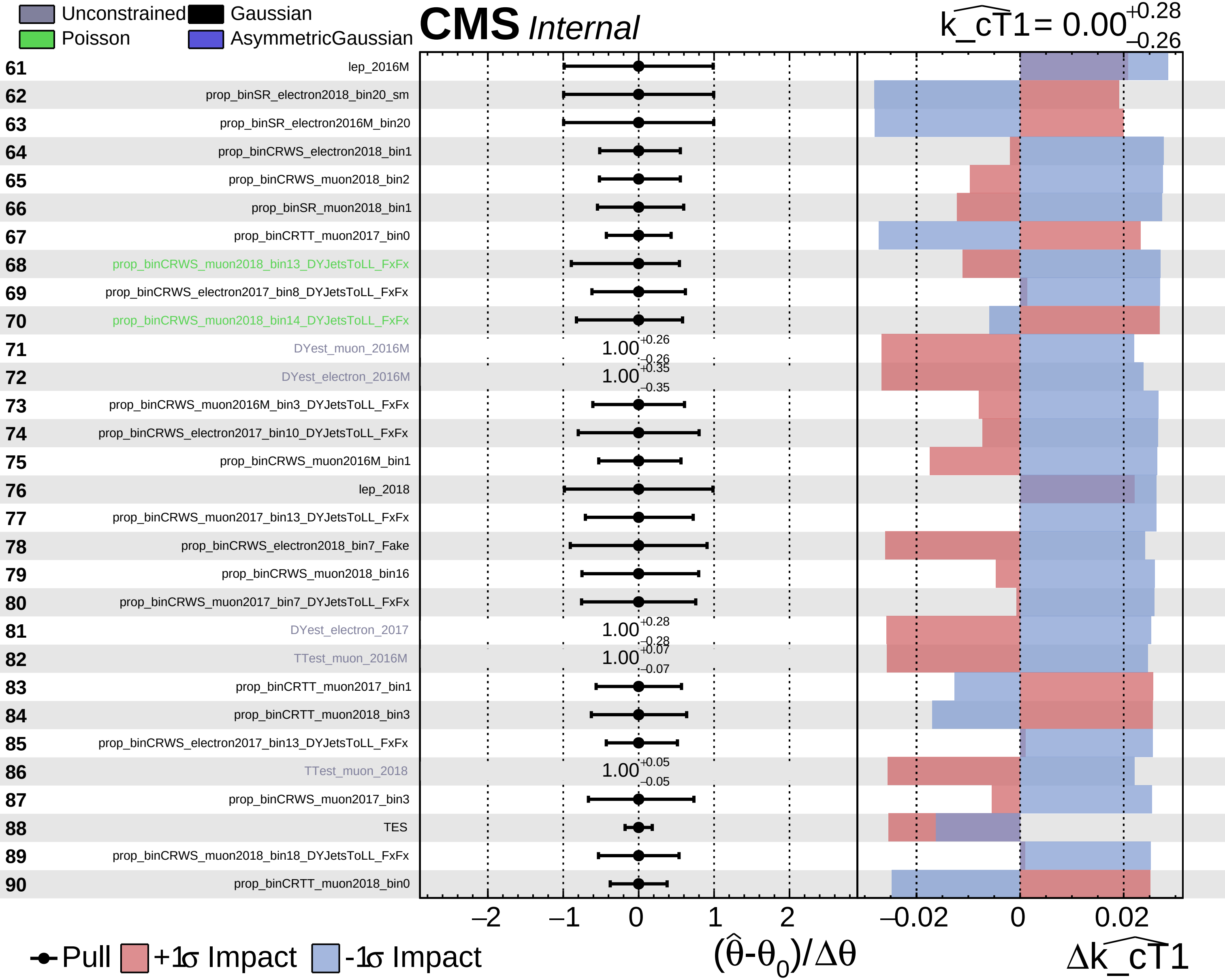


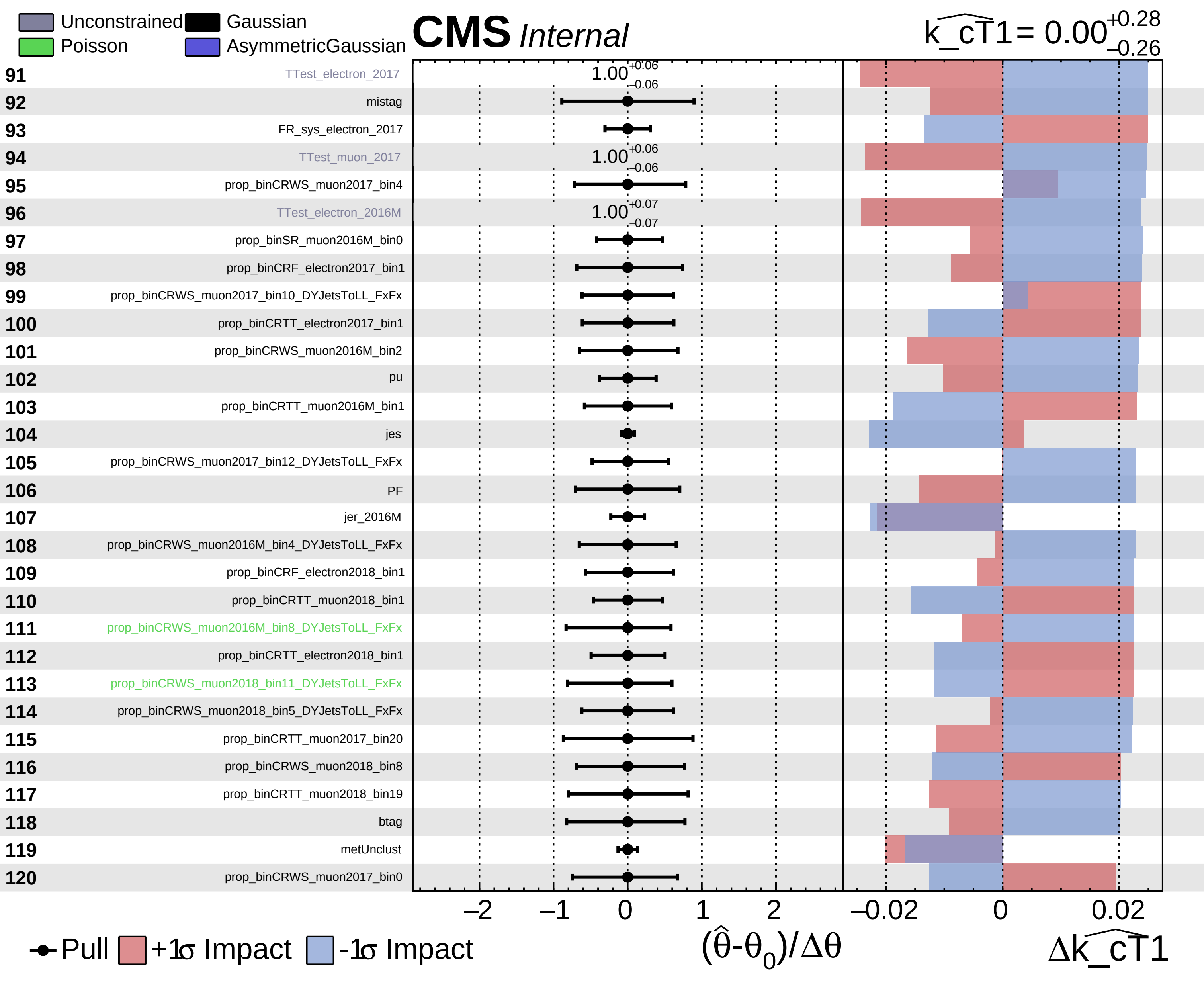
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1 pdf_total
2 DYest_muon_2017
1.00 +0.32 −0.32
3 prop_binSR_muon2018_bin20
4 prop_binSR_electron2018_bin18
5 prop_binCRWS_muon2018_bin0
6 TTest_electron_2018
1.00 +0.06 −0.06
7 FES
8 prop_binSR_electron2018_bin0
9 prop_binCRF_electron2018_bin0
10 prop_binCRWS_muon2016M_bin0
11 prop_binCRF_electron2017_bin0
12 FR_sys_electron_2018
13 prop_binCRWS_electron2018_bin7_DYJetsToLL_FxFx
14 prop_binSR_muon2018_bin18
15 prop_binCRTT_muon2016M_bin0
16 prop_binCRTT_electron2016M_bin0
17 prop_binSR_muon2018_bin19
18 prop_binSR_electron2017_bin20
19 prop_binCRWS_electron2018_bin0
20 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
21 prop_binCRF_muon2017_bin0
22 prop_binSR_muon2017_bin18
23 FR_sys_muon_2018
24 puID
25 prop_binSR_muon2018_bin3
26 tau_vsjet_2016M
27 prop_binCRF_muon2018_bin15
28 prop_binCRF_electron2016M_bin0
29 prop_binCRWS_electron2017_bin0
30 prop_binCRWS_electron2018_bin9
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



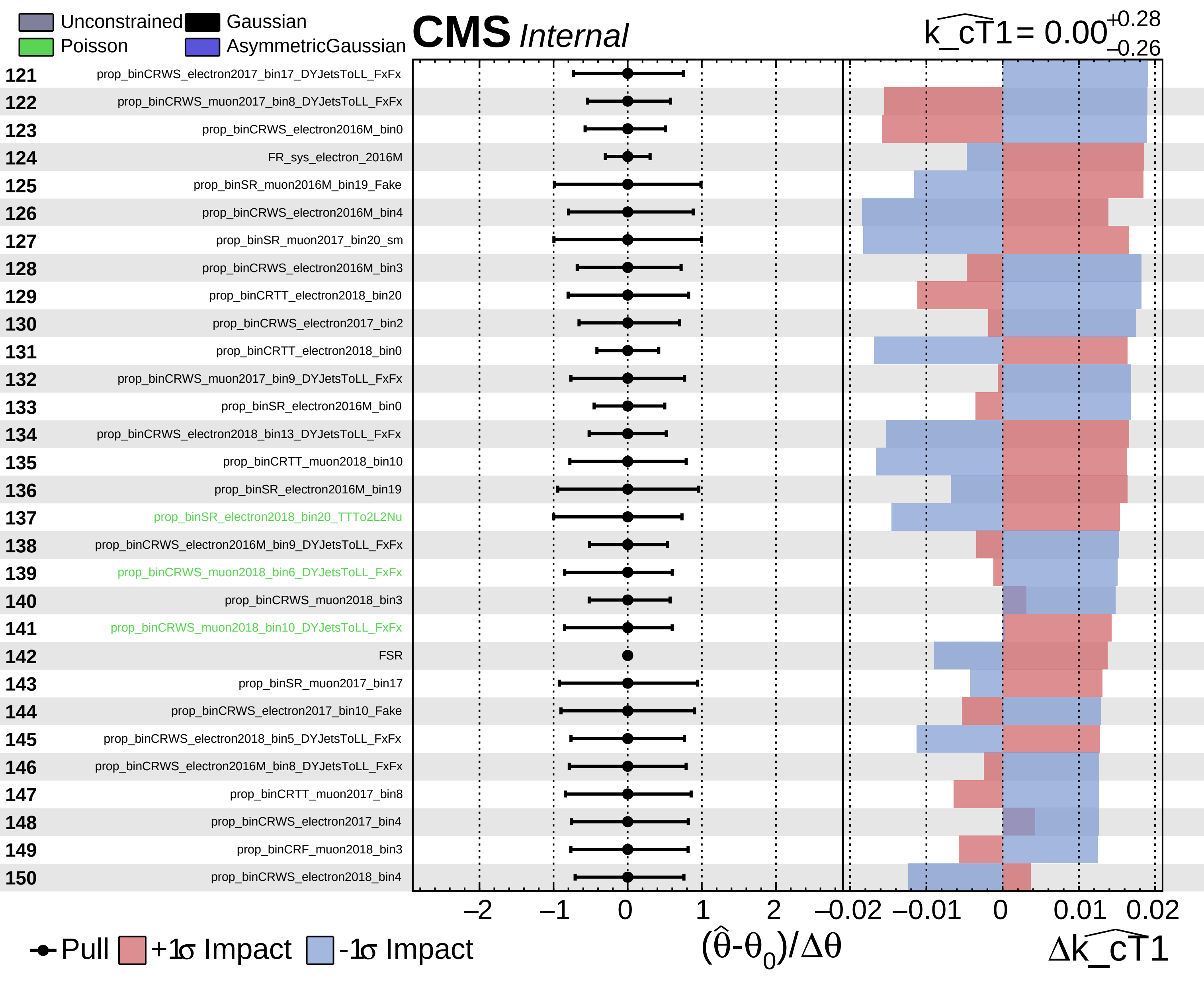
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
31 ISR
32 prop_binSR_electron2018_bin19
33 prop_binCRF_muon2018_bin1
34 prop_binCRF_muon2018_bin0
35 prop_binSR_muon2018_bin0
36 prop_binCRWS_muon2018_bin4
37 prop_binCRWS_electron2018_bin2
38 prop_binSR_muon2016M_bin20
39 prop_binCRF_muon2016M_bin0
40 DYest_electron_2018
1.00 +0.27 −0.27
41 DYest_muon_2018
1.00 +0.20 −0.20
42 prop_binCRWS_electron2018_bin3
43 prop_binCRWS_muon2018_bin1
44 prop_binCRWS_muon2017_bin2
45 lumi_2017
46 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
47 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
48 prop_binCRWS_electron2016M_bin1
49 prop_binCRTT_electron2018_bin8
50 lep_2017
51 FR_sys_muon_2016M
52 tau_vsjet_2017
53 prop_binCRWS_muon2016M_bin9_DYJetsToLL_FxFx
54 prop_binCRWS_muon2017_bin19_DYJetsToLL_FxFx
55 prop_binSR_muon2017_bin20_VG
56 prop_binCRTT_muon2018_bin2
57 prop_binCRF_electron2018_bin7_DYJetsToLL_FxFx
58 prop_binSR_electron2017_bin19
59 prop_binCRTT_electron2017_bin0
60 prop_binSR_electron2017_bin0
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

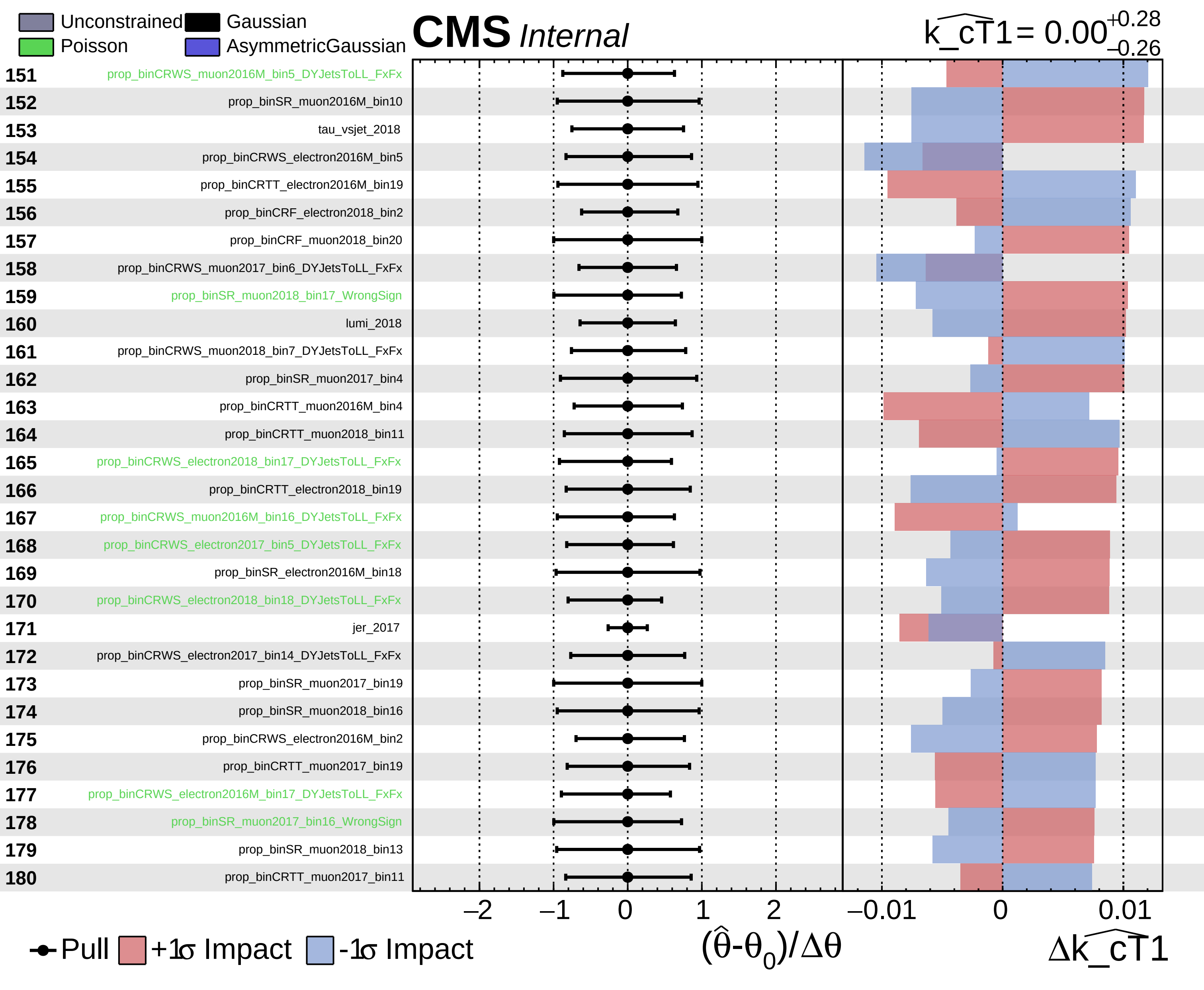


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
61 lep_2016M
62 prop_binSR_electron2018_bin20_sm
63 prop_binSR_electron2016M_bin20
64 prop_binCRWS_electron2018_bin1
65 prop_binCRWS_muon2018_bin2
66 prop_binSR_muon2018_bin1
67 prop_binCRTT_muon2017_bin0
68 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
69 prop_binCRWS_electron2017_bin8_DYJetsToLL_FxFx
70 prop_binCRWS_muon2018_bin14_DYJetsToLL_FxFx
71 DYest_muon_2016M $1.00^{+0.26}_{-0.26}$
72 DYest_electron_2016M $1.00^{+0.35}_{-0.35}$
73 prop_binCRWS_muon2016M_bin3_DYJetsToLL_FxFx
74 prop_binCRWS_electron2017_bin10_DYJetsToLL_FxFx
75 prop_binCRWS_muon2016M_bin1
76 lep_2018
77 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
78 prop_binCRWS_electron2018_bin7_Fake
79 prop_binCRWS_muon2018_bin16
80 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
81 DYest_electron_2017 $1.00^{+0.28}_{-0.28}$
82 TTest_muon_2016M $1.00^{+0.07}_{-0.07}$
83 prop_binCRTT_muon2017_bin1
84 prop_binCRTT_muon2018_bin3
85 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
86 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
87 prop_binCRWS_muon2017_bin3
88 TES
89 prop_binCRWS_muon2018_bin18_DYJetsToLL_FxFx
90 prop_binCRTT_muon2018_bin0
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

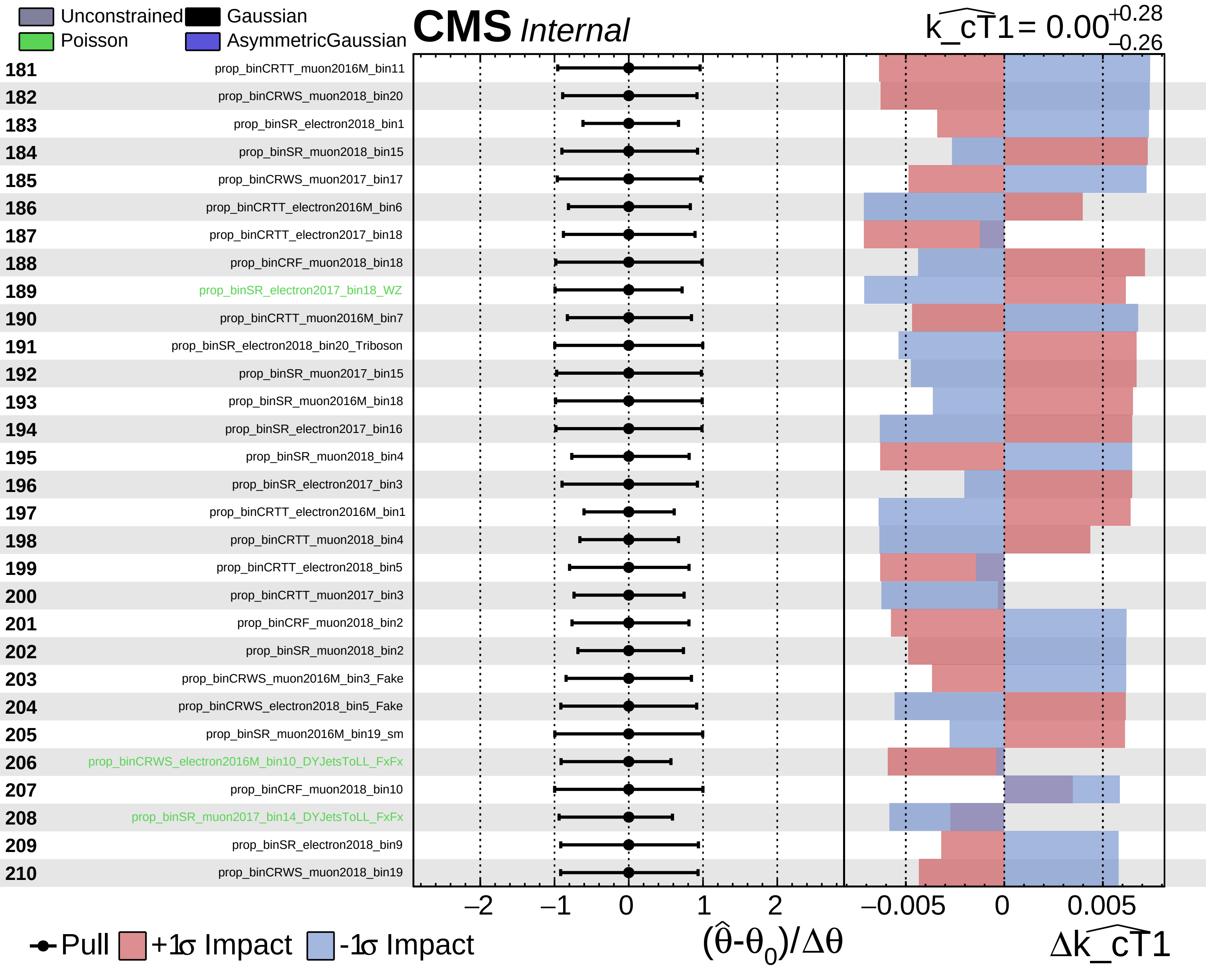


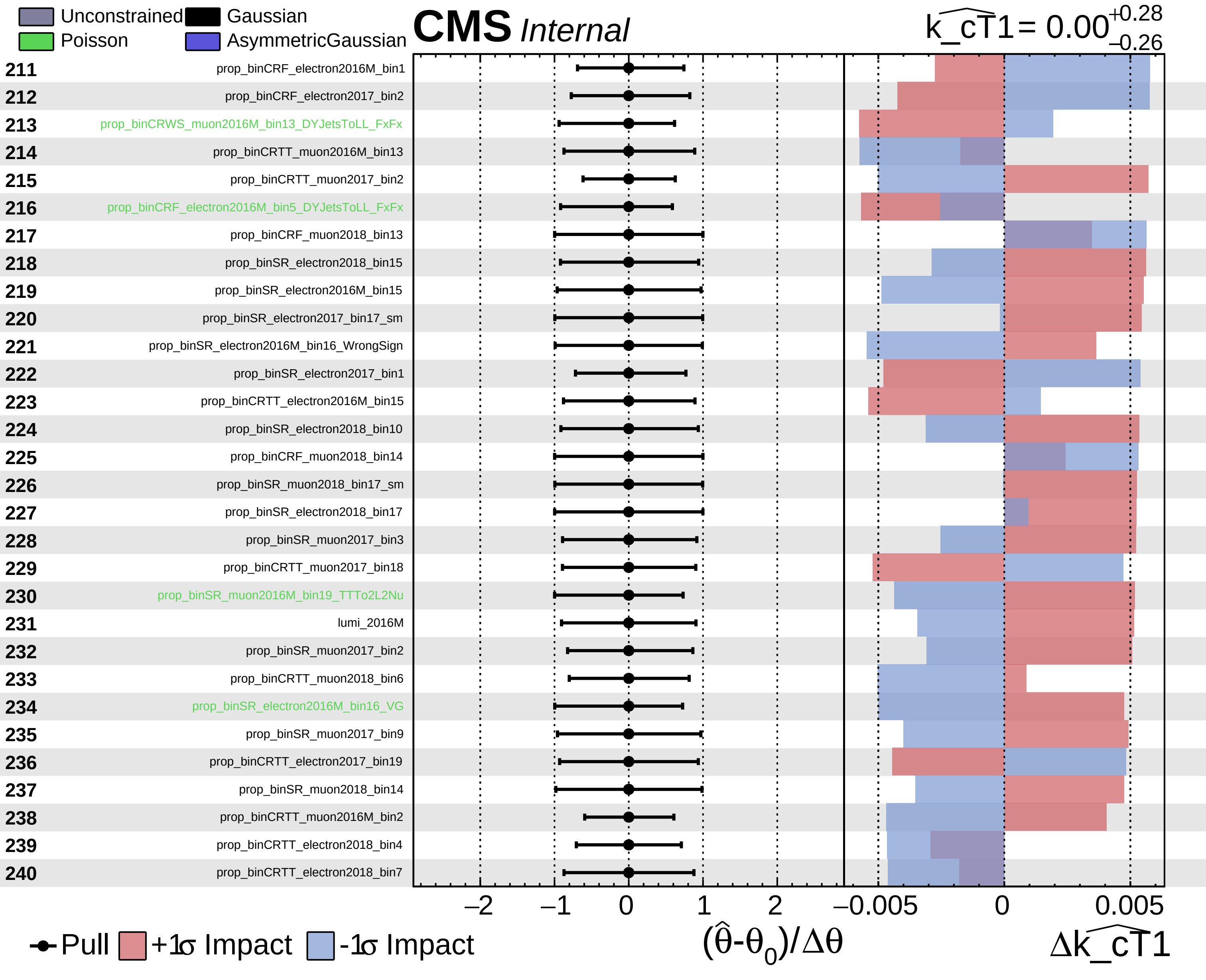
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
91 TTest_electron_2017
92 mistag
93 FR_sys_electron_2017
94 TTest_muon_2017
95 prop_binCRWS_muon2017_bin4
96 TTest_electron_2016M
97 prop_binSR_muon2016M_bin0
98 prop_binCRF_electron2017_bin1
99 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
100 prop_binCRTT_electron2017_bin1
101 prop_binCRWS_muon2016M_bin2
102 pu
103 prop_binCRTT_muon2016M_bin1
104 jes
105 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
106 PF
107 jer_2016M
108 prop_binCRWS_muon2016M_bin4_DYJetsToLL_FxFx
109 prop_binCRF_electron2018_bin1
110 prop_binCRTT_muon2018_bin1
111 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
112 prop_binCRTT_electron2018_bin1
113 prop_binCRWS_muon2018_bin11_DYJetsToLL_FxFx
114 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
115 prop_binCRTT_muon2017_bin20
116 prop_binCRWS_muon2018_bin8
117 prop_binCRTT_muon2018_bin19
118 btag
119 metUnclust
120 prop_binCRWS_muon2017_bin0
1.00 +0.06 −0.06
1.00 +0.06 −0.06
1.00 +0.07 −0.07
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
121 prop_binCRWS_electron2017_bin17_DYJetsToLL_FxFx
122 prop_binCRWS_muon2017_bin8_DYJetsToLL_FxFx
123 prop_binCRWS_electron2016M_bin0
124 FR_sys_electron_2016M
125 prop_binSR_muon2016M_bin19_Fake
126 prop_binCRWS_electron2016M_bin4
127 prop_binSR_muon2017_bin20_sm
128 prop_binCRWS_electron2016M_bin3
129 prop_binCRTT_electron2018_bin20
130 prop_binCRWS_electron2017_bin2
131 prop_binCRTT_electron2018_bin0
132 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
133 prop_binSR_electron2016M_bin0
134 prop_binCRWS_electron2018_bin13_DYJetsToLL_FxFx
135 prop_binCRTT_muon2018_bin10
136 prop_binSR_electron2016M_bin19
137 prop_binSR_electron2018_bin20_TTTo2L2Nu
138 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
139 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
140 prop_binCRWS_muon2018_bin3
141 prop_binCRWS_muon2018_bin10_DYJetsToLL_FxFx
142 FSR
143 prop_binSR_muon2017_bin17
144 prop_binCRWS_electron2017_bin10_Fake
145 prop_binCRWS_electron2018_bin5_DYJetsToLL_FxFx
146 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
147 prop_binCRTT_muon2017_bin8
148 prop_binCRWS_electron2017_bin4
149 prop_binCRF_muon2018_bin3
150 prop_binCRWS_electron2018_bin4
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1

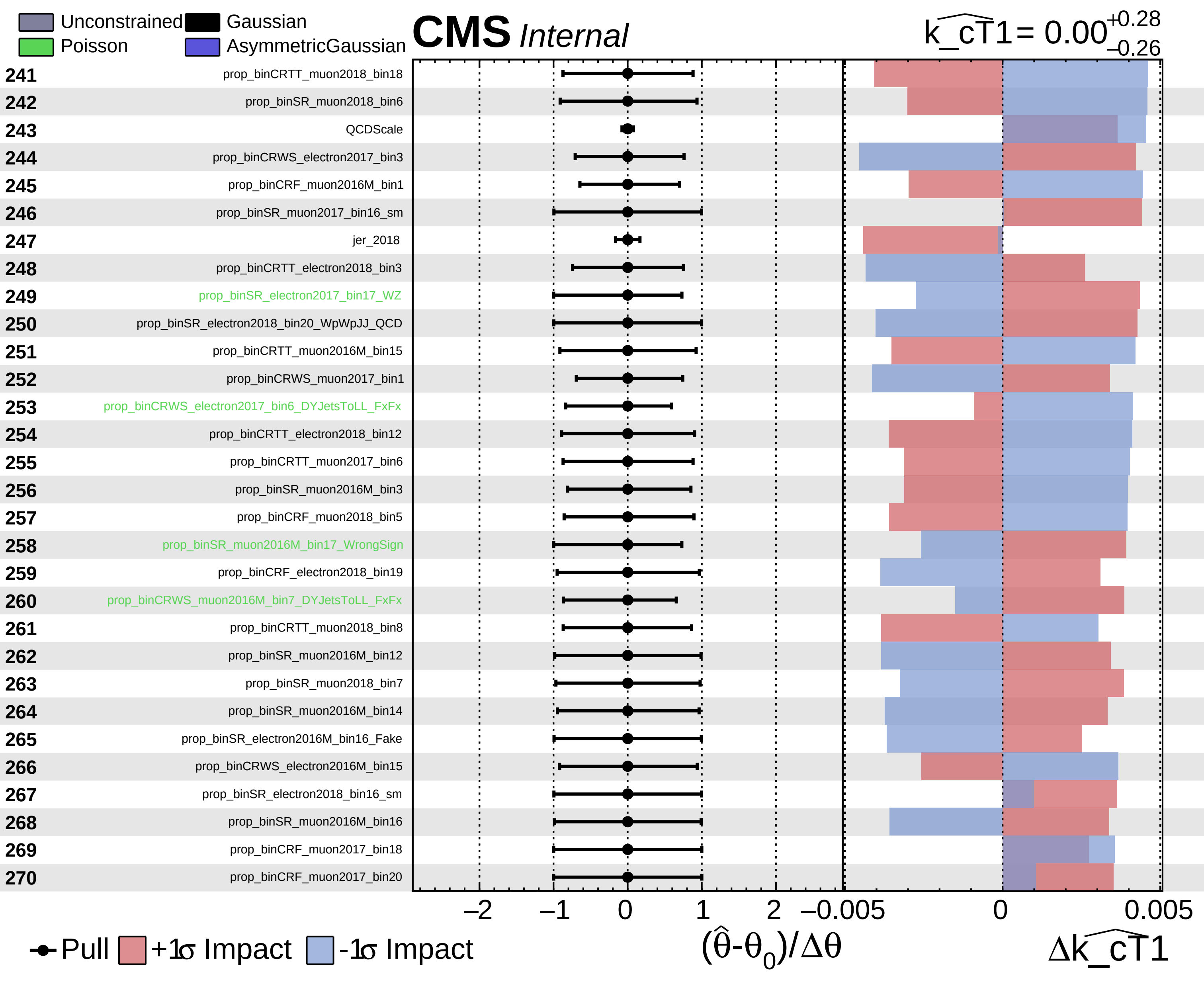


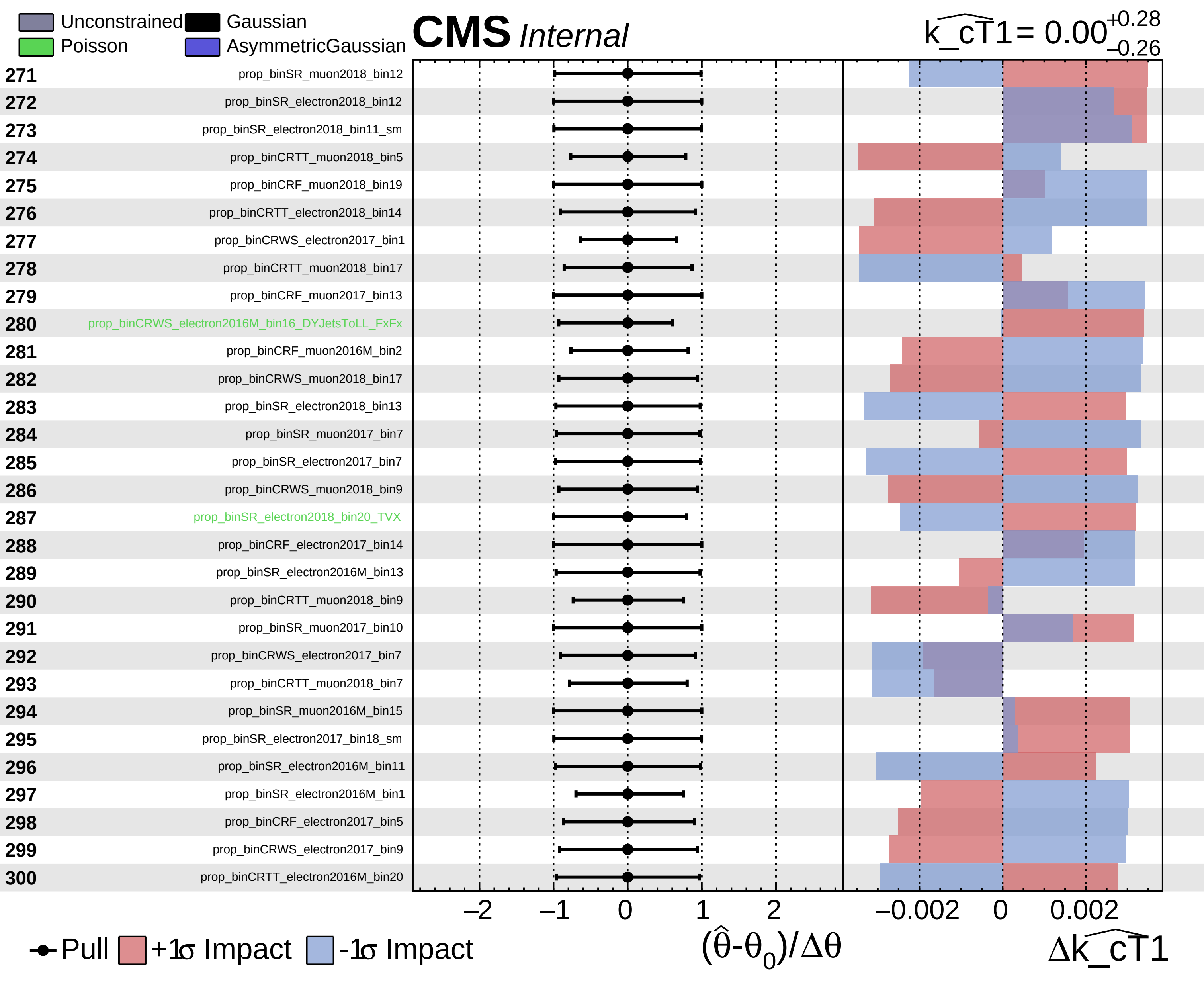
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
151 prop_binCRWS_muon2016M_bin5_DYJetsToLL_FxFx
152 prop_binSR_muon2016M_bin10
153 tau_vsjet_2018
154 prop_binCRWS_electron2016M_bin5
155 prop_binCRTT_electron2016M_bin19
156 prop_binCRF_electron2018_bin2
157 prop_binCRF_muon2018_bin20
158 prop_binCRWS_muon2017_bin6_DYJetsToLL_FxFx
159 prop_binSR_muon2018_bin17_WrongSign
160 lumi_2018
161 prop_binCRWS_muon2018_bin7_DYJetsToLL_FxFx
162 prop_binSR_muon2017_bin4
163 prop_binCRTT_muon2016M_bin4
164 prop_binCRTT_muon2018_bin11
165 prop_binCRWS_electron2018_bin17_DYJetsToLL_FxFx
166 prop_binCRTT_electron2018_bin19
167 prop_binCRWS_muon2016M_bin16_DYJetsToLL_FxFx
168 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
169 prop_binSR_electron2016M_bin18
170 prop_binCRWS_electron2018_bin18_DYJetsToLL_FxFx
171 jer_2017
172 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
173 prop_binSR_muon2017_bin19
174 prop_binSR_muon2018_bin16
175 prop_binCRWS_electron2016M_bin2
176 prop_binCRTT_muon2017_bin19
177 prop_binCRWS_electron2016M_bin17_DYJetsToLL_FxFx
178 prop_binSR_muon2017_bin16_WrongSign
179 prop_binSR_muon2018_bin13
180 prop_binCRTT_muon2017_bin11
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
181 prop_binCRTT_muon2016M_bin11
182 prop_binCRWS_muon2018_bin20
183 prop_binSR_electron2018_bin1
184 prop_binSR_muon2018_bin15
185 prop_binCRWS_muon2017_bin17
186 prop_binCRTT_electron2016M_bin6
187 prop_binCRTT_electron2017_bin18
188 prop_binCRF_muon2018_bin18
189 prop_binSR_electron2017_bin18_WZ
190 prop_binCRTT_muon2016M_bin7
191 prop_binSR_electron2018_bin20_Triboson
192 prop_binSR_muon2017_bin15
193 prop_binSR_muon2016M_bin18
194 prop_binSR_electron2017_bin16
195 prop_binSR_muon2018_bin4
196 prop_binSR_electron2017_bin3
197 prop_binCRTT_electron2016M_bin1
198 prop_binCRTT_muon2018_bin4
199 prop_binCRTT_electron2018_bin5
200 prop_binCRTT_muon2017_bin3
201 prop_binCRF_muon2018_bin2
202 prop_binSR_muon2018_bin2
203 prop_binCRWS_muon2016M_bin3_Fake
204 prop_binCRWS_electron2018_bin5_Fake
205 prop_binSR_muon2016M_bin19_sm
206 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
207 prop_binCRF_muon2018_bin10
208 prop_binSR_muon2017_bin14_DYJetsToLL_FxFx
209 prop_binSR_electron2018_bin9
210 prop_binCRWS_muon2018_bin19
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

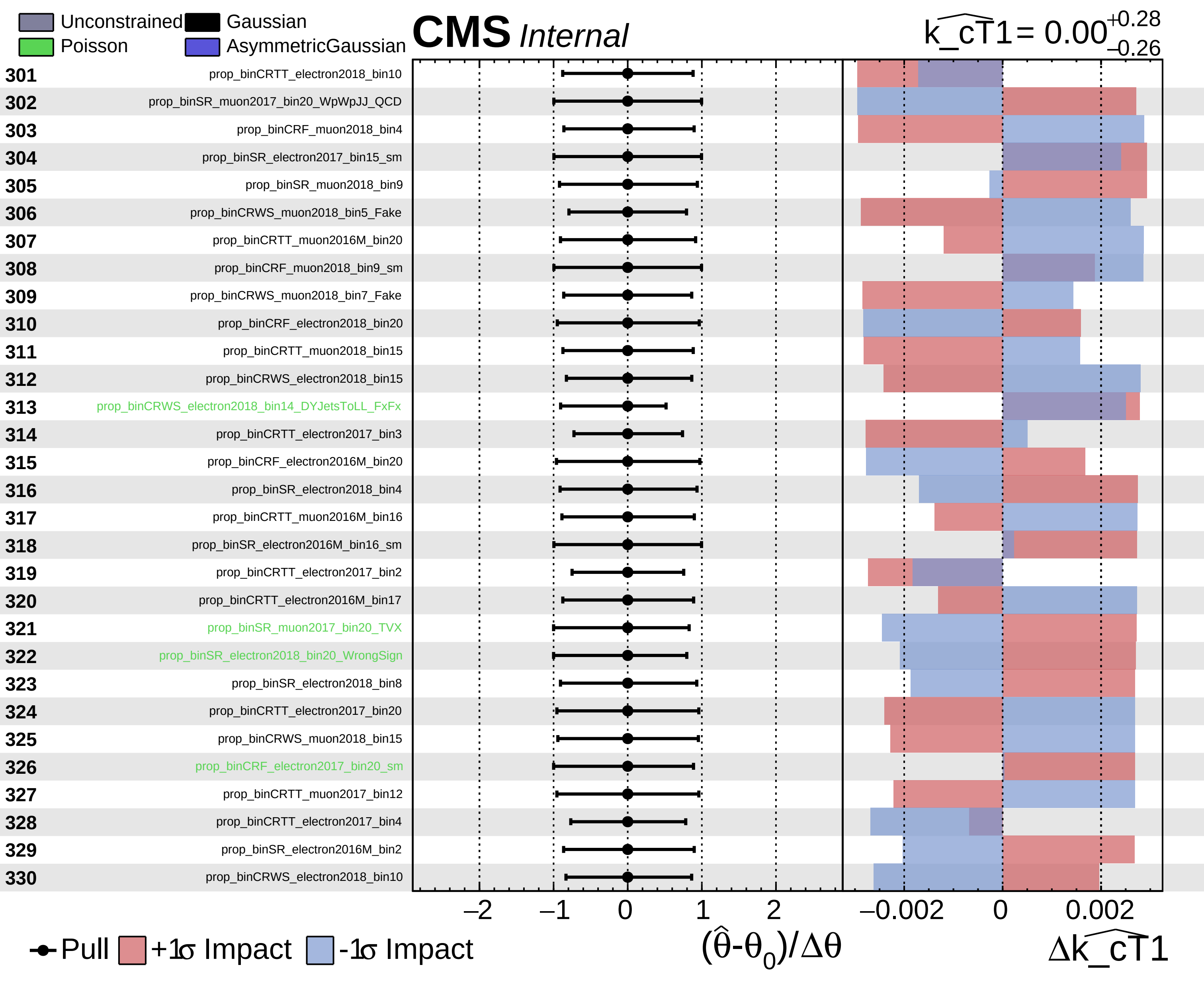


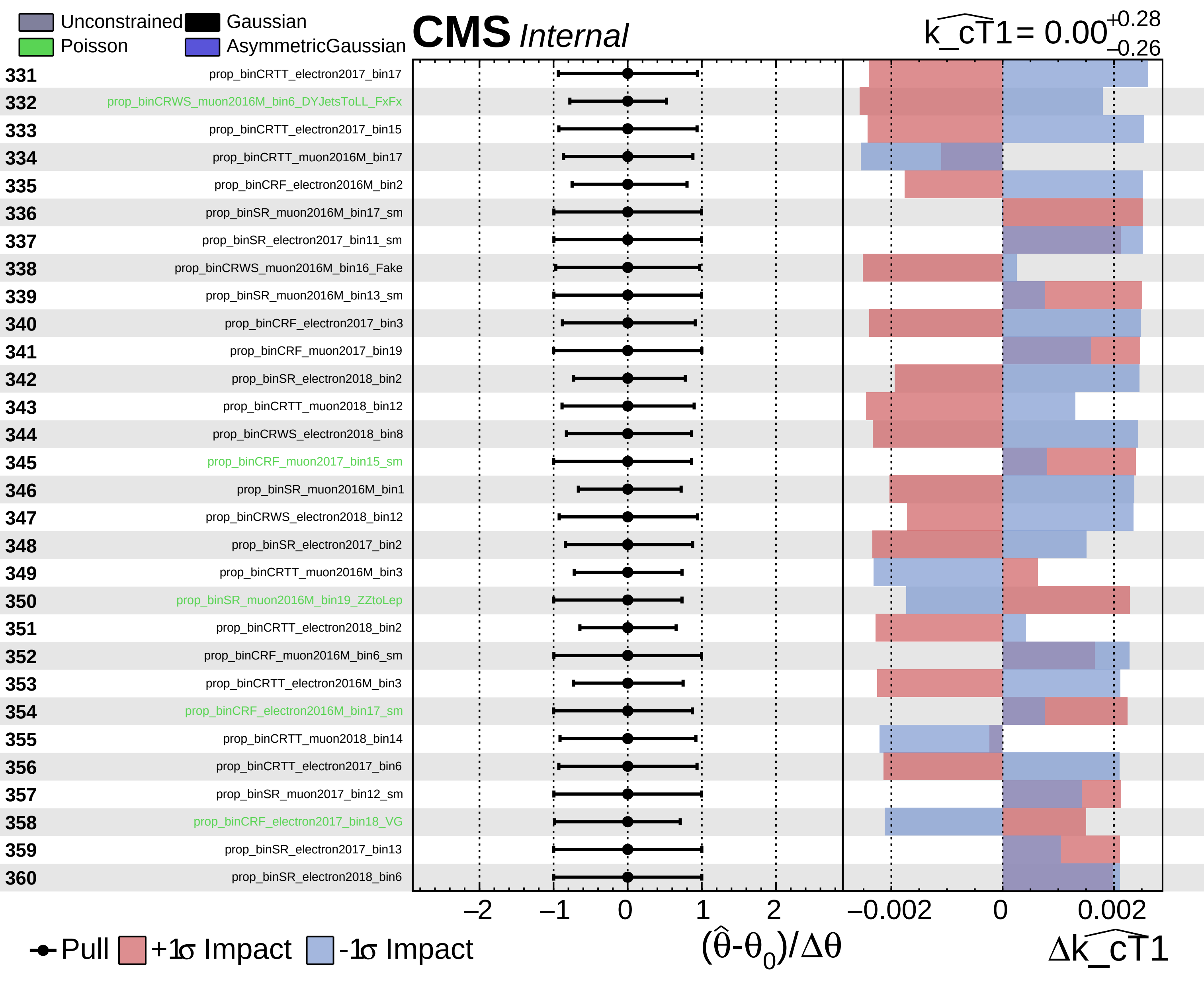
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
211 prop_binCRF_electron2016M_bin1
212 prop_binCRF_electron2017_bin2
213 prop_binCRWS_muon2016M_bin13_DYJetsToLL_FxFx
214 prop_binCRTT_muon2016M_bin13
215 prop_binCRTT_muon2017_bin2
216 prop_binCRF_electron2016M_bin5_DYJetsToLL_FxFx
217 prop_binCRF_muon2018_bin13
218 prop_binSR_electron2018_bin15
219 prop_binSR_electron2016M_bin15
220 prop_binSR_electron2017_bin17_sm
221 prop_binSR_electron2016M_bin16_WrongSign
222 prop_binSR_electron2017_bin1
223 prop_binCRTT_electron2016M_bin15
224 prop_binSR_electron2018_bin10
225 prop_binCRF_muon2018_bin14
226 prop_binSR_muon2018_bin17_sm
227 prop_binSR_electron2018_bin17
228 prop_binSR_muon2017_bin3
229 prop_binCRTT_muon2017_bin18
230 prop_binSR_muon2016M_bin19_TTTo2L2Nu
231 lumi_2016M
232 prop_binSR_muon2017_bin2
233 prop_binCRTT_muon2018_bin6
234 prop_binSR_electron2016M_bin16_VG
235 prop_binSR_muon2017_bin9
236 prop_binCRTT_electron2017_bin19
237 prop_binSR_muon2018_bin14
238 prop_binCRTT_muon2016M_bin2
239 prop_binCRTT_electron2018_bin4
240 prop_binCRTT_electron2018_bin7
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

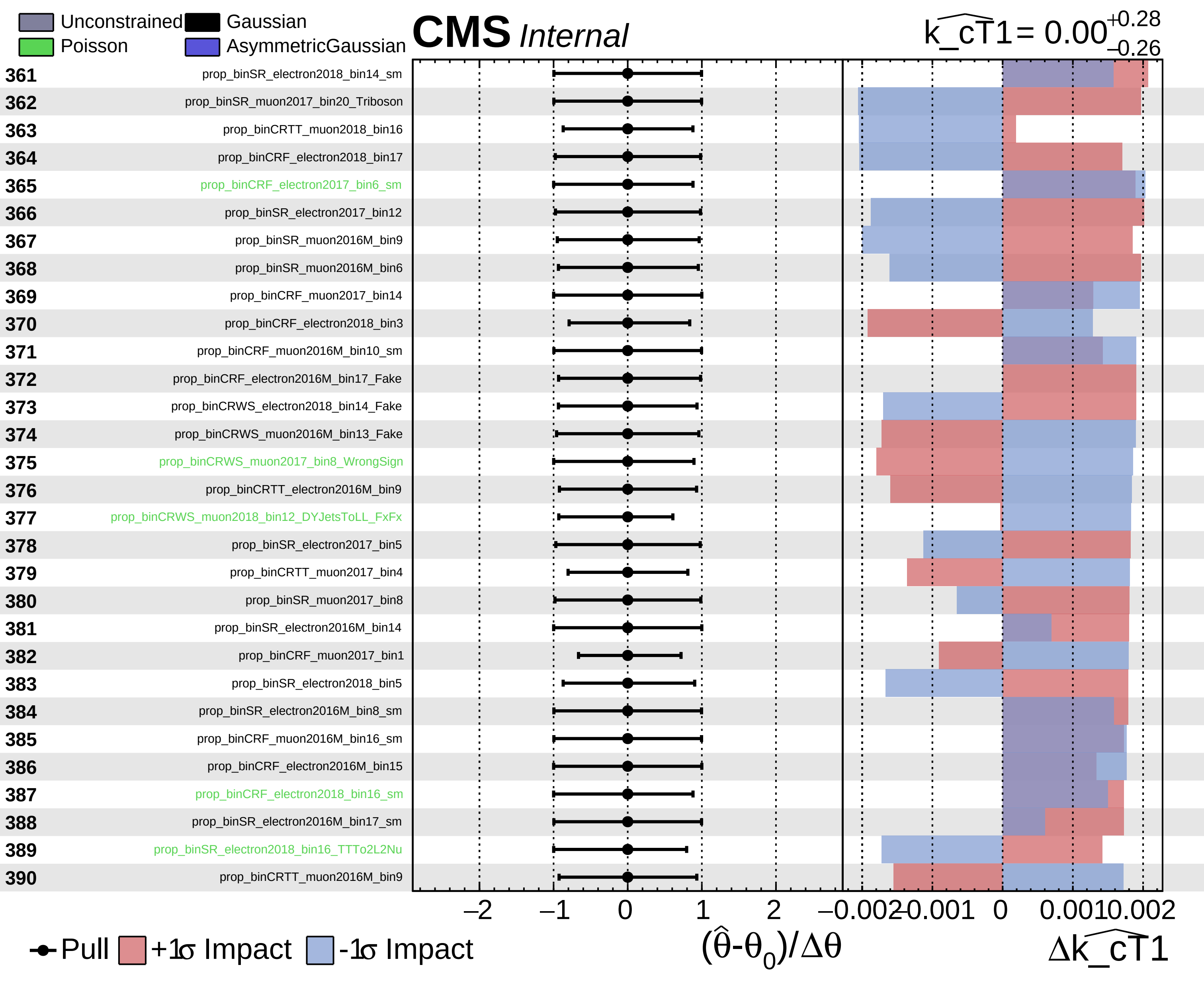

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
241 prop_binCRTT_muon2018_bin18
242 prop_binSR_muon2018_bin6
243 QCDScale
244 prop_binCRWS_electron2017_bin3
245 prop_binCRF_muon2016M_bin1
246 prop_binSR_muon2017_bin16_sm
247 jer_2018
248 prop_binCRTT_electron2018_bin3
249 prop_binSR_electron2017_bin17_WZ
250 prop_binSR_electron2018_bin20_WpWpJJ_QCD
251 prop_binCRTT_muon2016M_bin15
252 prop_binCRWS_muon2017_bin1
253 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
254 prop_binCRTT_electron2018_bin12
255 prop_binCRTT_muon2017_bin6
256 prop_binSR_muon2016M_bin3
257 prop_binCRF_muon2018_bin5
258 prop_binSR_muon2016M_bin17_WrongSign
259 prop_binCRF_electron2018_bin19
260 prop_binCRWS_muon2016M_bin7_DYJetsToLL_FxFx
261 prop_binCRTT_muon2018_bin8
262 prop_binSR_muon2016M_bin12
263 prop_binSR_muon2018_bin7
264 prop_binSR_muon2016M_bin14
265 prop_binSR_electron2016M_bin16_Fake
266 prop_binCRWS_electron2016M_bin15
267 prop_binSR_electron2018_bin16_sm
268 prop_binSR_muon2016M_bin16
269 prop_binCRF_muon2017_bin18
270 prop_binCRF_muon2017_bin20
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



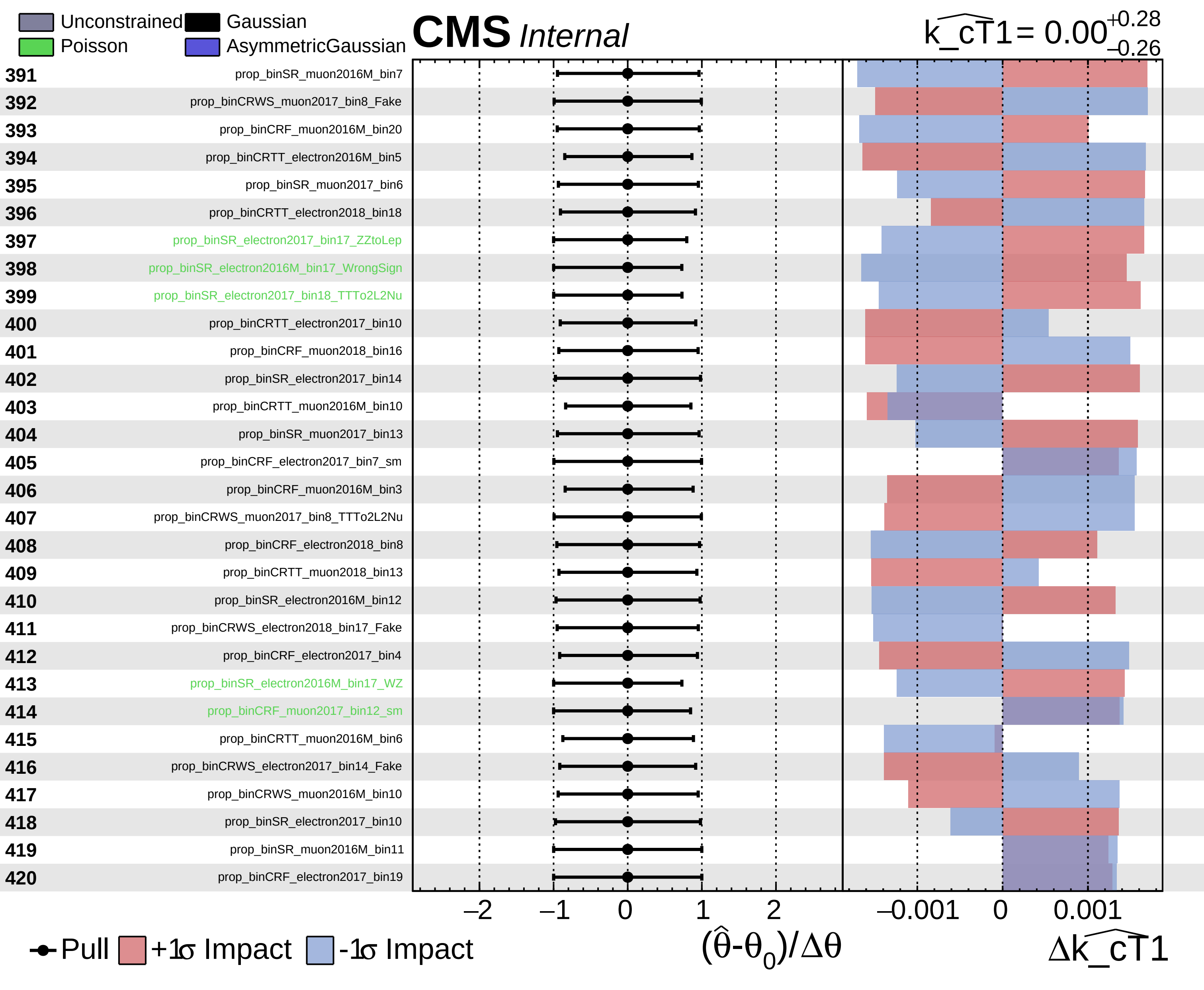
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
271 prop_binSR_muon2018_bin12
272 prop_binSR_electron2018_bin12
273 prop_binSR_electron2018_bin11_sm
274 prop_binCRTT_muon2018_bin5
275 prop_binCRF_muon2018_bin19
276 prop_binCRTT_electron2018_bin14
277 prop_binCRWS_electron2017_bin1
278 prop_binCRTT_muon2018_bin17
279 prop_binCRF_muon2017_bin13
280 prop_binCRWS_electron2016M_bin16_DYJetsToLL_FxFx
281 prop_binCRF_muon2016M_bin2
282 prop_binCRWS_muon2018_bin17
283 prop_binSR_electron2018_bin13
284 prop_binSR_muon2017_bin7
285 prop_binSR_electron2017_bin7
286 prop_binCRWS_muon2018_bin9
287 prop_binSR_electron2018_bin20_TVX
288 prop_binCRF_electron2017_bin14
289 prop_binSR_electron2016M_bin13
290 prop_binCRTT_muon2018_bin9
291 prop_binSR_muon2017_bin10
292 prop_binCRWS_electron2017_bin7
293 prop_binCRTT_muon2018_bin7
294 prop_binSR_muon2016M_bin15
295 prop_binSR_electron2017_bin18_sm
296 prop_binSR_electron2016M_bin11
297 prop_binSR_electron2016M_bin1
298 prop_binCRF_electron2017_bin5
299 prop_binCRWS_electron2017_bin9
300 prop_binCRTT_electron2016M_bin20
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
301 prop_binCRTT_electron2018_bin10
302 prop_binSR_muon2017_bin20_WpWpJJ_QCD
303 prop_binCRF_muon2018_bin4
304 prop_binSR_electron2017_bin15_sm
305 prop_binSR_muon2018_bin9
306 prop_binCRWS_muon2018_bin5_Fake
307 prop_binCRTT_muon2016M_bin20
308 prop_binCRF_muon2018_bin9_sm
309 prop_binCRWS_muon2018_bin7_Fake
310 prop_binCRF_electron2018_bin20
311 prop_binCRTT_muon2018_bin15
312 prop_binCRWS_electron2018_bin15
313 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
314 prop_binCRTT_electron2017_bin3
315 prop_binCRF_electron2016M_bin20
316 prop_binSR_electron2018_bin4
317 prop_binCRTT_muon2016M_bin16
318 prop_binSR_electron2016M_bin16_sm
319 prop_binCRTT_electron2017_bin2
320 prop_binCRTT_electron2016M_bin17
321 prop_binSR_muon2017_bin20_TVX
322 prop_binSR_electron2018_bin20_WrongSign
323 prop_binSR_electron2018_bin8
324 prop_binCRTT_electron2017_bin20
325 prop_binCRWS_muon2018_bin15
326 prop_binCRF_electron2017_bin20_sm
327 prop_binCRTT_muon2017_bin12
328 prop_binCRTT_electron2017_bin4
329 prop_binSR_electron2016M_bin2
330 prop_binCRWS_electron2018_bin10
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

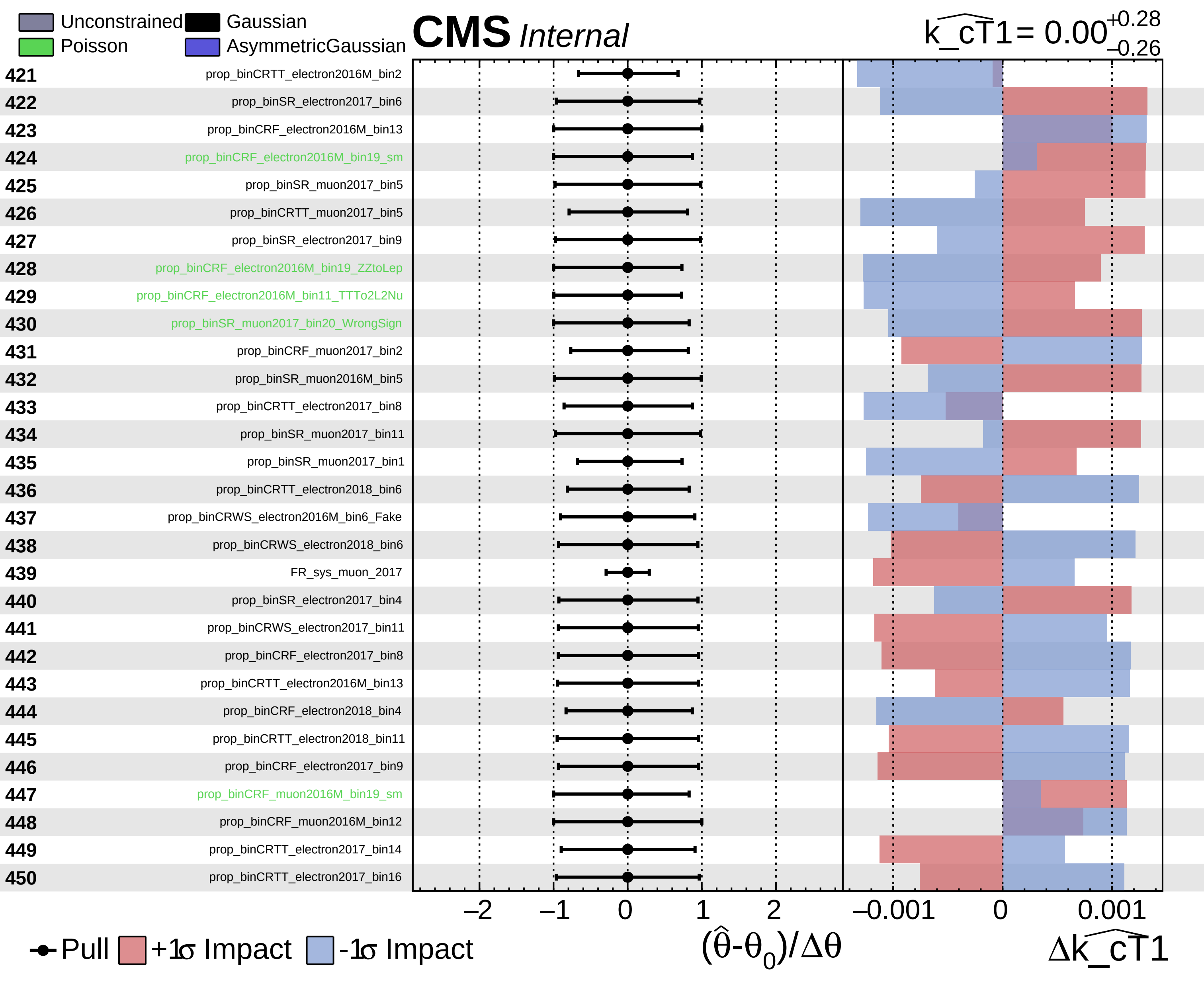

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
331 prop_binCRTT_electron2017_bin17
332 prop_binCRWS_muon2016M_bin6_DYJetsToLL_FxFx
333 prop_binCRTT_electron2017_bin15
334 prop_binCRTT_muon2016M_bin17
335 prop_binCRF_electron2016M_bin2
336 prop_binSR_muon2016M_bin17_sm
337 prop_binSR_electron2017_bin11_sm
338 prop_binCRWS_muon2016M_bin16_Fake
339 prop_binSR_muon2016M_bin13_sm
340 prop_binCRF_electron2017_bin3
341 prop_binCRF_muon2017_bin19
342 prop_binSR_electron2018_bin2
343 prop_binCRTT_muon2018_bin12
344 prop_binCRWS_electron2018_bin8
345 prop_binCRF_muon2017_bin15_sm
346 prop_binSR_muon2016M_bin1
347 prop_binCRWS_electron2018_bin12
348 prop_binSR_electron2017_bin2
349 prop_binCRTT_muon2016M_bin3
350 prop_binSR_muon2016M_bin19_ZZtoLep
351 prop_binCRTT_electron2018_bin2
352 prop_binCRF_muon2016M_bin6_sm
353 prop_binCRTT_electron2016M_bin3
354 prop_binCRF_electron2016M_bin17_sm
355 prop_binCRTT_muon2018_bin14
356 prop_binCRTT_electron2017_bin6
357 prop_binSR_muon2017_bin12_sm
358 prop_binCRF_electron2017_bin18_VG
359 prop_binSR_electron2017_bin13
360 prop_binSR_electron2018_bin6
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

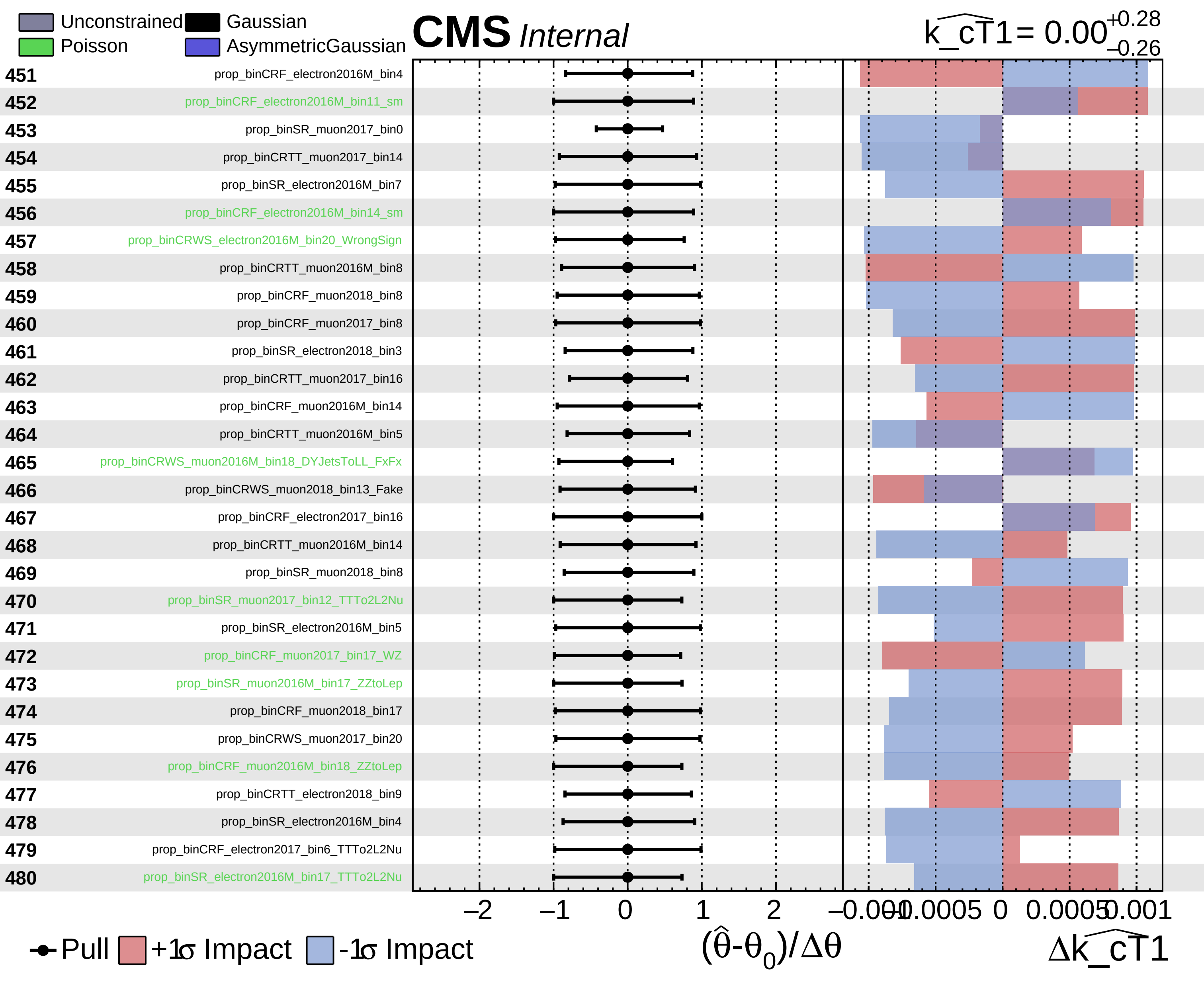


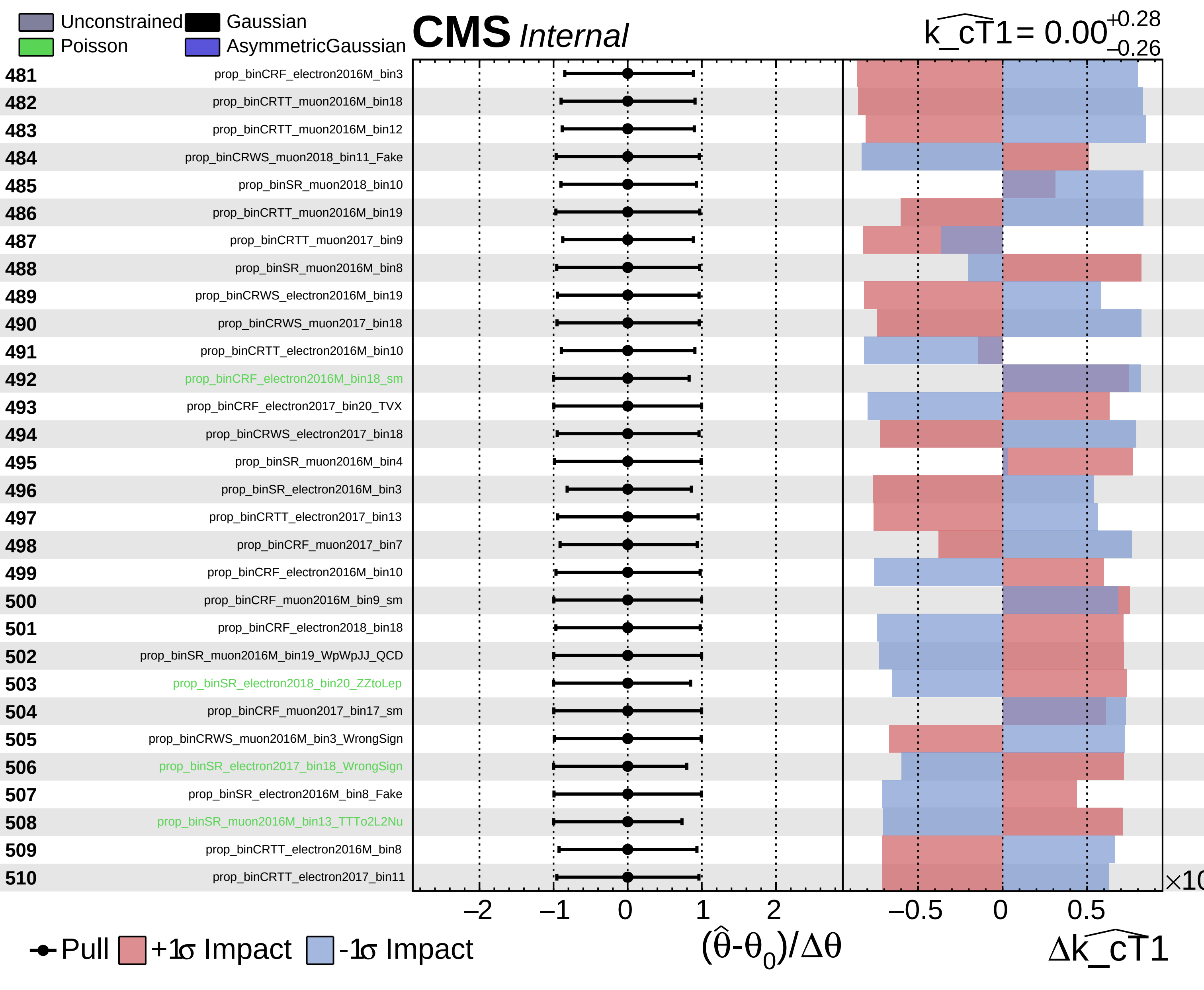
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
361 prop_binSR_electron2018_bin14_sm
362 prop_binSR_muon2017_bin20_Triboson
363 prop_binCRTT_muon2018_bin16
364 prop_binCRF_electron2018_bin17
365 prop_binCRF_electron2017_bin6_sm
366 prop_binSR_electron2017_bin12
367 prop_binSR_muon2016M_bin9
368 prop_binSR_muon2016M_bin6
369 prop_binCRF_muon2017_bin14
370 prop_binCRF_electron2018_bin3
371 prop_binCRF_muon2016M_bin10_sm
372 prop_binCRF_electron2016M_bin17_Fake
373 prop_binCRWS_electron2018_bin14_Fake
374 prop_binCRWS_muon2016M_bin13_Fake
375 prop_binCRWS_muon2017_bin8_WrongSign
376 prop_binCRTT_electron2016M_bin9
377 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
378 prop_binSR_electron2017_bin5
379 prop_binCRTT_muon2017_bin4
380 prop_binSR_muon2017_bin8
381 prop_binSR_electron2016M_bin14
382 prop_binCRF_muon2017_bin1
383 prop_binSR_electron2018_bin5
384 prop_binSR_electron2016M_bin8_sm
385 prop_binCRF_muon2016M_bin16_sm
386 prop_binCRF_electron2016M_bin15
387 prop_binCRF_electron2018_bin16_sm
388 prop_binSR_electron2016M_bin17_sm
389 prop_binSR_electron2018_bin16_TTTo2L2Nu
390 prop_binCRTT_muon2016M_bin9
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



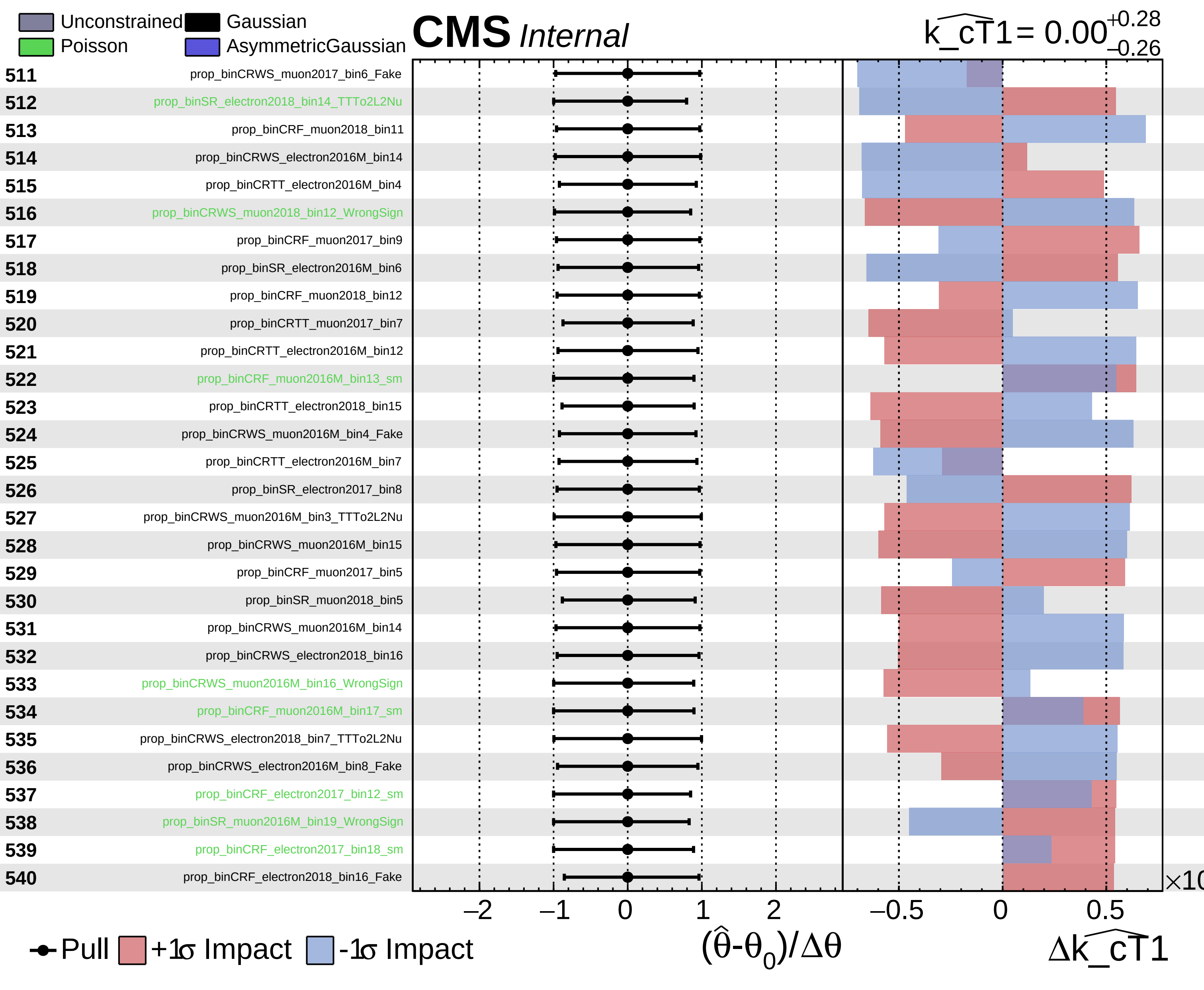
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
391 prop_binSR_muon2016M_bin7
392 prop_binCRWS_muon2017_bin8_Fake
393 prop_binCRF_muon2016M_bin20
394 prop_binCRTT_electron2016M_bin5
395 prop_binSR_muon2017_bin6
396 prop_binCRTT_electron2018_bin18
397 prop_binSR_electron2017_bin17_ZZtoLep
398 prop_binSR_electron2016M_bin17_WrongSign
399 prop_binSR_electron2017_bin18_TTTo2L2Nu
400 prop_binCRTT_electron2017_bin10
401 prop_binCRF_muon2018_bin16
402 prop_binSR_electron2017_bin14
403 prop_binCRTT_muon2016M_bin10
404 prop_binSR_muon2017_bin13
405 prop_binCRF_electron2017_bin7_sm
406 prop_binCRF_muon2016M_bin3
407 prop_binCRWS_muon2017_bin8_TTTo2L2Nu
408 prop_binCRF_electron2018_bin8
409 prop_binCRTT_muon2018_bin13
410 prop_binSR_electron2016M_bin12
411 prop_binCRWS_electron2018_bin17_Fake
412 prop_binCRF_electron2017_bin4
413 prop_binSR_electron2016M_bin17_WZ
414 prop_binCRF_muon2017_bin12_sm
415 prop_binCRTT_muon2016M_bin6
416 prop_binCRWS_electron2017_bin14_Fake
417 prop_binCRWS_muon2016M_bin10
418 prop_binSR_electron2017_bin10
419 prop_binSR_muon2016M_bin11
420 prop_binCRF_electron2017_bin19
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
421 prop_binCRTT_electron2016M_bin2
422 prop_binSR_electron2017_bin6
423 prop_binCRF_electron2016M_bin13
424 prop_binCRF_electron2016M_bin19_sm
425 prop_binSR_muon2017_bin5
426 prop_binCRTT_muon2017_bin5
427 prop_binSR_electron2017_bin9
428 prop_binCRF_electron2016M_bin19_ZZtoLep
429 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
430 prop_binSR_muon2017_bin20_WrongSign
431 prop_binCRF_muon2017_bin2
432 prop_binSR_muon2016M_bin5
433 prop_binCRTT_electron2017_bin8
434 prop_binSR_muon2017_bin11
435 prop_binSR_muon2017_bin1
436 prop_binCRTT_electron2018_bin6
437 prop_binCRWS_electron2016M_bin6_Fake
438 prop_binCRWS_electron2018_bin6
439 FR_sys_muon_2017
440 prop_binSR_electron2017_bin4
441 prop_binCRWS_electron2017_bin11
442 prop_binCRF_electron2017_bin8
443 prop_binCRTT_electron2016M_bin13
444 prop_binCRF_electron2018_bin4
445 prop_binCRTT_electron2018_bin11
446 prop_binCRF_electron2017_bin9
447 prop_binCRF_muon2016M_bin19_sm
448 prop_binCRF_muon2016M_bin12
449 prop_binCRTT_electron2017_bin14
450 prop_binCRTT_electron2017_bin16
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1

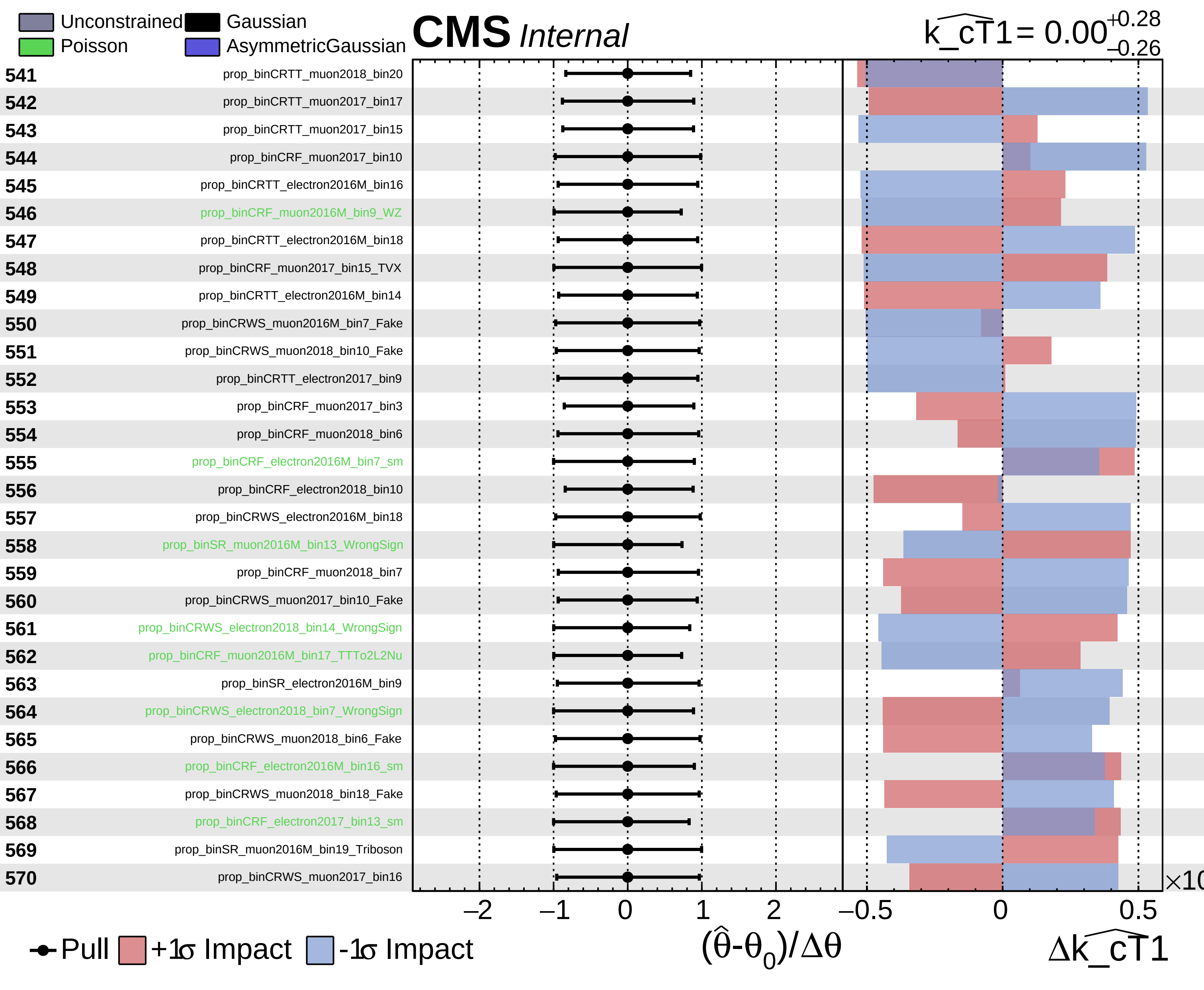

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
451 prop_binCRF_electron2016M_bin4
452 prop_binCRF_electron2016M_bin11_sm
453 prop_binSR_muon2017_bin0
454 prop_binCRTT_muon2017_bin14
455 prop_binSR_electron2016M_bin7
456 prop_binCRF_electron2016M_bin14_sm
457 prop_binCRWS_electron2016M_bin20_WrongSign
458 prop_binCRTT_muon2016M_bin8
459 prop_binCRF_muon2018_bin8
460 prop_binCRF_muon2017_bin8
461 prop_binSR_electron2018_bin3
462 prop_binCRTT_muon2017_bin16
463 prop_binCRF_muon2016M_bin14
464 prop_binCRTT_muon2016M_bin5
465 prop_binCRWS_muon2016M_bin18_DYJetsToLL_FxFx
466 prop_binCRWS_muon2018_bin13_Fake
467 prop_binCRF_electron2017_bin16
468 prop_binCRTT_muon2016M_bin14
469 prop_binSR_muon2018_bin8
470 prop_binSR_muon2017_bin12_TTTo2L2Nu
471 prop_binSR_electron2016M_bin5
472 prop_binCRF_muon2017_bin17_WZ
473 prop_binSR_muon2016M_bin17_ZZtoLep
474 prop_binCRF_muon2018_bin17
475 prop_binCRWS_muon2017_bin20
476 prop_binCRF_muon2016M_bin18_ZZtoLep
477 prop_binCRTT_electron2018_bin9
478 prop_binSR_electron2016M_bin4
479 prop_binCRF_electron2017_bin6_TTTo2L2Nu
480 prop_binSR_electron2016M_bin17_TTTo2L2Nu
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1

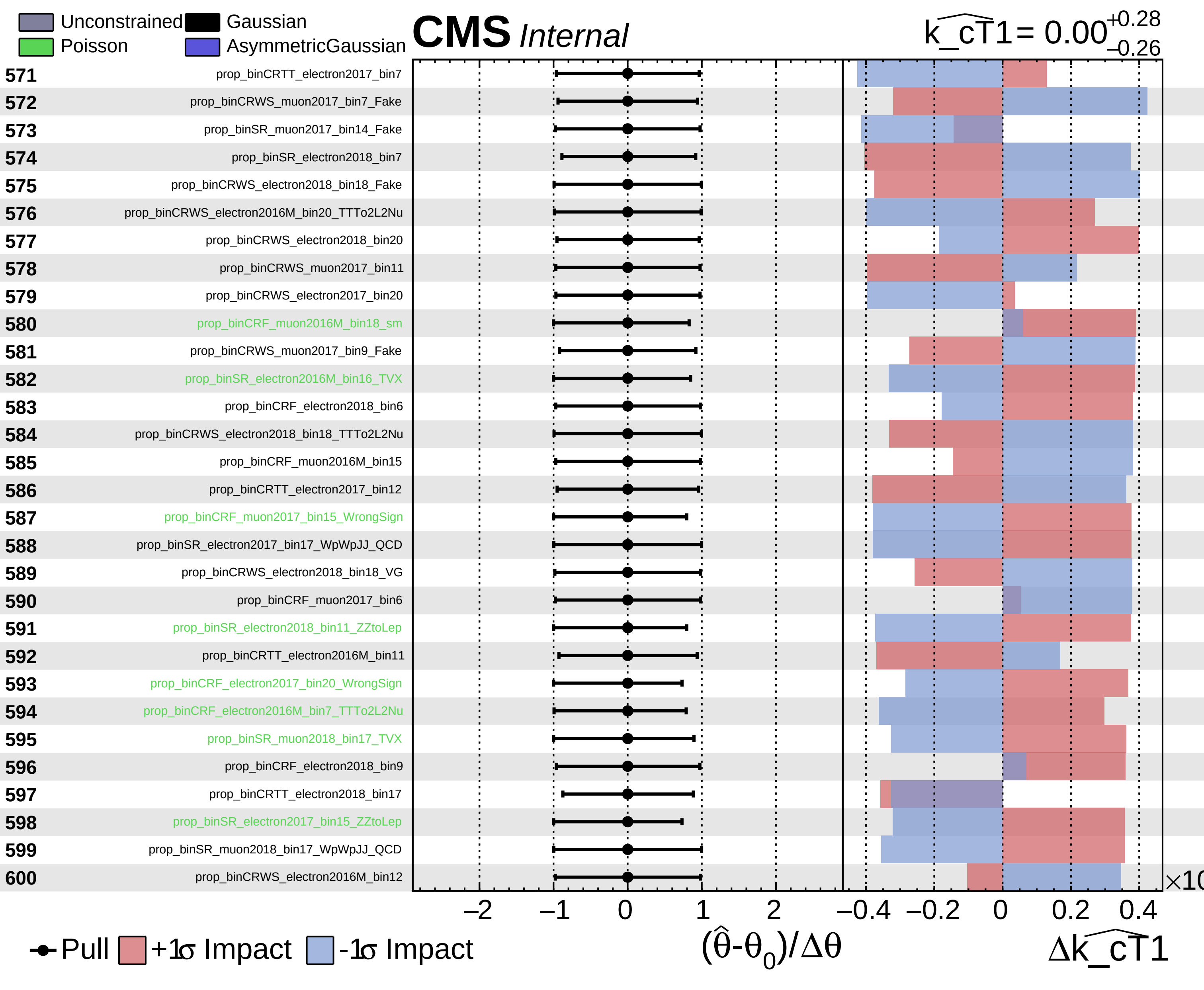


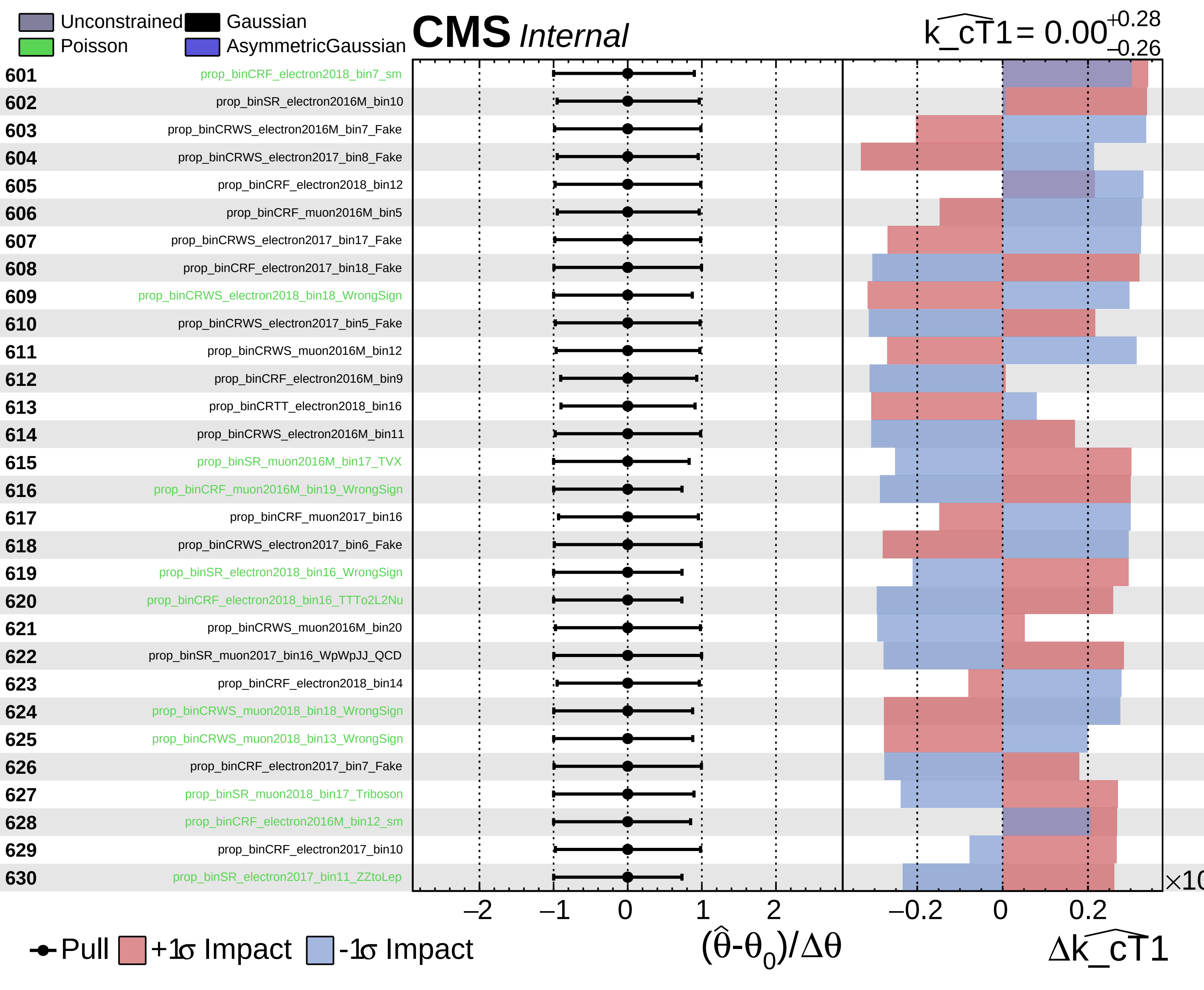
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
481 prop_binCRF_electron2016M_bin3
482 prop_binCRTT_muon2016M_bin18
483 prop_binCRTT_muon2016M_bin12
484 prop_binCRWS_muon2018_bin11_Fake
485 prop_binSR_muon2018_bin10
486 prop_binCRTT_muon2016M_bin19
487 prop_binCRTT_muon2017_bin9
488 prop_binSR_muon2016M_bin8
489 prop_binCRWS_electron2016M_bin19
490 prop_binCRWS_muon2017_bin18
491 prop_binCRTT_electron2016M_bin10
492 prop_binCRF_electron2016M_bin18_sm
493 prop_binCRF_electron2017_bin20_TVX
494 prop_binCRWS_electron2017_bin18
495 prop_binSR_muon2016M_bin4
496 prop_binSR_electron2016M_bin3
497 prop_binCRTT_electron2017_bin13
498 prop_binCRF_muon2017_bin7
499 prop_binCRF_electron2016M_bin10
500 prop_binCRF_muon2016M_bin9_sm
501 prop_binCRF_electron2018_bin18
502 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
503 prop_binSR_electron2018_bin20_ZZtoLep
504 prop_binCRF_muon2017_bin17_sm
505 prop_binCRWS_muon2016M_bin3_WrongSign
506 prop_binSR_electron2017_bin18_WrongSign
507 prop_binSR_electron2016M_bin8_Fake
508 prop_binSR_muon2016M_bin13_TTTo2L2Nu
509 prop_binCRTT_electron2016M_bin8
510 prop_binCRTT_electron2017_bin11
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



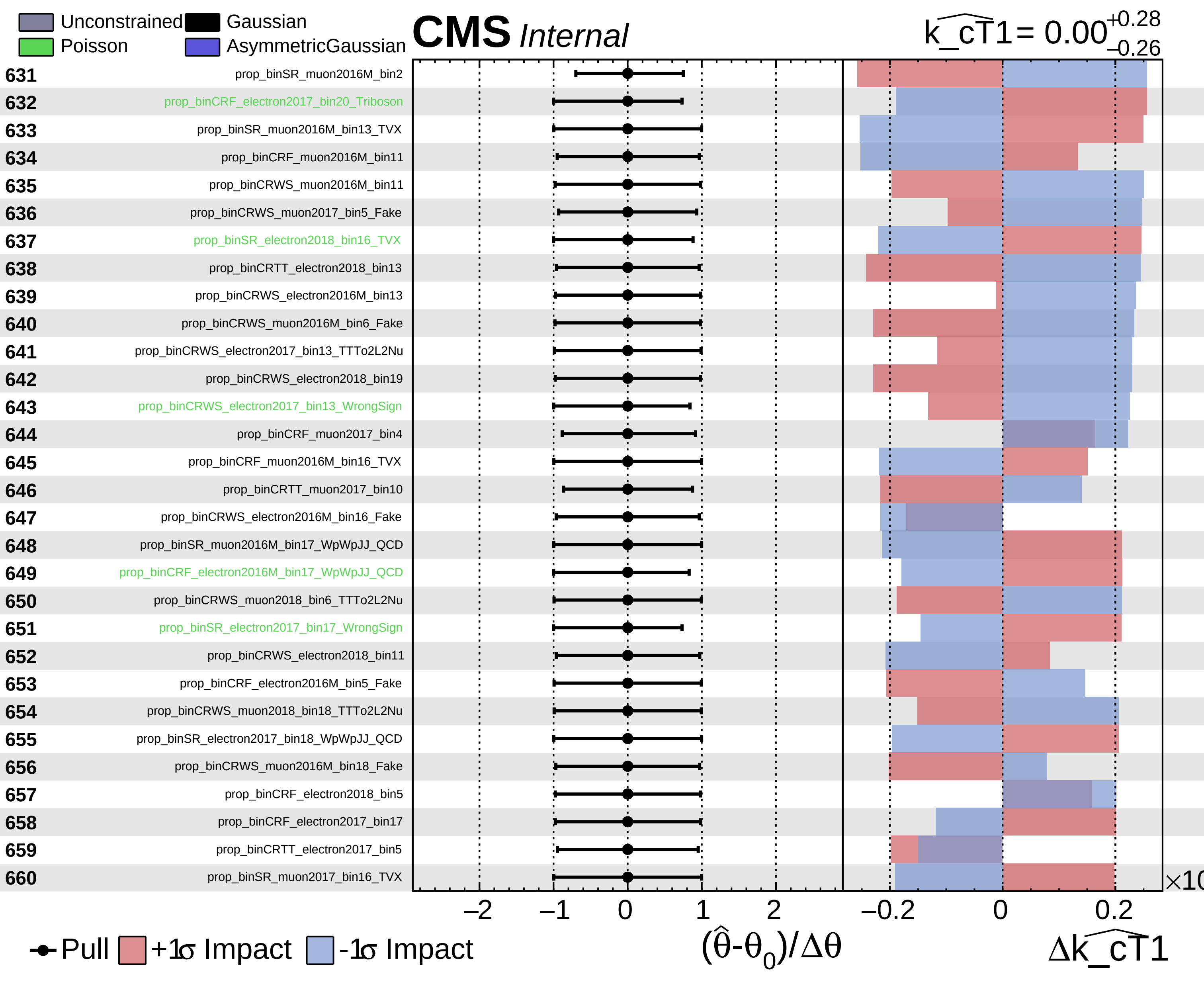
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
511 prop_binCRWS_muon2017_bin6_Fake
512 prop_binSR_electron2018_bin14_TTTo2L2Nu
513 prop_binCRF_muon2018_bin11
514 prop_binCRWS_electron2016M_bin14
515 prop_binCRTT_electron2016M_bin4
516 prop_binCRWS_muon2018_bin12_WrongSign
517 prop_binCRF_muon2017_bin9
518 prop_binSR_electron2016M_bin6
519 prop_binCRF_muon2018_bin12
520 prop_binCRTT_muon2017_bin7
521 prop_binCRTT_electron2016M_bin12
522 prop_binCRF_muon2016M_bin13_sm
523 prop_binCRTT_electron2018_bin15
524 prop_binCRWS_muon2016M_bin4_Fake
525 prop_binCRTT_electron2016M_bin7
526 prop_binSR_electron2017_bin8
527 prop_binCRWS_muon2016M_bin3_TTTo2L2Nu
528 prop_binCRWS_muon2016M_bin15
529 prop_binCRF_muon2017_bin5
530 prop_binSR_muon2018_bin5
531 prop_binCRWS_muon2016M_bin14
532 prop_binCRWS_electron2018_bin16
533 prop_binCRWS_muon2016M_bin16_WrongSign
534 prop_binCRF_muon2016M_bin17_sm
535 prop_binCRWS_electron2018_bin7_TTTo2L2Nu
536 prop_binCRWS_electron2016M_bin8_Fake
537 prop_binCRF_electron2017_bin12_sm
538 prop_binSR_muon2016M_bin19_WrongSign
539 prop_binCRF_electron2017_bin18_sm
540 prop_binCRF_electron2018_bin16_Fake
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
541 prop_binCRTT_muon2018_bin20
542 prop_binCRTT_muon2017_bin17
543 prop_binCRTT_muon2017_bin15
544 prop_binCRF_muon2017_bin10
545 prop_binCRTT_electron2016M_bin16
546 prop_binCRF_muon2016M_bin9_WZ
547 prop_binCRTT_electron2016M_bin18
548 prop_binCRF_muon2017_bin15_TVX
549 prop_binCRTT_electron2016M_bin14
550 prop_binCRWS_muon2016M_bin7_Fake
551 prop_binCRWS_muon2018_bin10_Fake
552 prop_binCRTT_electron2017_bin9
553 prop_binCRF_muon2017_bin3
554 prop_binCRF_muon2018_bin6
555 prop_binCRF_electron2016M_bin7_sm
556 prop_binCRF_electron2018_bin10
557 prop_binCRWS_electron2016M_bin18
558 prop_binSR_muon2016M_bin13_WrongSign
559 prop_binCRF_muon2018_bin7
560 prop_binCRWS_muon2017_bin10_Fake
561 prop_binCRWS_electron2018_bin14_WrongSign
562 prop_binCRF_muon2016M_bin17_TTTo2L2Nu
563 prop_binSR_electron2016M_bin9
564 prop_binCRWS_electron2018_bin7_WrongSign
565 prop_binCRWS_muon2018_bin6_Fake
566 prop_binCRF_electron2016M_bin16_sm
567 prop_binCRWS_muon2018_bin18_Fake
568 prop_binCRF_electron2017_bin13_sm
569 prop_binSR_muon2016M_bin19_Triboson
570 prop_binCRWS_muon2017_bin16
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

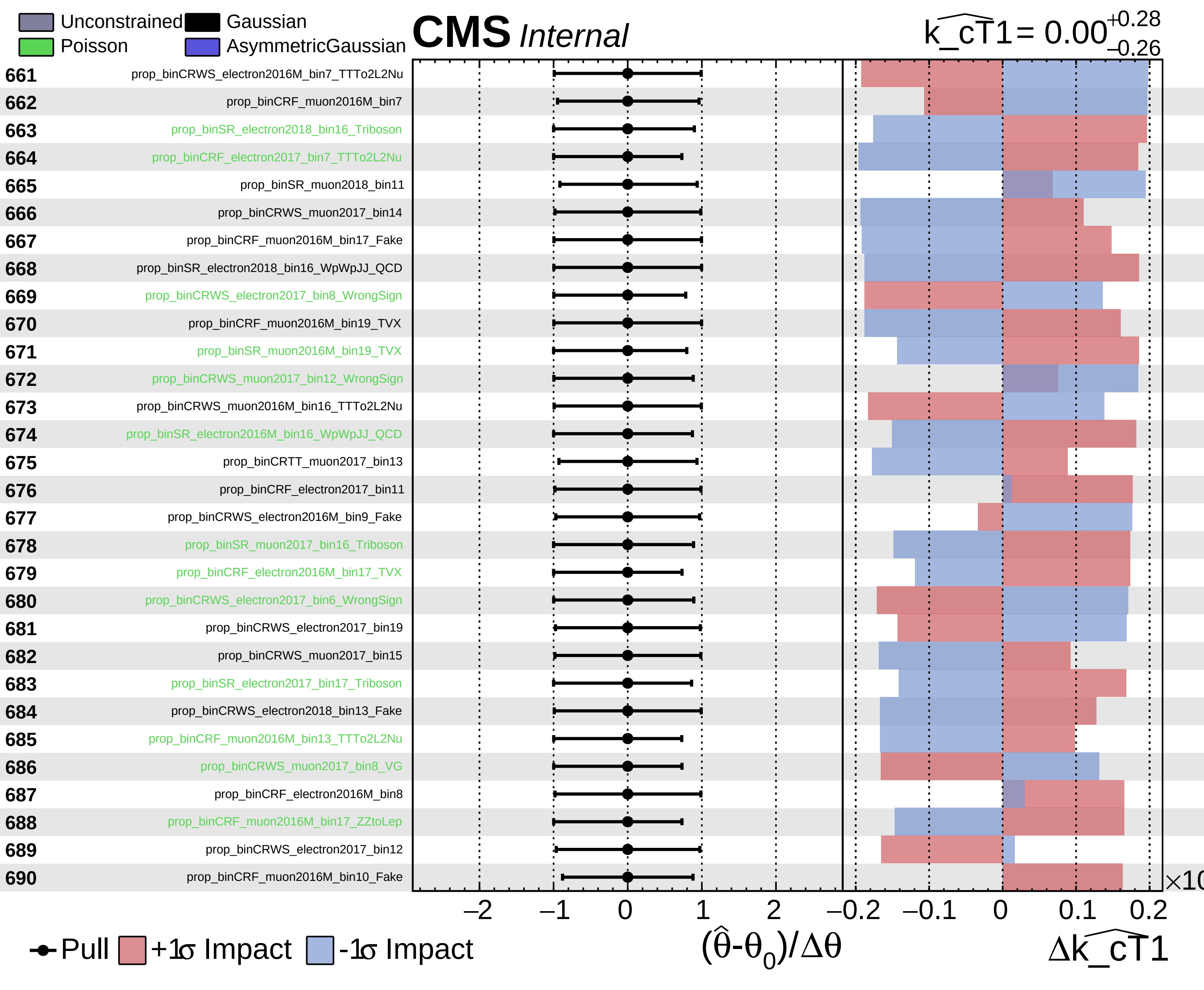

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
571 prop_binCRTT_electron2017_bin7
572 prop_binCRWS_muon2017_bin7_Fake
573 prop_binSR_muon2017_bin14_Fake
574 prop_binSR_electron2018_bin7
575 prop_binCRWS_electron2018_bin18_Fake
576 prop_binCRWS_electron2016M_bin20_TTTo2L2Nu
577 prop_binCRWS_electron2018_bin20
578 prop_binCRWS_muon2017_bin11
579 prop_binCRWS_electron2017_bin20
580 prop_binCRF_muon2016M_bin18_sm
581 prop_binCRWS_muon2017_bin9_Fake
582 prop_binSR_electron2016M_bin16_TVX
583 prop_binCRF_electron2018_bin6
584 prop_binCRWS_electron2018_bin18_TTTo2L2Nu
585 prop_binCRF_muon2016M_bin15
586 prop_binCRTT_electron2017_bin12
587 prop_binCRF_muon2017_bin15_WrongSign
588 prop_binSR_electron2017_bin17_WpWpJJ_QCD
589 prop_binCRWS_electron2018_bin18_VG
590 prop_binCRF_muon2017_bin6
591 prop_binSR_electron2018_bin11_ZZtoLep
592 prop_binCRTT_electron2016M_bin11
593 prop_binCRF_electron2017_bin20_WrongSign
594 prop_binCRF_electron2016M_bin7_TTTo2L2Nu
595 prop_binSR_muon2018_bin17_TVX
596 prop_binCRF_electron2018_bin9
597 prop_binCRTT_electron2018_bin17
598 prop_binSR_electron2017_bin15_ZZtoLep
599 prop_binSR_muon2018_bin17_WpWpJJ_QCD
600 prop_binCRWS_electron2016M_bin12
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



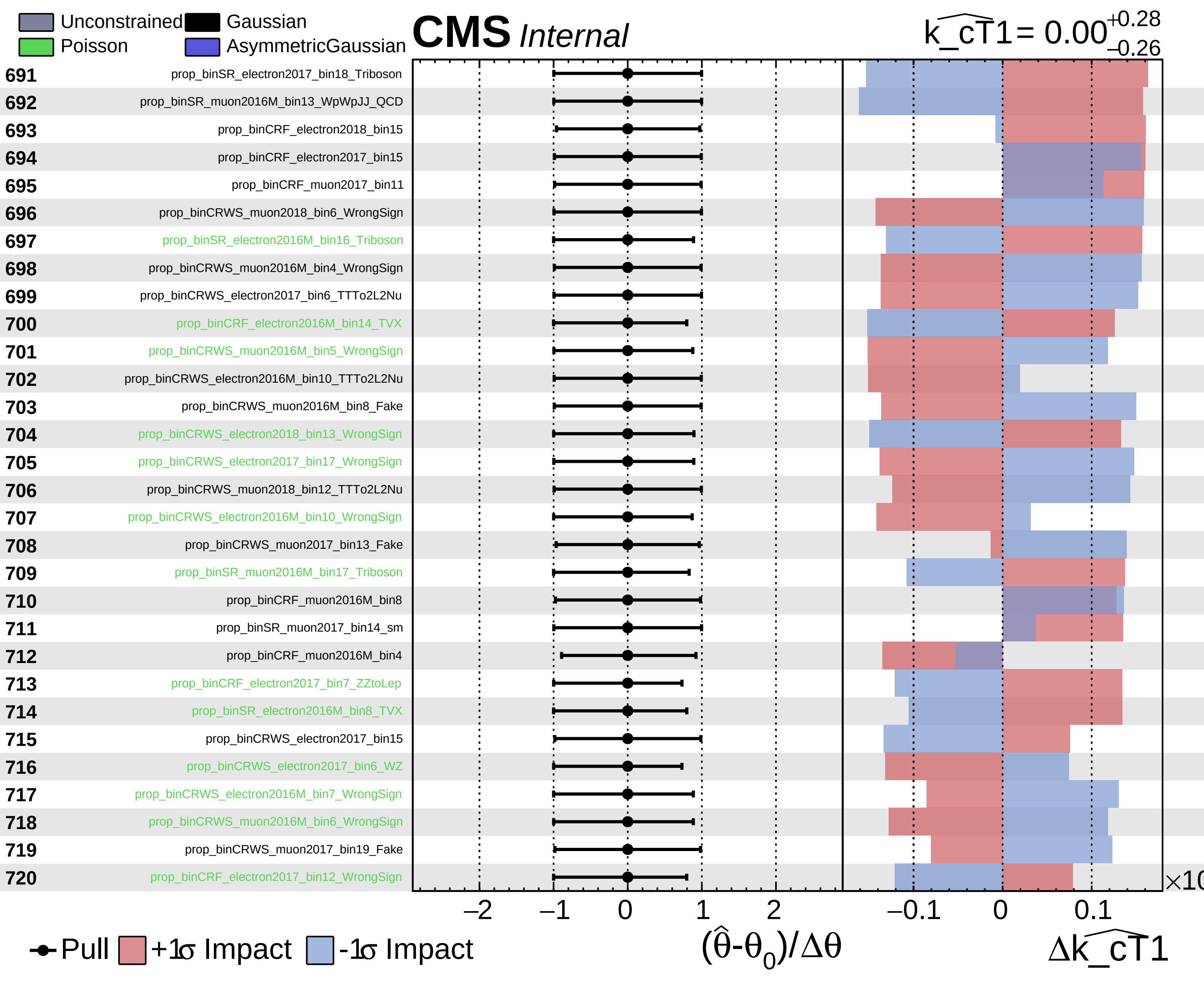
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
601 prop_binCRF_electron2018_bin7_sm
602 prop_binSR_electron2016M_bin10
603 prop_binCRWS_electron2016M_bin7_Fake
604 prop_binCRWS_electron2017_bin8_Fake
605 prop_binCRF_electron2018_bin12
606 prop_binCRF_muon2016M_bin5
607 prop_binCRWS_electron2017_bin17_Fake
608 prop_binCRF_electron2017_bin18_Fake
609 prop_binCRWS_electron2018_bin18_WrongSign
610 prop_binCRWS_electron2017_bin5_Fake
611 prop_binCRWS_muon2016M_bin12
612 prop_binCRF_electron2016M_bin9
613 prop_binCRTT_electron2018_bin16
614 prop_binCRWS_electron2016M_bin11
615 prop_binSR_muon2016M_bin17_TVX
616 prop_binCRF_muon2016M_bin19_WrongSign
617 prop_binCRF_muon2017_bin16
618 prop_binCRWS_electron2017_bin6_Fake
619 prop_binSR_electron2018_bin16_WrongSign
620 prop_binCRF_electron2018_bin16_TTTo2L2Nu
621 prop_binCRWS_muon2016M_bin20
622 prop_binSR_muon2017_bin16_WpWpJJ_QCD
623 prop_binCRF_electron2018_bin14
624 prop_binCRWS_muon2018_bin18_WrongSign
625 prop_binCRWS_muon2018_bin13_WrongSign
626 prop_binCRF_electron2017_bin7_Fake
627 prop_binSR_muon2018_bin17_Triboson
628 prop_binCRF_electron2016M_bin12_sm
629 prop_binCRF_electron2017_bin10
630 prop_binSR_electron2017_bin11_ZZtoLep
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



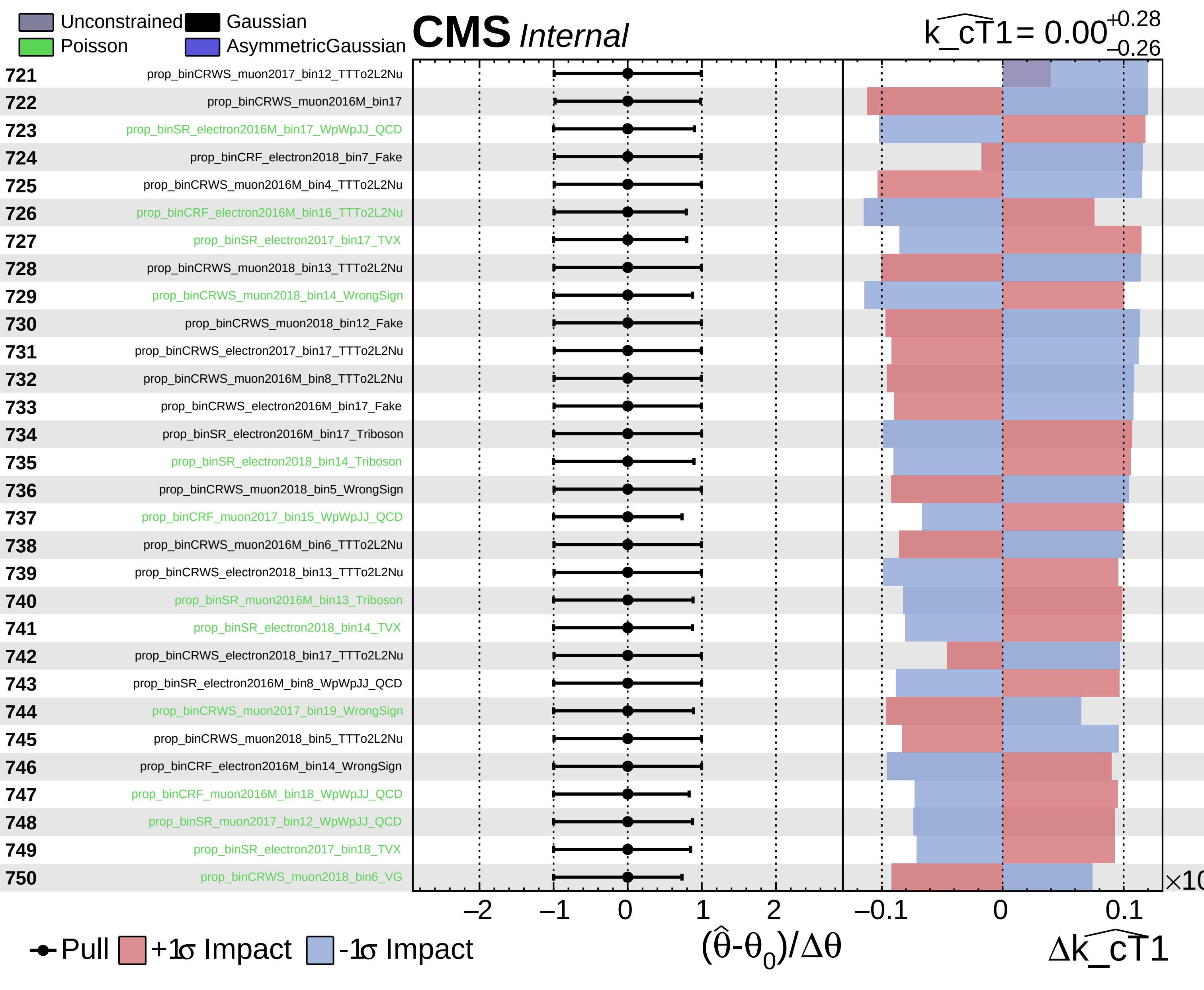
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
631 prop_binSR_muon2016M_bin2
632 prop_binCRF_electron2017_bin20_Triboson
633 prop_binSR_muon2016M_bin13_TVX
634 prop_binCRF_muon2016M_bin11
635 prop_binCRWS_muon2016M_bin11
636 prop_binCRWS_muon2017_bin5_Fake
637 prop_binSR_electron2018_bin16_TVX
638 prop_binCRTT_electron2018_bin13
639 prop_binCRWS_electron2016M_bin13
640 prop_binCRWS_muon2016M_bin6_Fake
641 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
642 prop_binCRWS_electron2018_bin19
643 prop_binCRWS_electron2017_bin13_WrongSign
644 prop_binCRF_muon2017_bin4
645 prop_binCRF_muon2016M_bin16_TVX
646 prop_binCRTT_muon2017_bin10
647 prop_binCRWS_electron2016M_bin16_Fake
648 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
649 prop_binCRF_electron2016M_bin17_WpWpJJ_QCD
650 prop_binCRWS_muon2018_bin6_TTTo2L2Nu
651 prop_binSR_electron2017_bin17_WrongSign
652 prop_binCRWS_electron2018_bin11
653 prop_binCRF_electron2016M_bin5_Fake
654 prop_binCRWS_muon2018_bin18_TTTo2L2Nu
655 prop_binSR_electron2017_bin18_WpWpJJ_QCD
656 prop_binCRWS_muon2016M_bin18_Fake
657 prop_binCRF_electron2018_bin5
658 prop_binCRF_electron2017_bin17
659 prop_binCRTT_electron2017_bin5
660 prop_binSR_muon2017_bin16_TVX
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

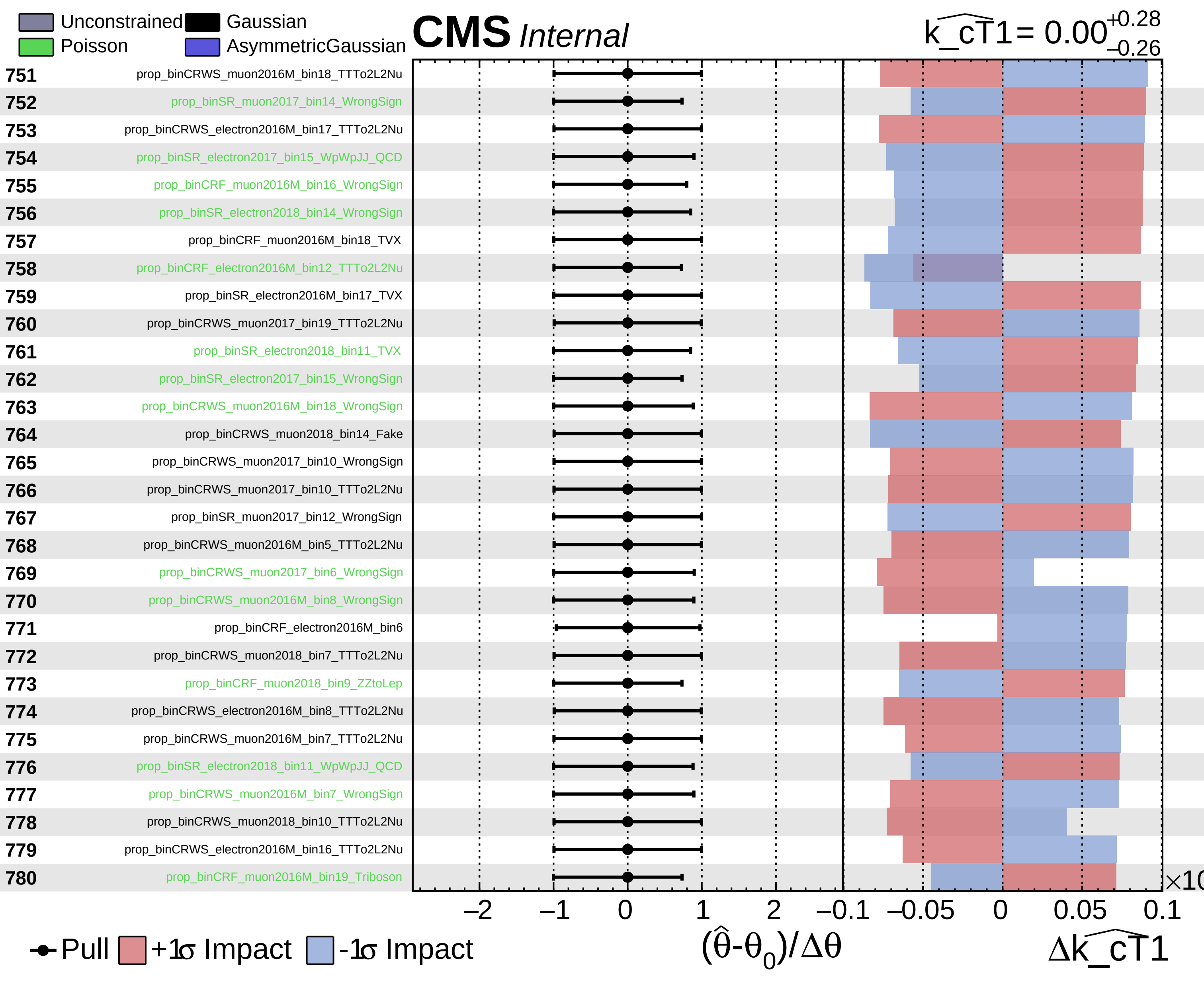


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
661 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
662 prop_binCRF_muon2016M_bin7
663 prop_binSR_electron2018_bin16_Triboson
664 prop_binCRF_electron2017_bin7_TTTo2L2Nu
665 prop_binSR_muon2018_bin11
666 prop_binCRWS_muon2017_bin14
667 prop_binCRF_muon2016M_bin17_Fake
668 prop_binSR_electron2018_bin16_WpWpJJ_QCD
669 prop_binCRWS_electron2017_bin8_WrongSign
670 prop_binCRF_muon2016M_bin19_TVX
671 prop_binSR_muon2016M_bin19_TVX
672 prop_binCRWS_muon2017_bin12_WrongSign
673 prop_binCRWS_muon2016M_bin16_TTTo2L2Nu
674 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
675 prop_binCRTT_muon2017_bin13
676 prop_binCRF_electron2017_bin11
677 prop_binCRWS_electron2016M_bin9_Fake
678 prop_binSR_muon2017_bin16_Triboson
679 prop_binCRF_electron2016M_bin17_TVX
680 prop_binCRWS_electron2017_bin6_WrongSign
681 prop_binCRWS_electron2017_bin19
682 prop_binCRWS_muon2017_bin15
683 prop_binSR_electron2017_bin17_Triboson
684 prop_binCRWS_electron2018_bin13_Fake
685 prop_binCRF_muon2016M_bin13_TTTo2L2Nu
686 prop_binCRWS_muon2017_bin8_VG
687 prop_binCRF_electron2016M_bin8
688 prop_binCRF_muon2016M_bin17_ZZtoLep
689 prop_binCRWS_electron2017_bin12
690 prop_binCRF_muon2016M_bin10_Fake
−2
−1
0
1
2
−0.2
−0.1
0
0.1
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
691 prop_binSR_electron2017_bin18_Triboson
692 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
693 prop_binCRF_electron2018_bin15
694 prop_binCRF_electron2017_bin15
695 prop_binCRF_muon2017_bin11
696 prop_binCRWS_muon2018_bin6_WrongSign
697 prop_binSR_electron2016M_bin16_Triboson
698 prop_binCRWS_muon2016M_bin4_WrongSign
699 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
700 prop_binCRF_electron2016M_bin14_TVX
701 prop_binCRWS_muon2016M_bin5_WrongSign
702 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
703 prop_binCRWS_muon2016M_bin8_Fake
704 prop_binCRWS_electron2018_bin13_WrongSign
705 prop_binCRWS_electron2017_bin17_WrongSign
706 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
707 prop_binCRWS_electron2016M_bin10_WrongSign
708 prop_binCRWS_muon2017_bin13_Fake
709 prop_binSR_muon2016M_bin17_Triboson
710 prop_binCRF_muon2016M_bin8
711 prop_binSR_muon2017_bin14_sm
712 prop_binCRF_muon2016M_bin4
713 prop_binCRF_electron2017_bin7_ZZtoLep
714 prop_binSR_electron2016M_bin8_TVX
715 prop_binCRWS_electron2017_bin15
716 prop_binCRWS_electron2017_bin6_WZ
717 prop_binCRWS_electron2016M_bin7_WrongSign
718 prop_binCRWS_muon2016M_bin6_WrongSign
719 prop_binCRWS_muon2017_bin19_Fake
720 prop_binCRF_electron2017_bin12_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
721 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
722 prop_binCRWS_muon2016M_bin17
723 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
724 prop_binCRF_electron2018_bin7_Fake
725 prop_binCRWS_muon2016M_bin4_TTTo2L2Nu
726 prop_binCRF_electron2016M_bin16_TTTo2L2Nu
727 prop_binSR_electron2017_bin17_TVX
728 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
729 prop_binCRWS_muon2018_bin14_WrongSign
730 prop_binCRWS_muon2018_bin12_Fake
731 prop_binCRWS_electron2017_bin17_TTTo2L2Nu
732 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
733 prop_binCRWS_electron2016M_bin17_Fake
734 prop_binSR_electron2016M_bin17_Triboson
735 prop_binSR_electron2018_bin14_Triboson
736 prop_binCRWS_muon2018_bin5_WrongSign
737 prop_binCRF_muon2017_bin15_WpWpJJ_QCD
738 prop_binCRWS_muon2016M_bin6_TTTo2L2Nu
739 prop_binCRWS_electron2018_bin13_TTTo2L2Nu
740 prop_binSR_muon2016M_bin13_Triboson
741 prop_binSR_electron2018_bin14_TVX
742 prop_binCRWS_electron2018_bin17_TTTo2L2Nu
743 prop_binSR_electron2016M_bin8_WpWpJJ_QCD
744 prop_binCRWS_muon2017_bin19_WrongSign
745 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
746 prop_binCRF_electron2016M_bin14_WrongSign
747 prop_binCRF_muon2016M_bin18_WpWpJJ_QCD
748 prop_binSR_muon2017_bin12_WpWpJJ_QCD
749 prop_binSR_electron2017_bin18_TVX
750 prop_binCRWS_muon2018_bin6_VG
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1



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Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
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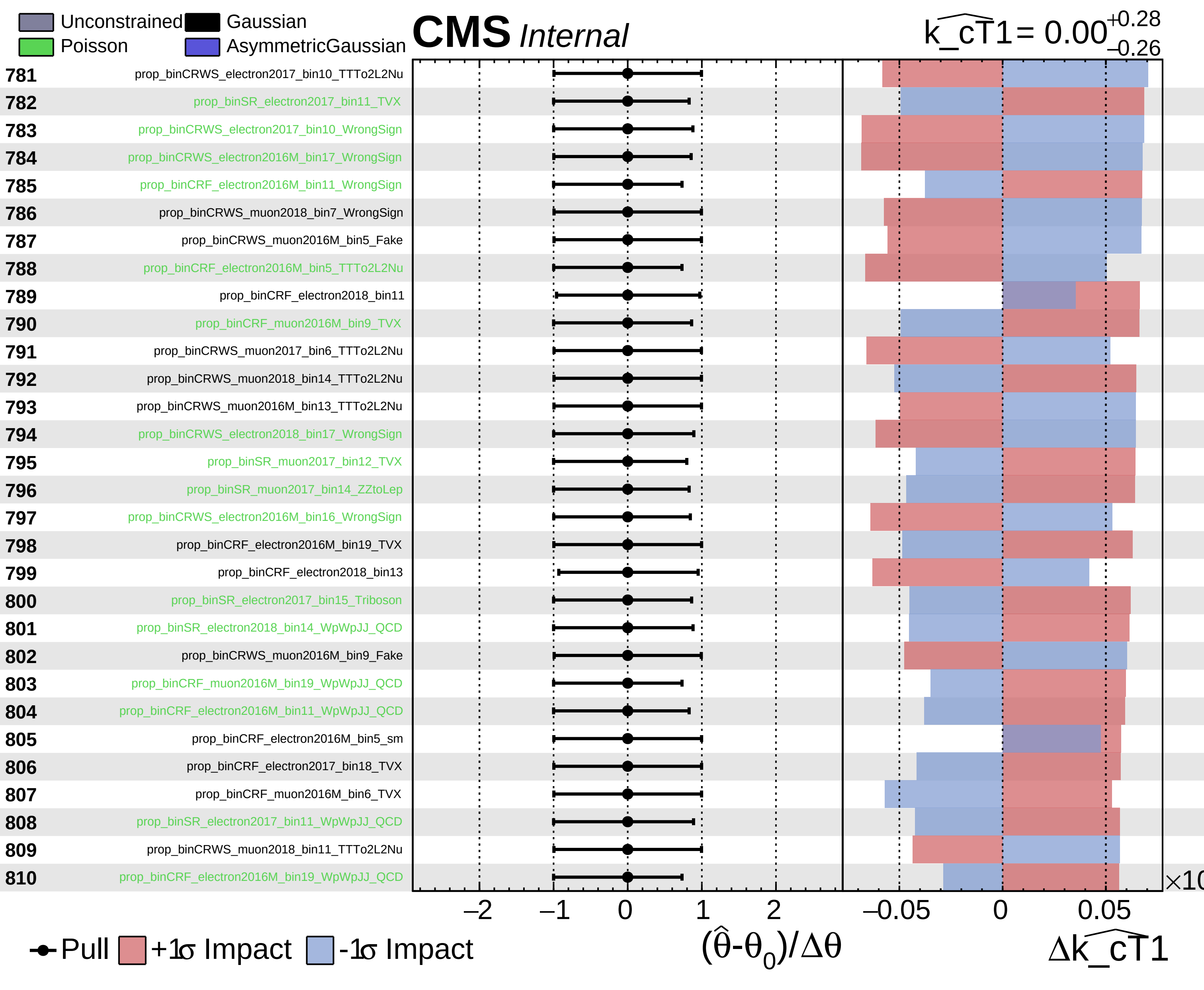
752 prop_binSR_muon2017_bin14_WrongSign
753 prop_binCRWS_electron2016M_bin17_TTTo2L2Nu
754 prop_binSR_electron2017_bin15_WpWpJJ_QCD
755 prop_binCRF_muon2016M_bin16_WrongSign
756 prop_binSR_electron2018_bin14_WrongSign
757 prop_binCRF_muon2016M_bin18_TVX
758 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
759 prop_binSR_electron2016M_bin17_TVX
760 prop_binCRWS_muon2017_bin19_TTTo2L2Nu
761 prop_binSR_electron2018_bin11_TVX
762 prop_binSR_electron2017_bin15_WrongSign
763 prop_binCRWS_muon2016M_bin18_WrongSign
764 prop_binCRWS_muon2018_bin14_Fake
765 prop_binCRWS_muon2017_bin10_WrongSign
766 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
767 prop_binSR_muon2017_bin12_WrongSign
768 prop_binCRWS_muon2016M_bin5_TTTo2L2Nu
769 prop_binCRWS_muon2017_bin6_WrongSign
770 prop_binCRWS_muon2016M_bin8_WrongSign
771 prop_binCRF_electron2016M_bin6
772 prop_binCRWS_muon2018_bin7_TTTo2L2Nu
773 prop_binCRF_muon2018_bin9_ZZtoLep
774 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
775 prop_binCRWS_muon2016M_bin7_TTTo2L2Nu
776 prop_binSR_electron2018_bin11_WpWpJJ_QCD
777 prop_binCRWS_muon2016M_bin7_WrongSign
778 prop_binCRWS_muon2018_bin10_TTTo2L2Nu
779 prop_binCRWS_electron2016M_bin16_TTTo2L2Nu
780 prop_binCRF_muon2016M_bin19_Triboson
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
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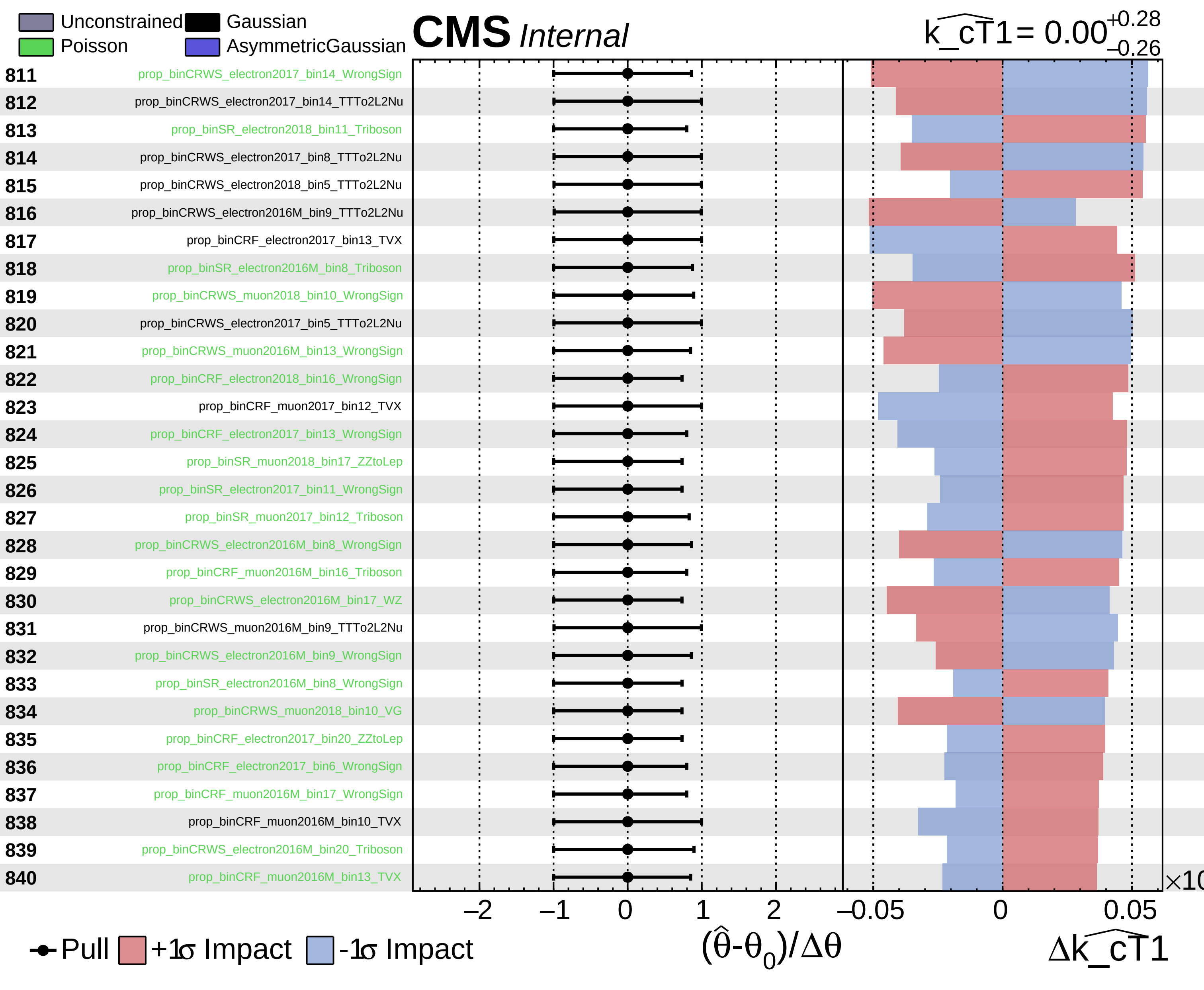
782 prop_binSR_electron2017_bin11_TVX
783 prop_binCRWS_electron2017_bin10_WrongSign
784 prop_binCRWS_electron2016M_bin17_WrongSign
785 prop_binCRF_electron2016M_bin11_WrongSign
786 prop_binCRWS_muon2018_bin7_WrongSign
787 prop_binCRWS_muon2016M_bin5_Fake
788 prop_binCRF_electron2016M_bin5_TTTo2L2Nu
789 prop_binCRF_electron2018_bin11
790 prop_binCRF_muon2016M_bin9_TVX
791 prop_binCRWS_muon2017_bin6_TTTo2L2Nu
792 prop_binCRWS_muon2018_bin14_TTTo2L2Nu
793 prop_binCRWS_muon2016M_bin13_TTTo2L2Nu
794 prop_binCRWS_electron2018_bin17_WrongSign
795 prop_binSR_muon2017_bin12_TVX
796 prop_binSR_muon2017_bin14_ZZtoLep
797 prop_binCRWS_electron2016M_bin16_WrongSign
798 prop_binCRF_electron2016M_bin19_TVX
799 prop_binCRF_electron2018_bin13
800 prop_binSR_electron2017_bin15_Triboson
801 prop_binSR_electron2018_bin14_WpWpJJ_QCD
802 prop_binCRWS_muon2016M_bin9_Fake
803 prop_binCRF_muon2016M_bin19_WpWpJJ_QCD
804 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
805 prop_binCRF_electron2016M_bin5_sm
806 prop_binCRF_electron2017_bin18_TVX
807 prop_binCRF_muon2016M_bin6_TVX
808 prop_binSR_electron2017_bin11_WpWpJJ_QCD
809 prop_binCRWS_muon2018_bin11_TTTo2L2Nu
810 prop_binCRF_electron2016M_bin19_WpWpJJ_QCD
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



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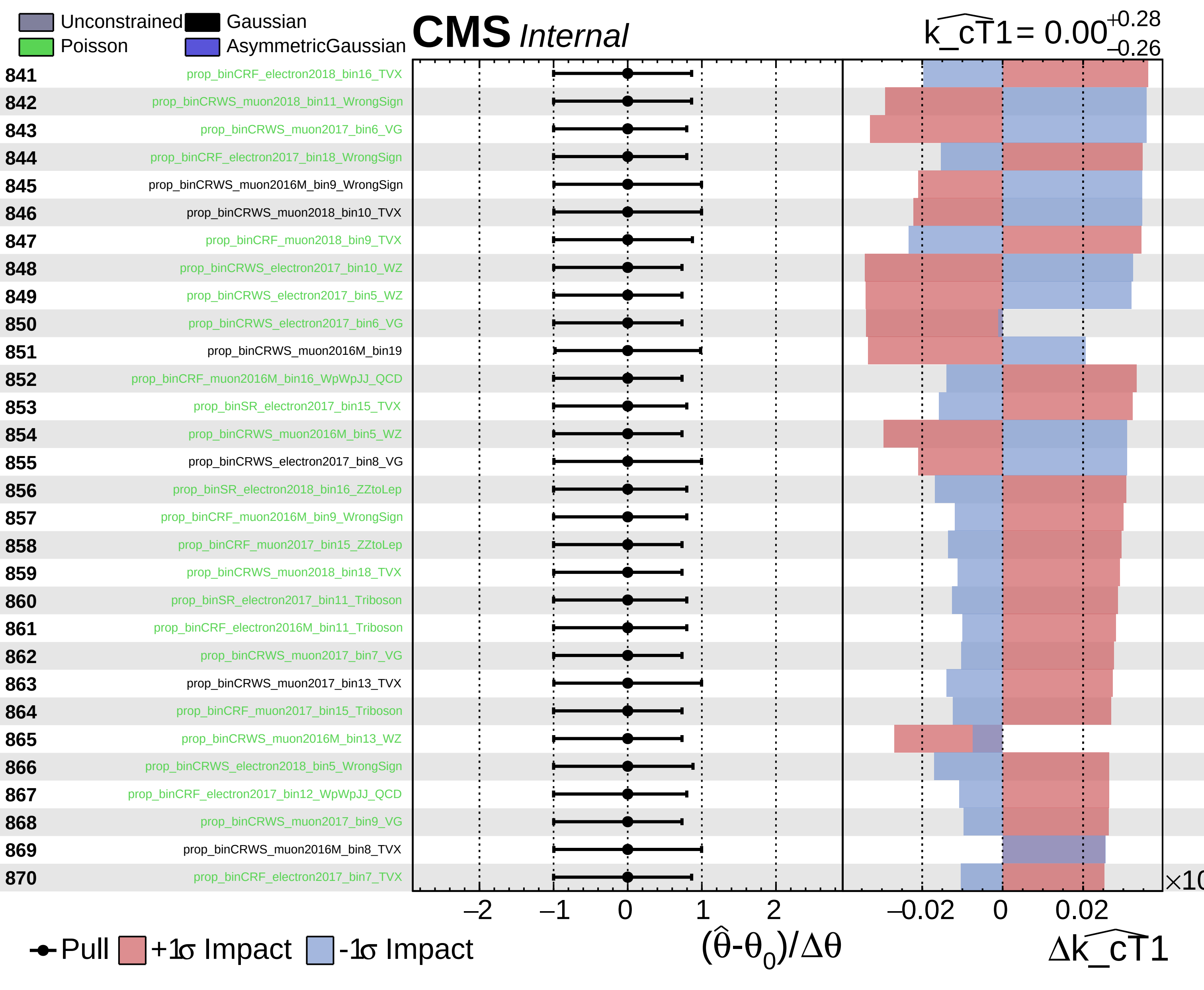
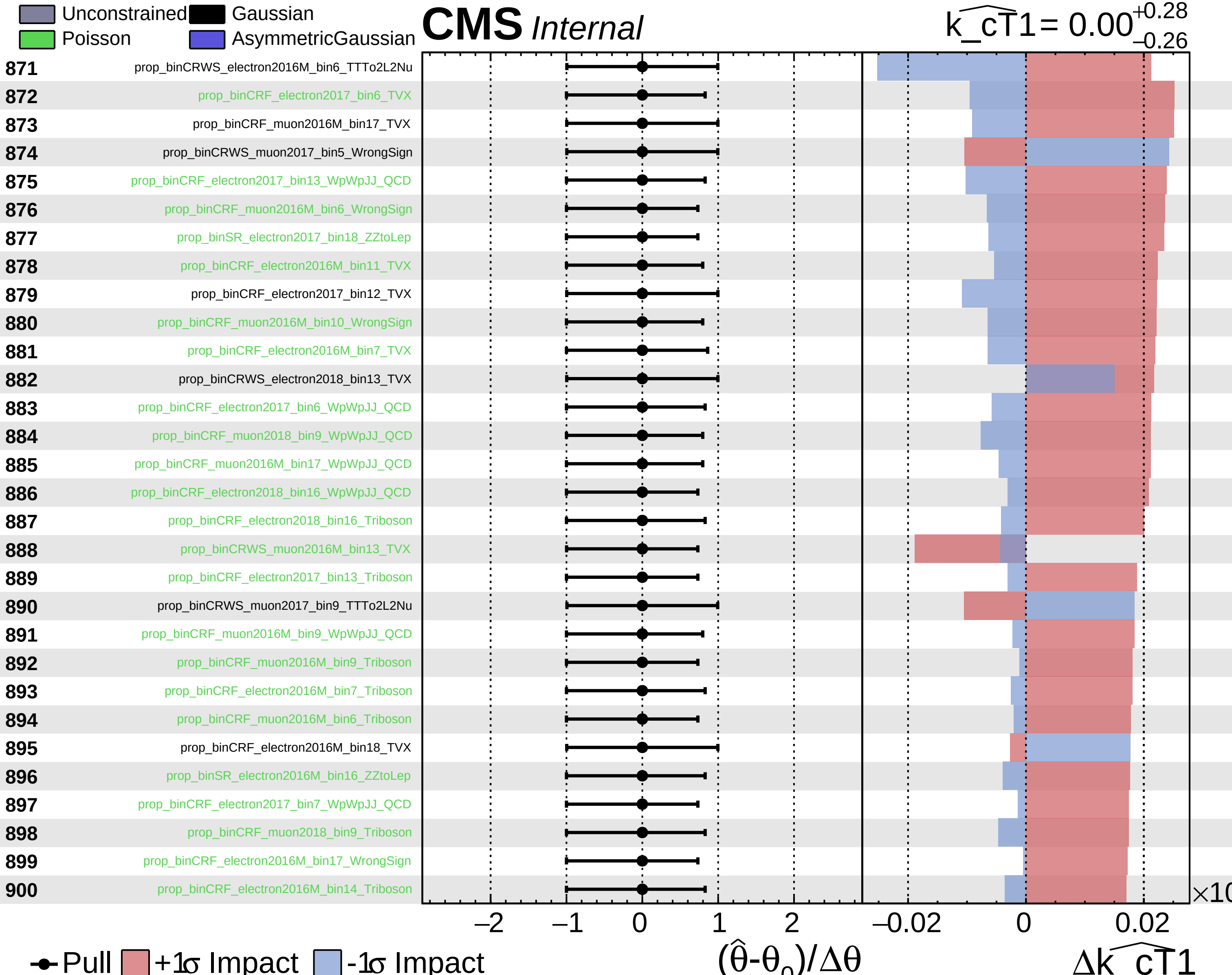
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
811 prop_binCRWS_electron2017_bin14_WrongSign
812 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
813 prop_binSR_electron2018_bin11_Triboson
814 prop_binCRWS_electron2017_bin8_TTTo2L2Nu
815 prop_binCRWS_electron2018_bin5_TTTo2L2Nu
816 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
817 prop_binCRF_electron2017_bin13_TVX
818 prop_binSR_electron2016M_bin8_Triboson
819 prop_binCRWS_muon2018_bin10_WrongSign
820 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
821 prop_binCRWS_muon2016M_bin13_WrongSign
822 prop_binCRF_electron2018_bin16_WrongSign
823 prop_binCRF_muon2017_bin12_TVX
824 prop_binCRF_electron2017_bin13_WrongSign
825 prop_binSR_muon2018_bin17_ZZtoLep
826 prop_binSR_electron2017_bin11_WrongSign
827 prop_binSR_muon2017_bin12_Triboson
828 prop_binCRWS_electron2016M_bin8_WrongSign
829 prop_binCRF_muon2016M_bin16_Triboson
830 prop_binCRWS_electron2016M_bin17_WZ
831 prop_binCRWS_muon2016M_bin9_TTTo2L2Nu
832 prop_binCRWS_electron2016M_bin9_WrongSign
833 prop_binSR_electron2016M_bin8_WrongSign
834 prop_binCRWS_muon2018_bin10_VG
835 prop_binCRF_electron2017_bin20_ZZtoLep
836 prop_binCRF_electron2017_bin6_WrongSign
837 prop_binCRF_muon2016M_bin17_WrongSign
838 prop_binCRF_muon2016M_bin10_TVX
839 prop_binCRWS_electron2016M_bin20_Triboson
840 prop_binCRF_muon2016M_bin13_TVX
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
841 prop_binCRF_electron2018_bin16_TVX
842 prop_binCRWS_muon2018_bin11_WrongSign
843 prop_binCRWS_muon2017_bin6_VG
844 prop_binCRF_electron2017_bin18_WrongSign
845 prop_binCRWS_muon2016M_bin9_WrongSign
846 prop_binCRWS_muon2018_bin10_TVX
847 prop_binCRF_muon2018_bin9_TVX
848 prop_binCRWS_electron2017_bin10_WZ
849 prop_binCRWS_electron2017_bin5_WZ
850 prop_binCRWS_electron2017_bin6_VG
851 prop_binCRWS_muon2016M_bin19
852 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
853 prop_binSR_electron2017_bin15_TVX
854 prop_binCRWS_muon2016M_bin5_WZ
855 prop_binCRWS_electron2017_bin8_VG
856 prop_binSR_electron2018_bin16_ZZtoLep
857 prop_binCRF_muon2016M_bin9_WrongSign
858 prop_binCRF_muon2017_bin15_ZZtoLep
859 prop_binCRWS_muon2018_bin18_TVX
860 prop_binSR_electron2017_bin11_Triboson
861 prop_binCRF_electron2016M_bin11_Triboson
862 prop_binCRWS_muon2017_bin7_VG
863 prop_binCRWS_muon2017_bin13_TVX
864 prop_binCRF_muon2017_bin15_Triboson
865 prop_binCRWS_muon2016M_bin13_WZ
866 prop_binCRWS_electron2018_bin5_WrongSign
867 prop_binCRF_electron2017_bin12_WpWpJJ_QCD
868 prop_binCRWS_muon2017_bin9_VG
869 prop_binCRWS_muon2016M_bin8_TVX
870 prop_binCRF_electron2017_bin7_TVX
×10
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1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Gaussian
Poisson
AsymmetricGaussian
CMS Internal
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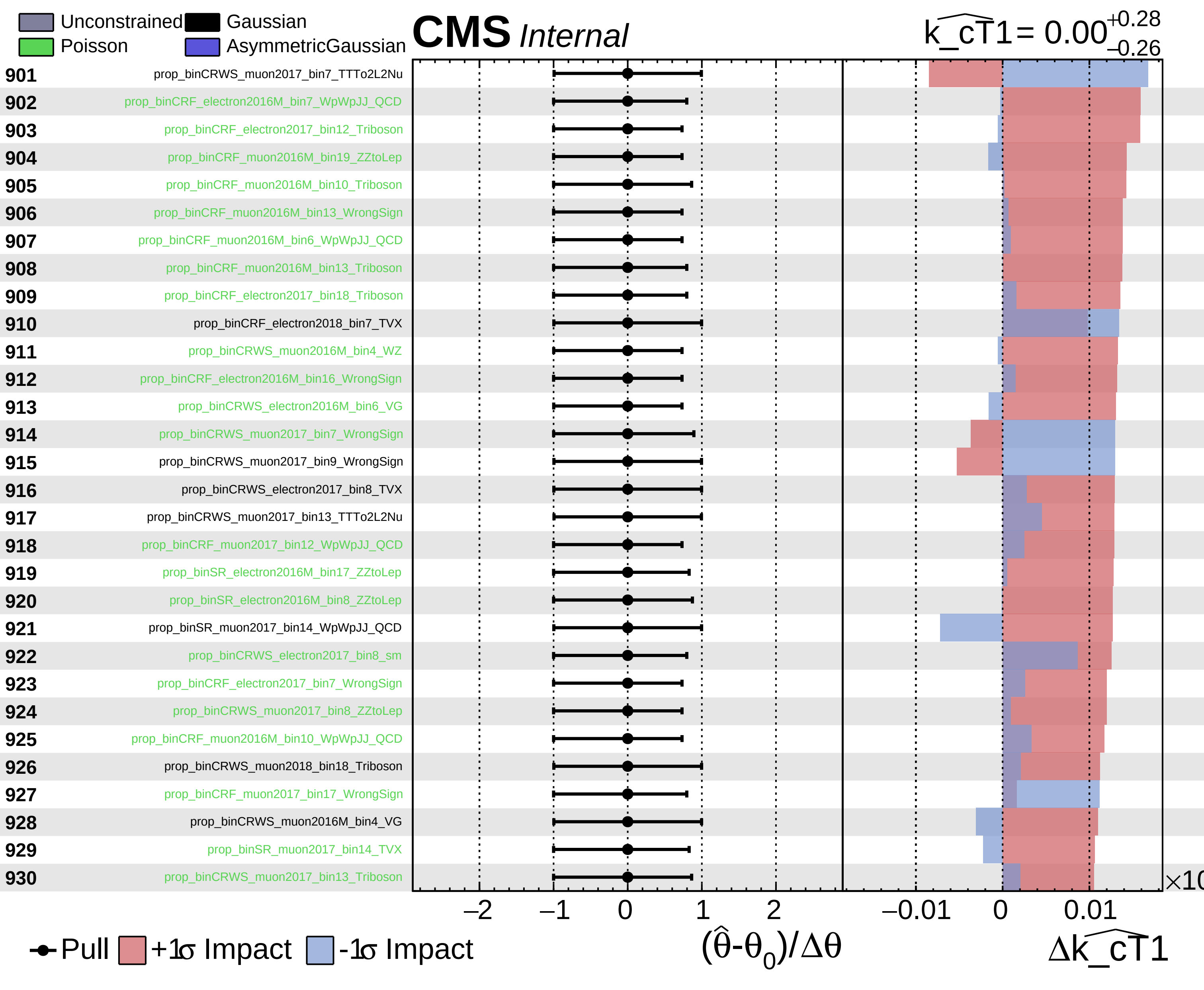
872 prop_binCRF_electron2017_bin6_TVX
873 prop_binCRF_muon2016M_bin17_TVX
874 prop_binCRWS_muon2017_bin5_WrongSign
875 prop_binCRF_electron2017_bin13_WpWpJJ_QCD
876 prop_binCRF_muon2016M_bin6_WrongSign
877 prop_binSR_electron2017_bin18_ZZtoLep
878 prop_binCRF_electron2016M_bin11_TVX
879 prop_binCRF_electron2017_bin12_TVX
880 prop_binCRF_muon2016M_bin10_WrongSign
881 prop_binCRF_electron2016M_bin7_TVX
882 prop_binCRWS_electron2018_bin13_TVX
883 prop_binCRF_electron2017_bin6_WpWpJJ_QCD
884 prop_binCRF_muon2018_bin9_WpWpJJ_QCD
885 prop_binCRF_muon2016M_bin17_WpWpJJ_QCD
886 prop_binCRF_electron2018_bin16_WpWpJJ_QCD
887 prop_binCRF_electron2018_bin16_Triboson
888 prop_binCRWS_muon2016M_bin13_TVX
889 prop_binCRF_electron2017_bin13_Triboson
890 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
891 prop_binCRF_muon2016M_bin9_WpWpJJ_QCD
892 prop_binCRF_muon2016M_bin9_Triboson
893 prop_binCRF_electron2016M_bin7_Triboson
894 prop_binCRF_muon2016M_bin6_Triboson
895 prop_binCRF_electron2016M_bin18_TVX
896 prop_binSR_electron2016M_bin16_ZZtoLep
897 prop_binCRF_electron2017_bin7_WpWpJJ_QCD
898 prop_binCRF_muon2018_bin9_Triboson
899 prop_binCRF_electron2016M_bin17_WrongSign
900 prop_binCRF_electron2016M_bin14_Triboson
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1



Unconstrained
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Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
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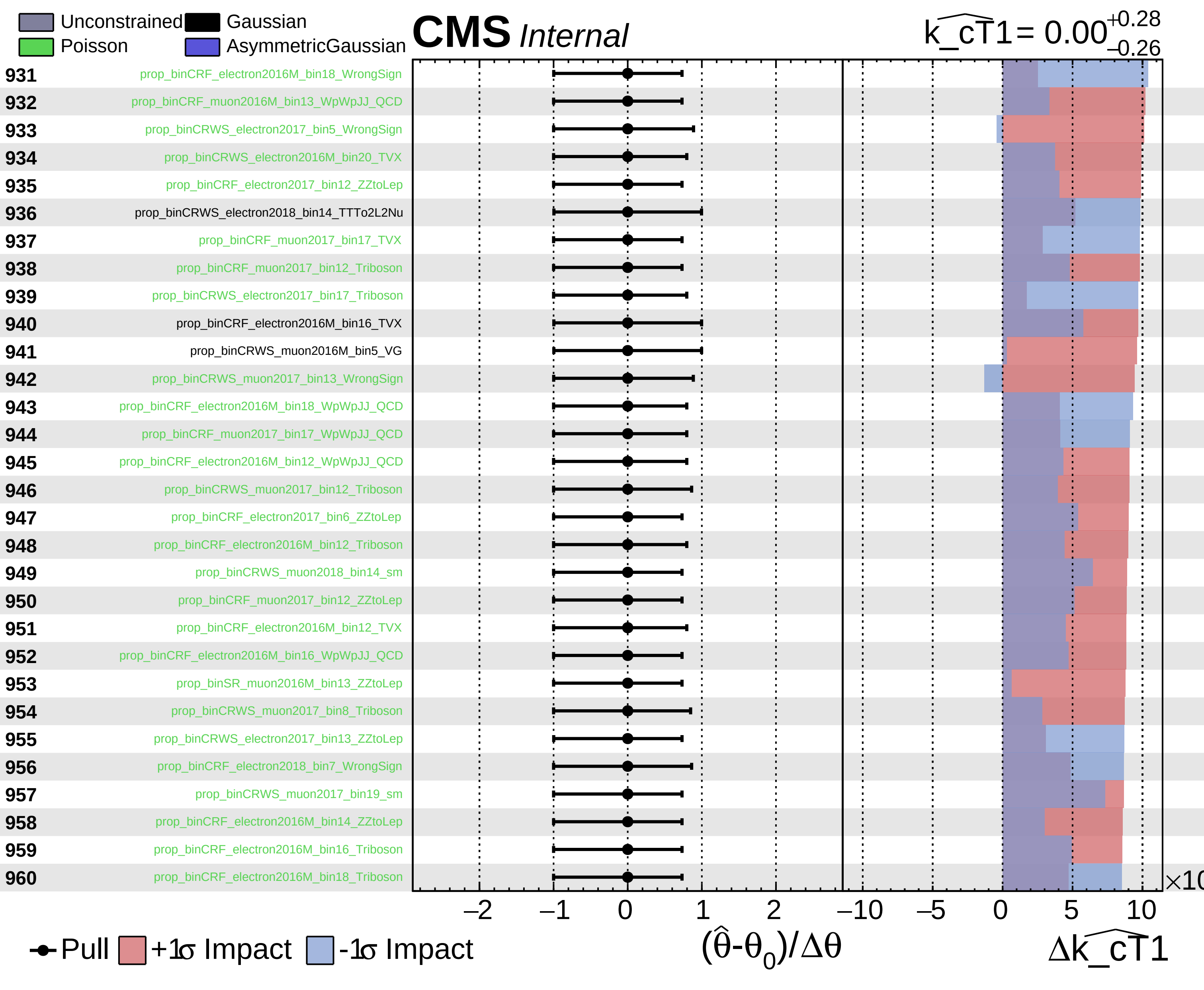
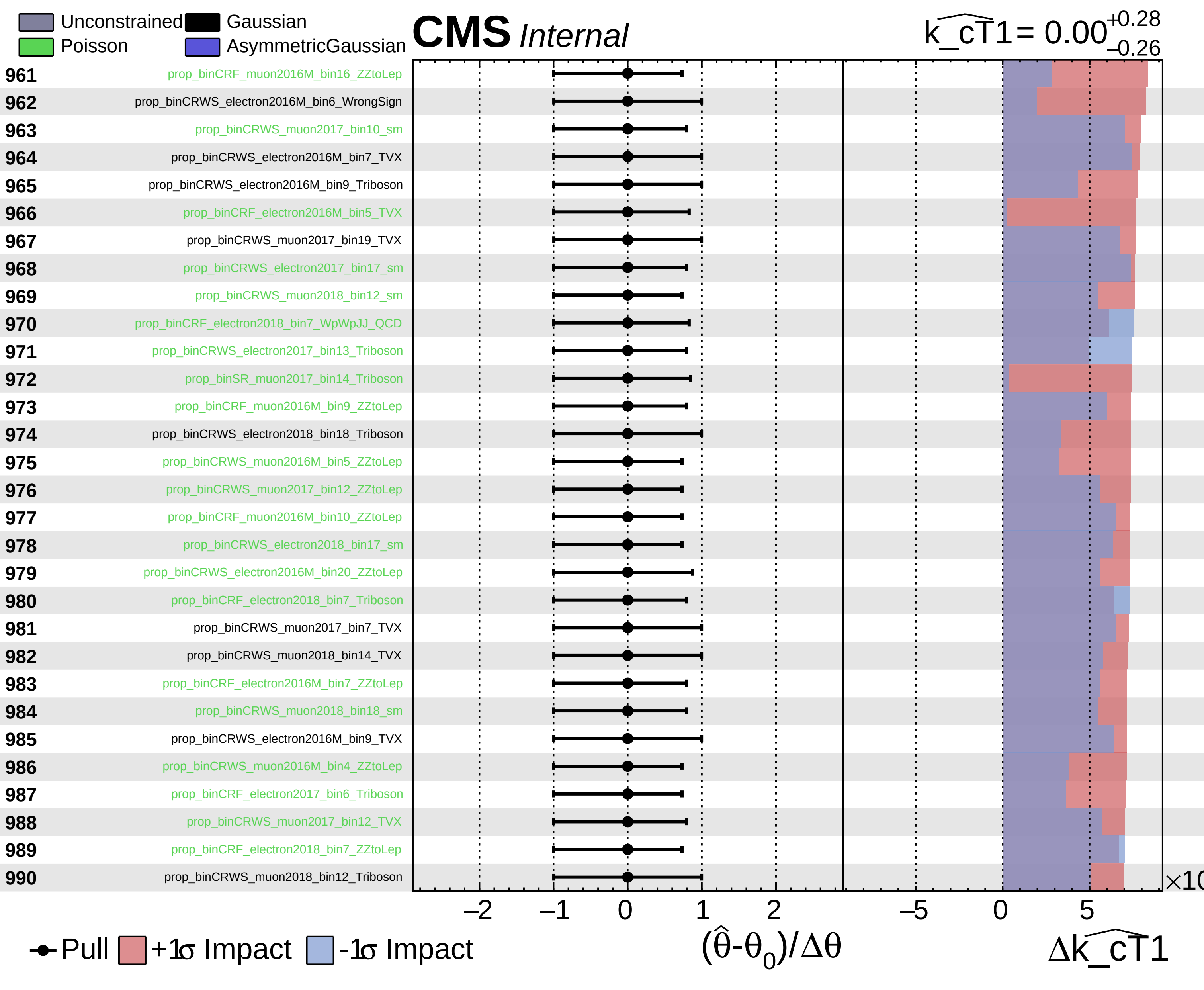
902 prop_binCRF_electron2016M_bin7_WpWpJJ_QCD
903 prop_binCRF_electron2017_bin12_Triboson
904 prop_binCRF_muon2016M_bin19_ZZtoLep
905 prop_binCRF_muon2016M_bin10_Triboson
906 prop_binCRF_muon2016M_bin13_WrongSign
907 prop_binCRF_muon2016M_bin6_WpWpJJ_QCD
908 prop_binCRF_muon2016M_bin13_Triboson
909 prop_binCRF_electron2017_bin18_Triboson
910 prop_binCRF_electron2018_bin7_TVX
911 prop_binCRWS_muon2016M_bin4_WZ
912 prop_binCRF_electron2016M_bin16_WrongSign
913 prop_binCRWS_electron2016M_bin6_VG
914 prop_binCRWS_muon2017_bin7_WrongSign
915 prop_binCRWS_muon2017_bin9_WrongSign
916 prop_binCRWS_electron2017_bin8_TVX
917 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
918 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
919 prop_binSR_electron2016M_bin17_ZZtoLep
920 prop_binSR_electron2016M_bin8_ZZtoLep
921 prop_binSR_muon2017_bin14_WpWpJJ_QCD
922 prop_binCRWS_electron2017_bin8_sm
923 prop_binCRF_electron2017_bin7_WrongSign
924 prop_binCRWS_muon2017_bin8_ZZtoLep
925 prop_binCRF_muon2016M_bin10_WpWpJJ_QCD
926 prop_binCRWS_muon2018_bin18_Triboson
927 prop_binCRF_muon2017_bin17_WrongSign
928 prop_binCRWS_muon2016M_bin4_VG
929 prop_binSR_muon2017_bin14_TVX
930 prop_binCRWS_muon2017_bin13_Triboson
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
931 prop_binCRF_electron2016M_bin18_WrongSign
932 prop_binCRF_muon2016M_bin13_WpWpJJ_QCD
933 prop_binCRWS_electron2017_bin5_WrongSign
934 prop_binCRWS_electron2016M_bin20_TVX
935 prop_binCRF_electron2017_bin12_ZZtoLep
936 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
937 prop_binCRF_muon2017_bin17_TVX
938 prop_binCRF_muon2017_bin12_Triboson
939 prop_binCRWS_electron2017_bin17_Triboson
940 prop_binCRF_electron2016M_bin16_TVX
941 prop_binCRWS_muon2016M_bin5_VG
942 prop_binCRWS_muon2017_bin13_WrongSign
943 prop_binCRF_electron2016M_bin18_WpWpJJ_QCD
944 prop_binCRF_muon2017_bin17_WpWpJJ_QCD
945 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
946 prop_binCRWS_muon2017_bin12_Triboson
947 prop_binCRF_electron2017_bin6_ZZtoLep
948 prop_binCRF_electron2016M_bin12_Triboson
949 prop_binCRWS_muon2018_bin14_sm
950 prop_binCRF_muon2017_bin12_ZZtoLep
951 prop_binCRF_electron2016M_bin12_TVX
952 prop_binCRF_electron2016M_bin16_WpWpJJ_QCD
953 prop_binSR_muon2016M_bin13_ZZtoLep
954 prop_binCRWS_muon2017_bin8_Triboson
955 prop_binCRWS_electron2017_bin13_ZZtoLep
956 prop_binCRF_electron2018_bin7_WrongSign
957 prop_binCRWS_muon2017_bin19_sm
958 prop_binCRF_electron2016M_bin14_ZZtoLep
959 prop_binCRF_electron2016M_bin16_Triboson
960 prop_binCRF_electron2016M_bin18_Triboson
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$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Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
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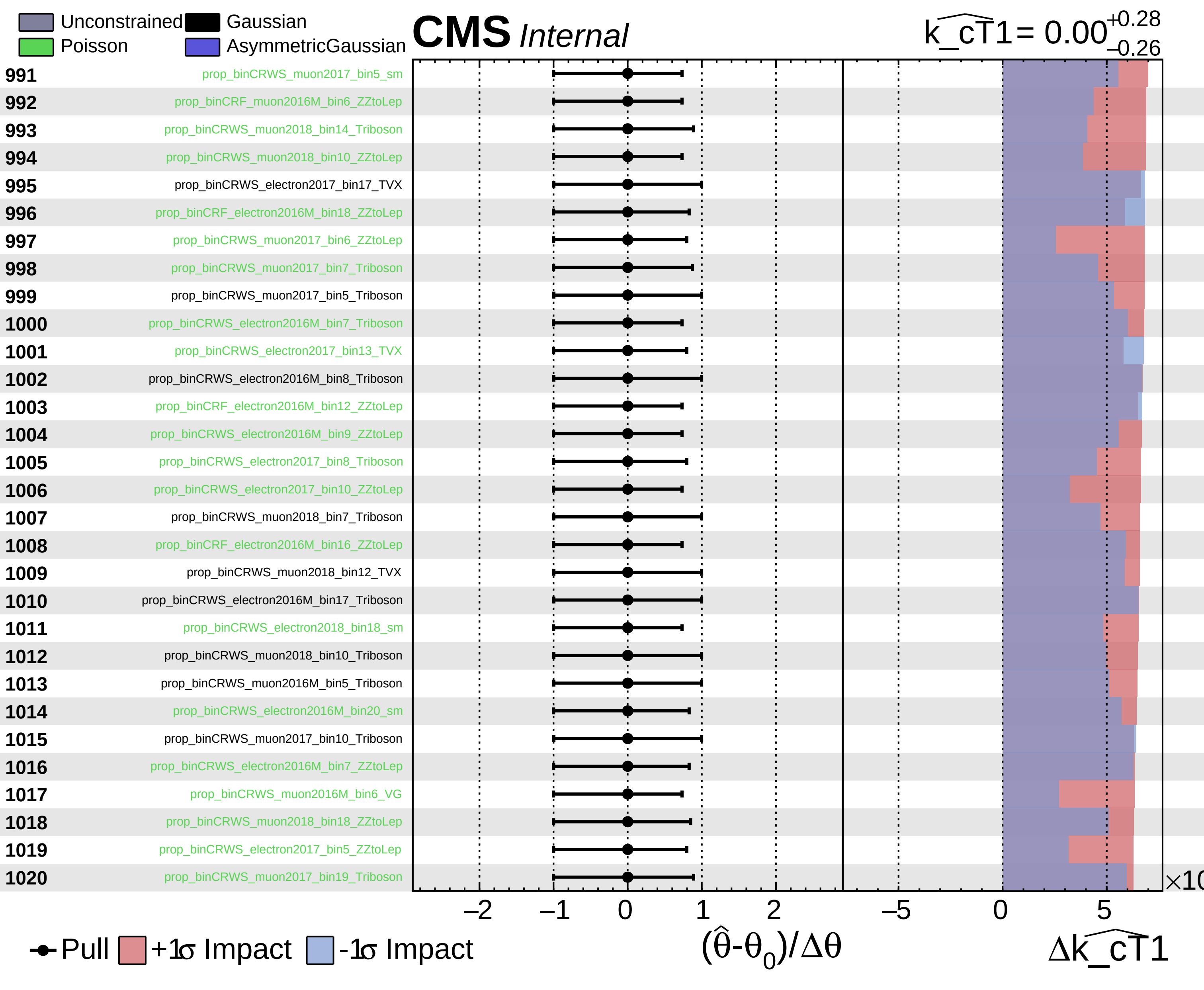
964 prop_binCRWS_electron2016M_bin7_TVX
965 prop_binCRWS_electron2016M_bin9_Triboson
966 prop_binCRF_electron2016M_bin5_TVX
967 prop_binCRWS_muon2017_bin19_TVX
968 prop_binCRWS_electron2017_bin17_sm
969 prop_binCRWS_muon2018_bin12_sm
970 prop_binCRF_electron2018_bin7_WpWpJJ_QCD
971 prop_binCRWS_electron2017_bin13_Triboson
972 prop_binSR_muon2017_bin14_Triboson
973 prop_binCRF_muon2016M_bin9_ZZtoLep
974 prop_binCRWS_electron2018_bin18_Triboson
975 prop_binCRWS_muon2016M_bin5_ZZtoLep
976 prop_binCRWS_muon2017_bin12_ZZtoLep
977 prop_binCRF_muon2016M_bin10_ZZtoLep
978 prop_binCRWS_electron2018_bin17_sm
979 prop_binCRWS_electron2016M_bin20_ZZtoLep
980 prop_binCRF_electron2018_bin7_Triboson
981 prop_binCRWS_muon2017_bin7_TVX
982 prop_binCRWS_muon2018_bin14_TVX
983 prop_binCRF_electron2016M_bin7_ZZtoLep
984 prop_binCRWS_muon2018_bin18_sm
985 prop_binCRWS_electron2016M_bin9_TVX
986 prop_binCRWS_muon2016M_bin4_ZZtoLep
987 prop_binCRF_electron2017_bin6_Triboson
988 prop_binCRWS_muon2017_bin12_TVX
989 prop_binCRF_electron2018_bin7_ZZtoLep
990 prop_binCRWS_muon2018_bin12_Triboson
−2 −1 0 1 2
−5 0 5
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
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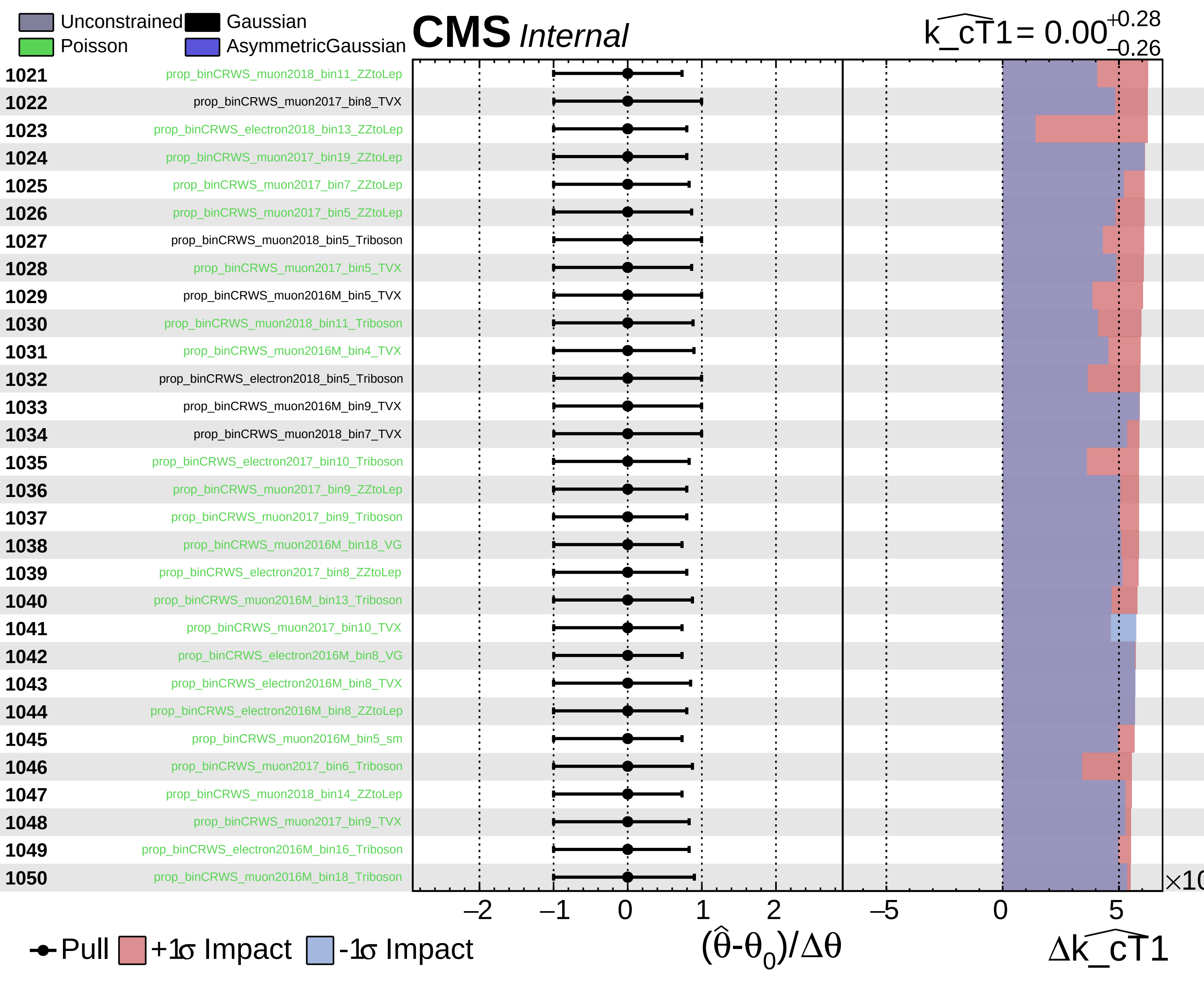
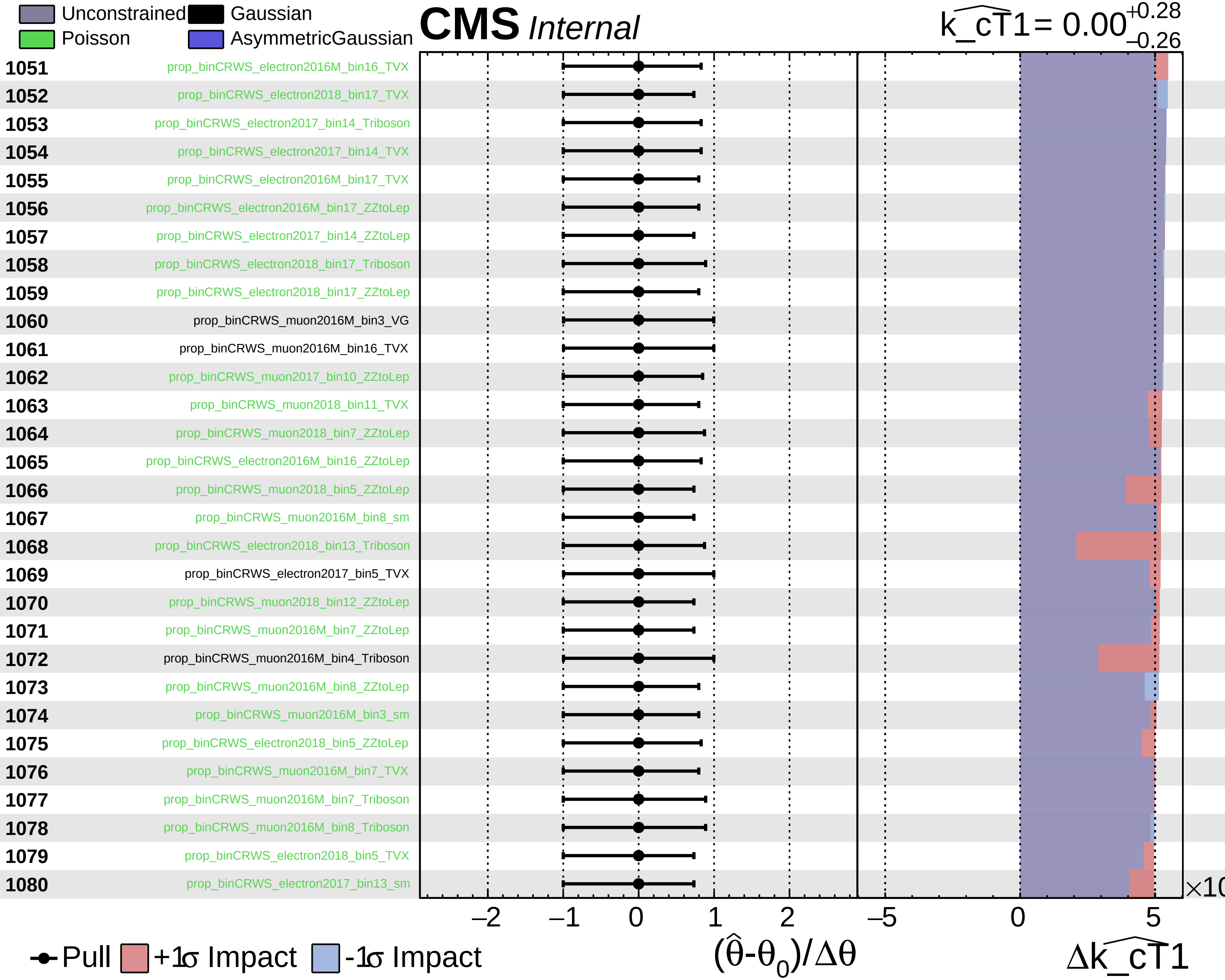
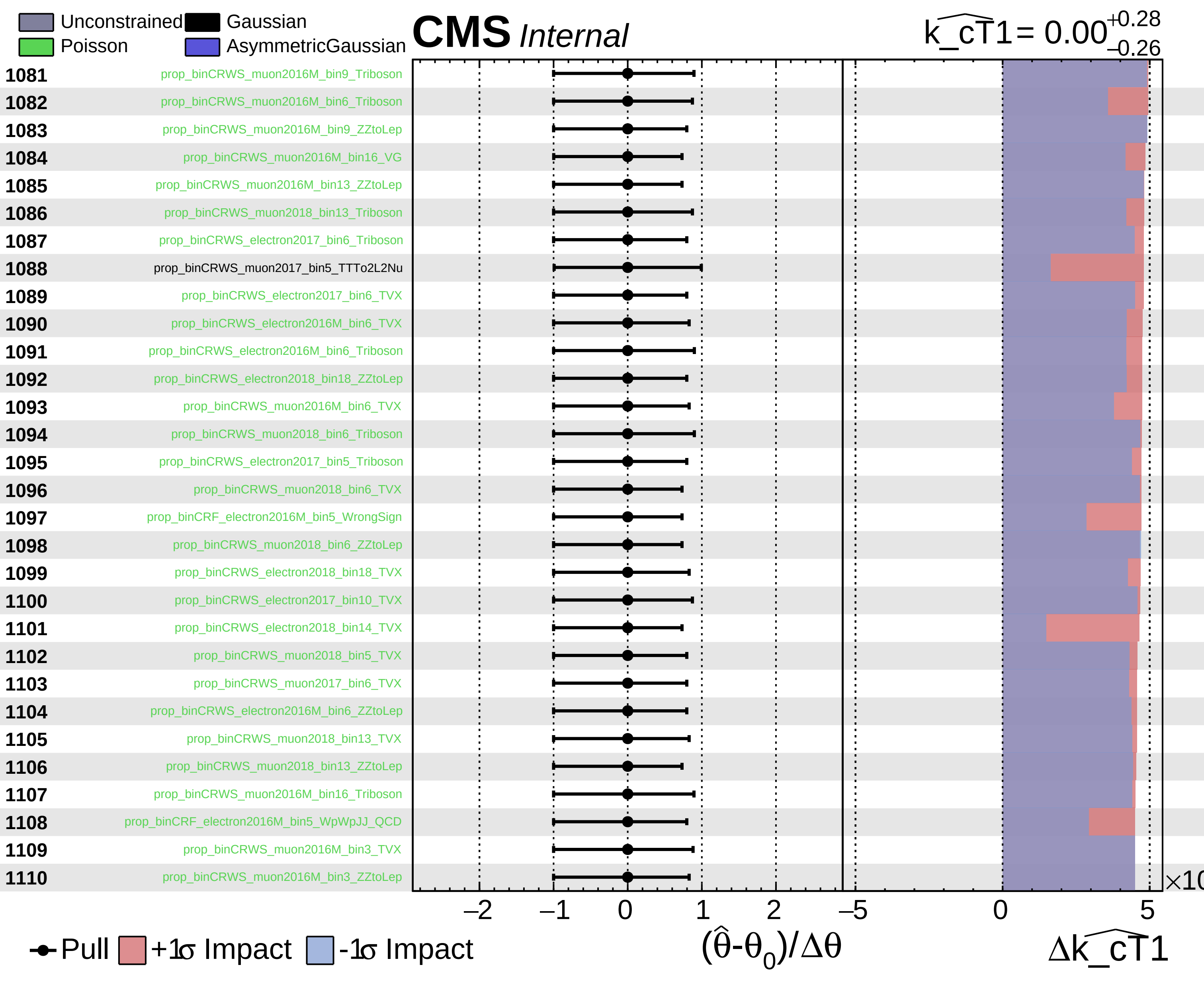
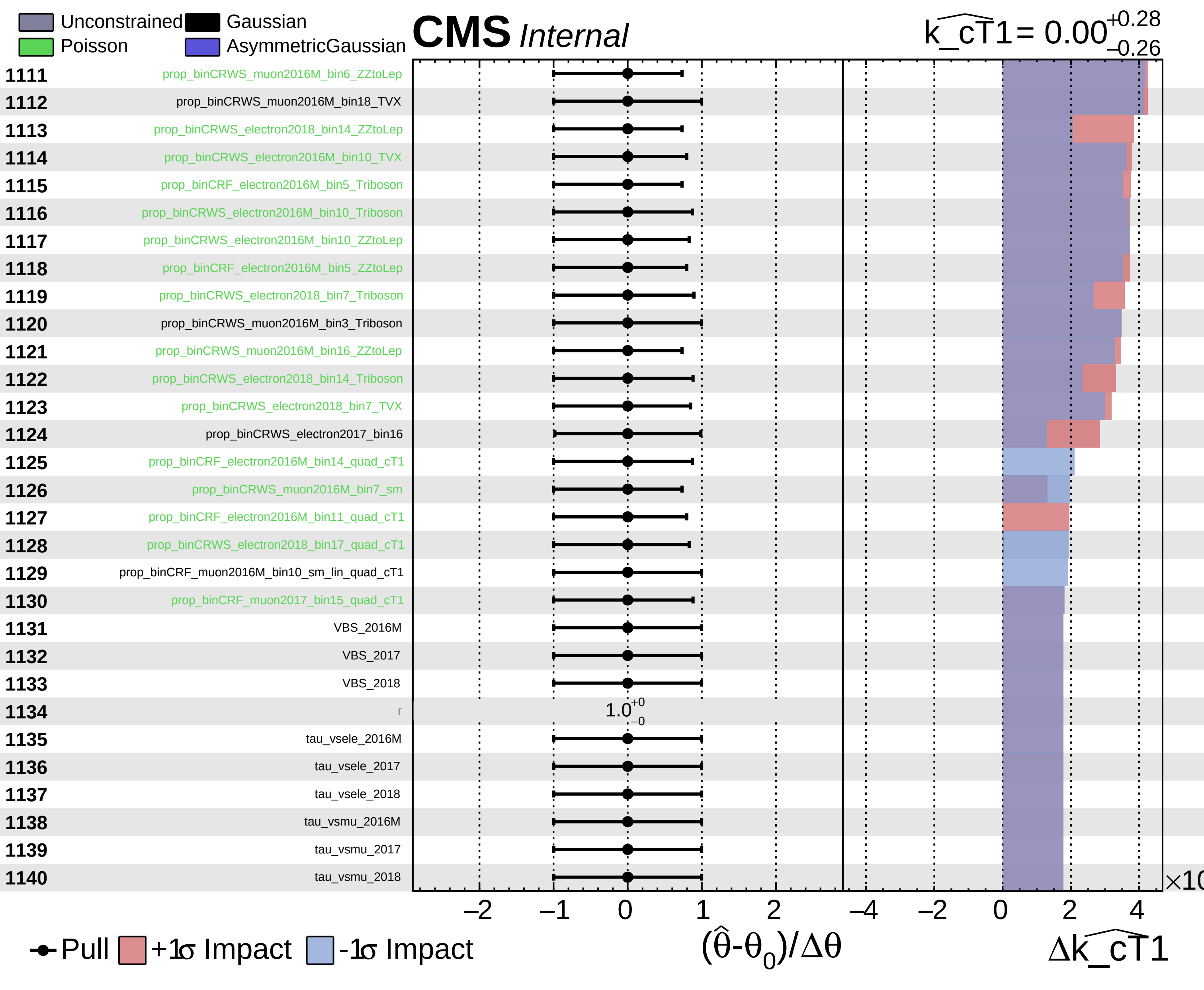
992 prop_binCRF_muon2016M_bin6_ZZtoLep
993 prop_binCRWS_muon2018_bin14_Triboson
994 prop_binCRWS_muon2018_bin10_ZZtoLep
995 prop_binCRWS_electron2017_bin17_TVX
996 prop_binCRF_electron2016M_bin18_ZZtoLep
997 prop_binCRWS_muon2017_bin6_ZZtoLep
998 prop_binCRWS_muon2017_bin7_Triboson
999 prop_binCRWS_muon2017_bin5_Triboson
1000 prop_binCRWS_electron2016M_bin7_Triboson
1001 prop_binCRWS_electron2017_bin13_TVX
1002 prop_binCRWS_electron2016M_bin8_Triboson
1003 prop_binCRF_electron2016M_bin12_ZZtoLep
1004 prop_binCRWS_electron2016M_bin9_ZZtoLep
1005 prop_binCRWS_electron2017_bin8_Triboson
1006 prop_binCRWS_electron2017_bin10_ZZtoLep
1007 prop_binCRWS_muon2018_bin7_Triboson
1008 prop_binCRF_electron2016M_bin16_ZZtoLep
1009 prop_binCRWS_muon2018_bin12_TVX
1010 prop_binCRWS_electron2016M_bin17_Triboson
1011 prop_binCRWS_electron2018_bin18_sm
1012 prop_binCRWS_muon2018_bin10_Triboson
1013 prop_binCRWS_muon2016M_bin5_Triboson
1014 prop_binCRWS_electron2016M_bin20_sm
1015 prop_binCRWS_muon2017_bin10_Triboson
1016 prop_binCRWS_electron2016M_bin7_ZZtoLep
1017 prop_binCRWS_muon2016M_bin6_VG
1018 prop_binCRWS_muon2018_bin18_ZZtoLep
1019 prop_binCRWS_electron2017_bin5_ZZtoLep
1020 prop_binCRWS_muon2017_bin19_Triboson
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

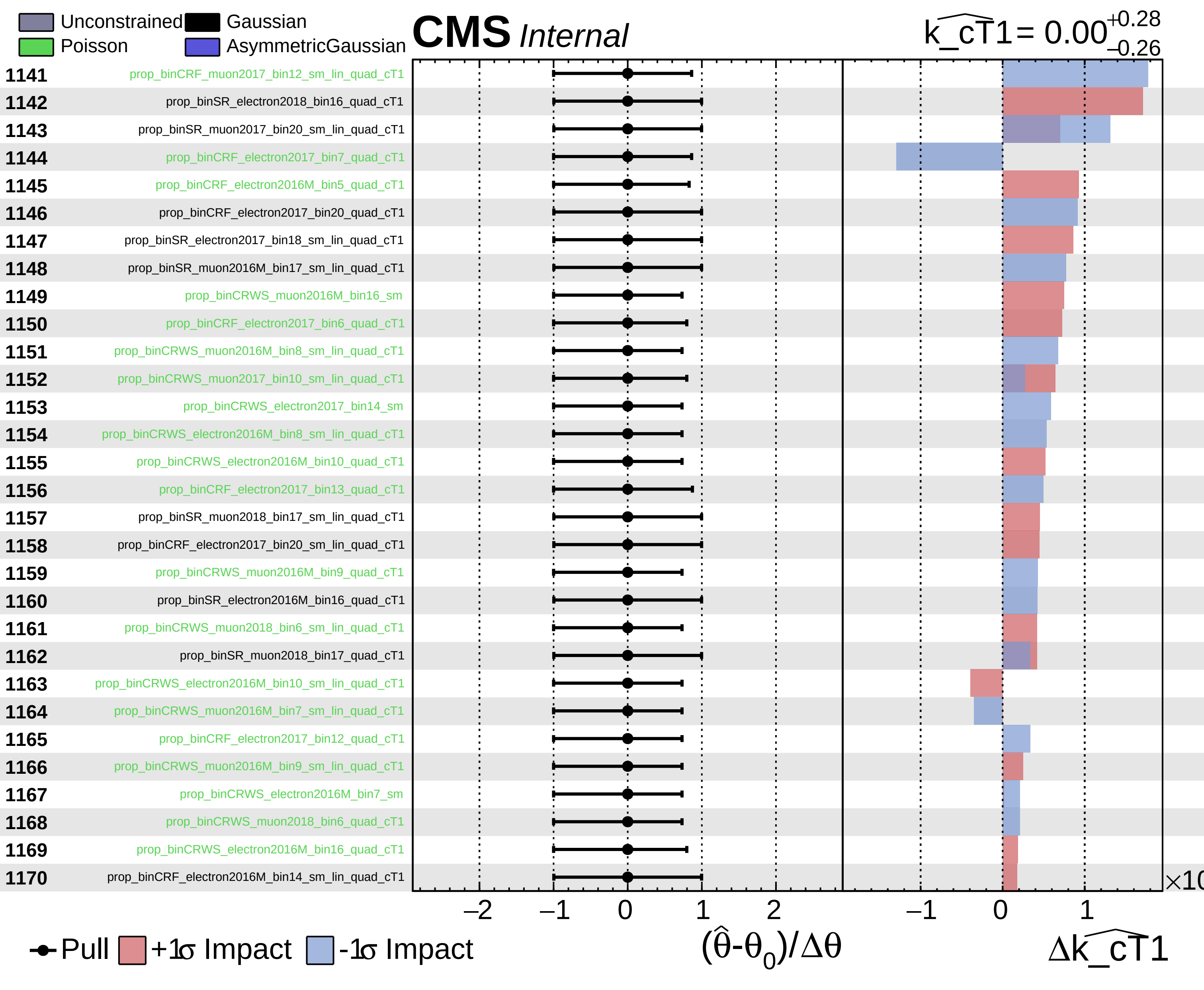


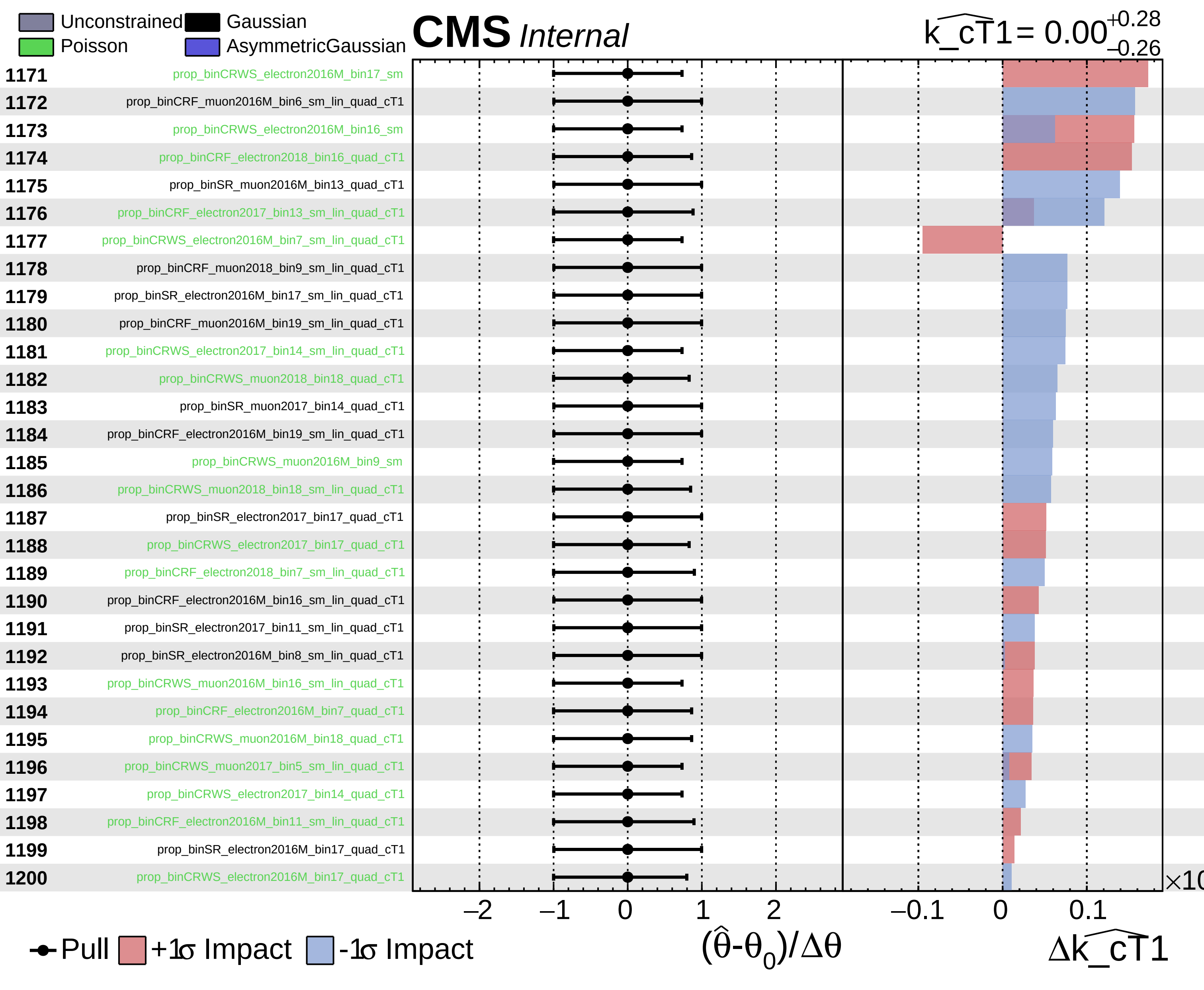
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1021 prop_binCRWS_muon2018_bin11_ZZtoLep
1022 prop_binCRWS_muon2017_bin8_TVX
1023 prop_binCRWS_electron2018_bin13_ZZtoLep
1024 prop_binCRWS_muon2017_bin19_ZZtoLep
1025 prop_binCRWS_muon2017_bin7_ZZtoLep
1026 prop_binCRWS_muon2017_bin5_ZZtoLep
1027 prop_binCRWS_muon2018_bin5_Triboson
1028 prop_binCRWS_muon2017_bin5_TVX
1029 prop_binCRWS_muon2016M_bin5_TVX
1030 prop_binCRWS_muon2018_bin11_Triboson
1031 prop_binCRWS_muon2016M_bin4_TVX
1032 prop_binCRWS_electron2018_bin5_Triboson
1033 prop_binCRWS_muon2016M_bin9_TVX
1034 prop_binCRWS_muon2018_bin7_TVX
1035 prop_binCRWS_electron2017_bin10_Triboson
1036 prop_binCRWS_muon2017_bin9_ZZtoLep
1037 prop_binCRWS_muon2017_bin9_Triboson
1038 prop_binCRWS_muon2016M_bin18_VG
1039 prop_binCRWS_electron2017_bin8_ZZtoLep
1040 prop_binCRWS_muon2016M_bin13_Triboson
1041 prop_binCRWS_muon2017_bin10_TVX
1042 prop_binCRWS_electron2016M_bin8_VG
1043 prop_binCRWS_electron2016M_bin8_TVX
1044 prop_binCRWS_electron2016M_bin8_ZZtoLep
1045 prop_binCRWS_muon2016M_bin5_sm
1046 prop_binCRWS_muon2017_bin6_Triboson
1047 prop_binCRWS_muon2018_bin14_ZZtoLep
1048 prop_binCRWS_muon2017_bin9_TVX
1049 prop_binCRWS_electron2016M_bin16_Triboson
1050 prop_binCRWS_muon2016M_bin18_Triboson
−2
−1
0
1
2
−5
0
5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1051 prop_binCRWS_electron2016M_bin16_TVX
1052 prop_binCRWS_electron2018_bin17_TVX
1053 prop_binCRWS_electron2017_bin14_Triboson
1054 prop_binCRWS_electron2017_bin14_TVX
1055 prop_binCRWS_electron2016M_bin17_TVX
1056 prop_binCRWS_electron2016M_bin17_ZZtoLep
1057 prop_binCRWS_electron2017_bin14_ZZtoLep
1058 prop_binCRWS_electron2018_bin17_Triboson
1059 prop_binCRWS_electron2018_bin17_ZZtoLep
1060 prop_binCRWS_muon2016M_bin3_VG
1061 prop_binCRWS_muon2016M_bin16_TVX
1062 prop_binCRWS_muon2017_bin10_ZZtoLep
1063 prop_binCRWS_muon2018_bin11_TVX
1064 prop_binCRWS_muon2018_bin7_ZZtoLep
1065 prop_binCRWS_electron2016M_bin16_ZZtoLep
1066 prop_binCRWS_muon2018_bin5_ZZtoLep
1067 prop_binCRWS_muon2016M_bin8_sm
1068 prop_binCRWS_electron2018_bin13_Triboson
1069 prop_binCRWS_electron2017_bin5_TVX
1070 prop_binCRWS_muon2018_bin12_ZZtoLep
1071 prop_binCRWS_muon2016M_bin7_ZZtoLep
1072 prop_binCRWS_muon2016M_bin4_Triboson
1073 prop_binCRWS_muon2016M_bin8_ZZtoLep
1074 prop_binCRWS_muon2016M_bin3_sm
1075 prop_binCRWS_electron2018_bin5_ZZtoLep
1076 prop_binCRWS_muon2016M_bin7_TVX
1077 prop_binCRWS_muon2016M_bin7_Triboson
1078 prop_binCRWS_muon2016M_bin8_Triboson
1079 prop_binCRWS_electron2018_bin5_TVX
1080 prop_binCRWS_electron2017_bin13_sm
−2
−1
0
1
2
−5
0
5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

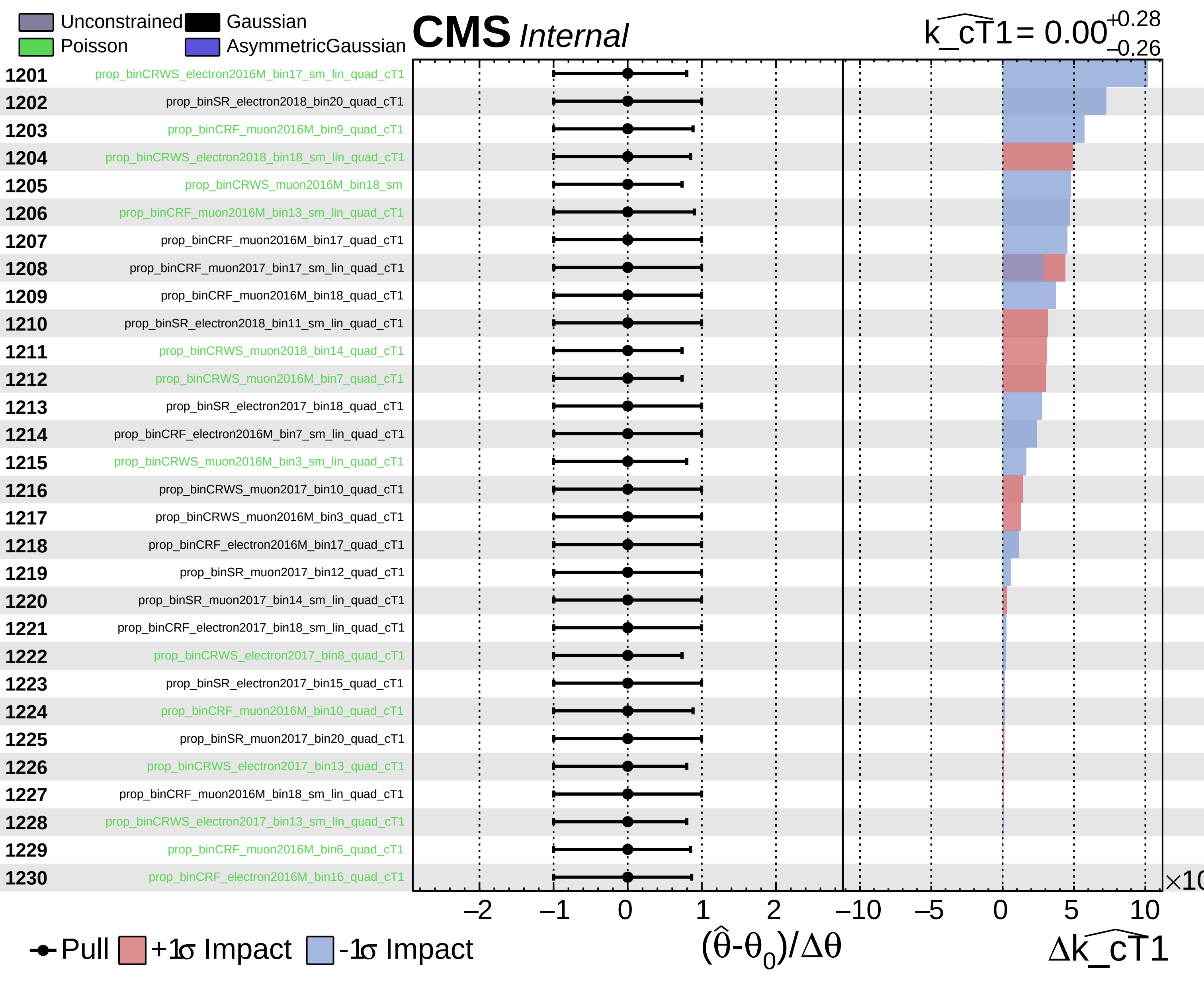

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1081 prop_binCRWS_muon2016M_bin9_Triboson
1082 prop_binCRWS_muon2016M_bin6_Triboson
1083 prop_binCRWS_muon2016M_bin9_ZZtoLep
1084 prop_binCRWS_muon2016M_bin16_VG
1085 prop_binCRWS_muon2016M_bin13_ZZtoLep
1086 prop_binCRWS_muon2018_bin13_Triboson
1087 prop_binCRWS_electron2017_bin6_Triboson
1088 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
1089 prop_binCRWS_electron2017_bin6_TVX
1090 prop_binCRWS_electron2016M_bin6_TVX
1091 prop_binCRWS_electron2016M_bin6_Triboson
1092 prop_binCRWS_electron2018_bin18_ZZtoLep
1093 prop_binCRWS_muon2016M_bin6_TVX
1094 prop_binCRWS_muon2018_bin6_Triboson
1095 prop_binCRWS_electron2017_bin5_Triboson
1096 prop_binCRWS_muon2018_bin6_TVX
1097 prop_binCRF_electron2016M_bin5_WrongSign
1098 prop_binCRWS_muon2018_bin6_ZZtoLep
1099 prop_binCRWS_electron2018_bin18_TVX
1100 prop_binCRWS_electron2017_bin10_TVX
1101 prop_binCRWS_electron2018_bin14_TVX
1102 prop_binCRWS_muon2018_bin5_TVX
1103 prop_binCRWS_muon2017_bin6_TVX
1104 prop_binCRWS_electron2016M_bin6_ZZtoLep
1105 prop_binCRWS_muon2018_bin13_TVX
1106 prop_binCRWS_muon2018_bin13_ZZtoLep
1107 prop_binCRWS_muon2016M_bin16_Triboson
1108 prop_binCRF_electron2016M_bin5_WpWpJJ_QCD
1109 prop_binCRWS_muon2016M_bin3_TVX
1110 prop_binCRWS_muon2016M_bin3_ZZtoLep
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

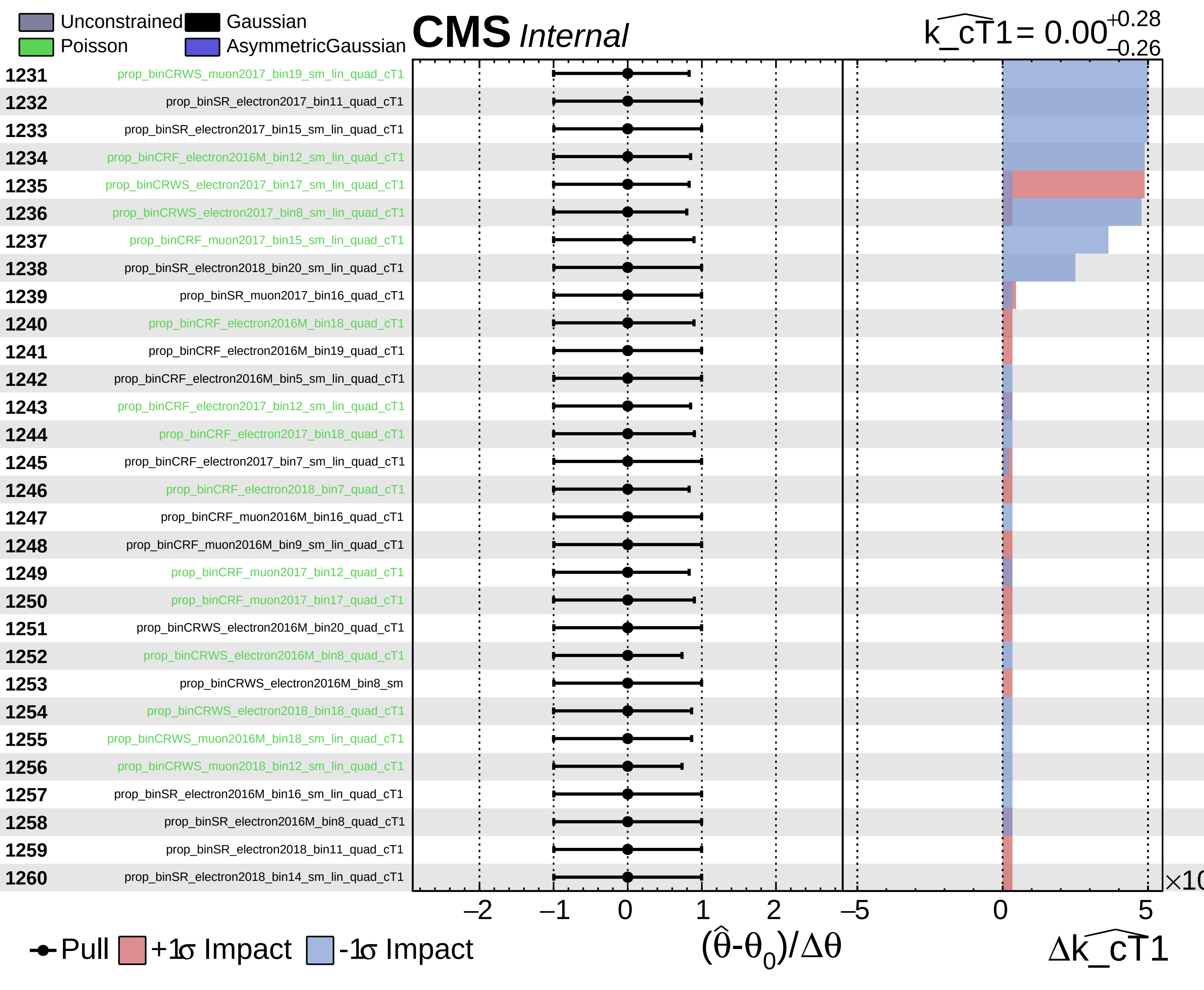

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1111 prop_binCRWS_muon2016M_bin6_ZZtoLep
1112 prop_binCRWS_muon2016M_bin18_TVX
1113 prop_binCRWS_electron2018_bin14_ZZtoLep
1114 prop_binCRWS_electron2016M_bin10_TVX
1115 prop_binCRF_electron2016M_bin5_Triboson
1116 prop_binCRWS_electron2016M_bin10_Triboson
1117 prop_binCRWS_electron2016M_bin10_ZZtoLep
1118 prop_binCRF_electron2016M_bin5_ZZtoLep
1119 prop_binCRWS_electron2018_bin7_Triboson
1120 prop_binCRWS_muon2016M_bin3_Triboson
1121 prop_binCRWS_muon2016M_bin16_ZZtoLep
1122 prop_binCRWS_electron2018_bin14_Triboson
1123 prop_binCRWS_electron2018_bin7_TVX
1124 prop_binCRWS_electron2017_bin16
1125 prop_binCRF_electron2016M_bin14_quad_cT1
1126 prop_binCRWS_muon2016M_bin7_sm
1127 prop_binCRF_electron2016M_bin11_quad_cT1
1128 prop_binCRWS_electron2018_bin17_quad_cT1
1129 prop_binCRF_muon2016M_bin10_sm_lin_quad_cT1
1130 prop_binCRF_muon2017_bin15_quad_cT1
1131 VBS_2016M
1132 VBS_2017
1133 VBS_2018
1134 r 1.0 +0 −0
1135 tau_vsele_2016M
1136 tau_vsele_2017
1137 tau_vsele_2018
1138 tau_vsmu_2016M
1139 tau_vsmu_2017
1140 tau_vsmu_2018
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1141 prop_binCRF_muon2017_bin12_sm_lin_quad_cT1
1142 prop_binSR_electron2018_bin16_quad_cT1
1143 prop_binSR_muon2017_bin20_sm_lin_quad_cT1
1144 prop_binCRF_electron2017_bin7_quad_cT1
1145 prop_binCRF_electron2016M_bin5_quad_cT1
1146 prop_binCRF_electron2017_bin20_quad_cT1
1147 prop_binSR_electron2017_bin18_sm_lin_quad_cT1
1148 prop_binSR_muon2016M_bin17_sm_lin_quad_cT1
1149 prop_binCRWS_muon2016M_bin16_sm
1150 prop_binCRF_electron2017_bin6_quad_cT1
1151 prop_binCRWS_muon2016M_bin8_sm_lin_quad_cT1
1152 prop_binCRWS_muon2017_bin10_sm_lin_quad_cT1
1153 prop_binCRWS_electron2017_bin14_sm
1154 prop_binCRWS_electron2016M_bin8_sm_lin_quad_cT1
1155 prop_binCRWS_electron2016M_bin10_quad_cT1
1156 prop_binCRF_electron2017_bin13_quad_cT1
1157 prop_binSR_muon2018_bin17_sm_lin_quad_cT1
1158 prop_binCRF_electron2017_bin20_sm_lin_quad_cT1
1159 prop_binCRWS_muon2016M_bin9_quad_cT1
1160 prop_binSR_electron2016M_bin16_quad_cT1
1161 prop_binCRWS_muon2018_bin6_sm_lin_quad_cT1
1162 prop_binSR_muon2018_bin17_quad_cT1
1163 prop_binCRWS_electron2016M_bin10_sm_lin_quad_cT1
1164 prop_binCRWS_muon2016M_bin7_sm_lin_quad_cT1
1165 prop_binCRF_electron2017_bin12_quad_cT1
1166 prop_binCRWS_muon2016M_bin9_sm_lin_quad_cT1
1167 prop_binCRWS_electron2016M_bin7_sm
1168 prop_binCRWS_muon2018_bin6_quad_cT1
1169 prop_binCRWS_electron2016M_bin16_quad_cT1
1170 prop_binCRF_electron2016M_bin14_sm_lin_quad_cT1
−2
−1
0
1
2
−1
0
1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1

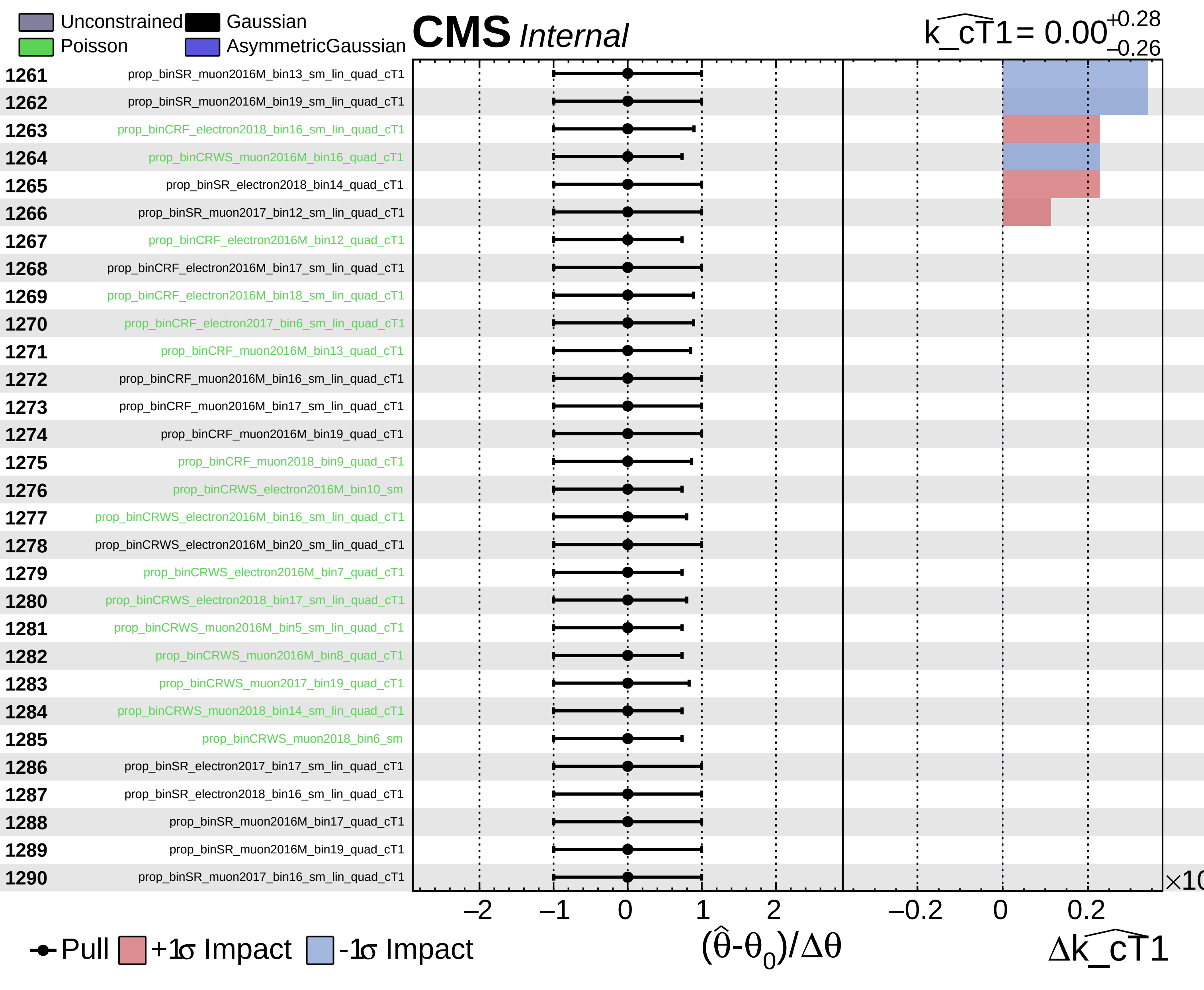


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1171 prop_binCRWS_electron2016M_bin17_sm
1172 prop_binCRF_muon2016M_bin6_sm_lin_quad_cT1
1173 prop_binCRWS_electron2016M_bin16_sm
1174 prop_binCRF_electron2018_bin16_quad_cT1
1175 prop_binSR_muon2016M_bin13_quad_cT1
1176 prop_binCRF_electron2017_bin13_sm_lin_quad_cT1
1177 prop_binCRWS_electron2016M_bin7_sm_lin_quad_cT1
1178 prop_binCRF_muon2018_bin9_sm_lin_quad_cT1
1179 prop_binSR_electron2016M_bin17_sm_lin_quad_cT1
1180 prop_binCRF_muon2016M_bin19_sm_lin_quad_cT1
1181 prop_binCRWS_electron2017_bin14_sm_lin_quad_cT1
1182 prop_binCRWS_muon2018_bin18_quad_cT1
1183 prop_binSR_muon2017_bin14_quad_cT1
1184 prop_binCRF_electron2016M_bin19_sm_lin_quad_cT1
1185 prop_binCRWS_muon2016M_bin9_sm
1186 prop_binCRWS_muon2018_bin18_sm_lin_quad_cT1
1187 prop_binSR_electron2017_bin17_quad_cT1
1188 prop_binCRWS_electron2017_bin17_quad_cT1
1189 prop_binCRF_electron2018_bin7_sm_lin_quad_cT1
1190 prop_binCRF_electron2016M_bin16_sm_lin_quad_cT1
1191 prop_binSR_electron2017_bin11_sm_lin_quad_cT1
1192 prop_binSR_electron2016M_bin8_sm_lin_quad_cT1
1193 prop_binCRWS_muon2016M_bin16_sm_lin_quad_cT1
1194 prop_binCRF_electron2016M_bin7_quad_cT1
1195 prop_binCRWS_muon2016M_bin18_quad_cT1
1196 prop_binCRWS_muon2017_bin5_sm_lin_quad_cT1
1197 prop_binCRWS_electron2017_bin14_quad_cT1
1198 prop_binCRF_electron2016M_bin11_sm_lin_quad_cT1
1199 prop_binSR_electron2016M_bin17_quad_cT1
1200 prop_binCRWS_electron2016M_bin17_quad_cT1
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1201 prop_binCRWS_electron2016M_bin17_sm_lin_quad_cT1
1202 prop_binSR_electron2018_bin20_quad_cT1
1203 prop_binCRF_muon2016M_bin9_quad_cT1
1204 prop_binCRWS_electron2018_bin18_sm_lin_quad_cT1
1205 prop_binCRWS_muon2016M_bin18_sm
1206 prop_binCRF_muon2016M_bin13_sm_lin_quad_cT1
1207 prop_binCRF_muon2016M_bin17_quad_cT1
1208 prop_binCRF_muon2017_bin17_sm_lin_quad_cT1
1209 prop_binCRF_muon2016M_bin18_quad_cT1
1210 prop_binSR_electron2018_bin11_sm_lin_quad_cT1
1211 prop_binCRWS_muon2018_bin14_quad_cT1
1212 prop_binCRWS_muon2016M_bin7_quad_cT1
1213 prop_binSR_electron2017_bin18_quad_cT1
1214 prop_binCRF_electron2016M_bin7_sm_lin_quad_cT1
1215 prop_binCRWS_muon2016M_bin3_sm_lin_quad_cT1
1216 prop_binCRWS_muon2017_bin10_quad_cT1
1217 prop_binCRWS_muon2016M_bin3_quad_cT1
1218 prop_binCRF_electron2016M_bin17_quad_cT1
1219 prop_binSR_muon2017_bin12_quad_cT1
1220 prop_binSR_muon2017_bin14_sm_lin_quad_cT1
1221 prop_binCRF_electron2017_bin18_sm_lin_quad_cT1
1222 prop_binCRWS_electron2017_bin8_quad_cT1
1223 prop_binSR_electron2017_bin15_quad_cT1
1224 prop_binCRF_muon2016M_bin10_quad_cT1
1225 prop_binSR_muon2017_bin20_quad_cT1
1226 prop_binCRWS_electron2017_bin13_quad_cT1
1227 prop_binCRF_muon2016M_bin18_sm_lin_quad_cT1
1228 prop_binCRWS_electron2017_bin13_sm_lin_quad_cT1
1229 prop_binCRF_muon2016M_bin6_quad_cT1
1230 prop_binCRF_electron2016M_bin16_quad_cT1
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.28}_{-0.26}$
1231 prop_binCRWS_muon2017_bin19_sm_lin_quad_cT1
1232 prop_binSR_electron2017_bin11_quad_cT1
1233 prop_binSR_electron2017_bin15_sm_lin_quad_cT1
1234 prop_binCRF_electron2016M_bin12_sm_lin_quad_cT1
1235 prop_binCRWS_electron2017_bin17_sm_lin_quad_cT1
1236 prop_binCRWS_electron2017_bin8_sm_lin_quad_cT1
1237 prop_binCRF_muon2017_bin15_sm_lin_quad_cT1
1238 prop_binSR_electron2018_bin20_sm_lin_quad_cT1
1239 prop_binSR_muon2017_bin16_quad_cT1
1240 prop_binCRF_electron2016M_bin18_quad_cT1
1241 prop_binCRF_electron2016M_bin19_quad_cT1
1242 prop_binCRF_electron2016M_bin5_sm_lin_quad_cT1
1243 prop_binCRF_electron2017_bin12_sm_lin_quad_cT1
1244 prop_binCRF_electron2017_bin18_quad_cT1
1245 prop_binCRF_electron2017_bin7_sm_lin_quad_cT1
1246 prop_binCRF_electron2018_bin7_quad_cT1
1247 prop_binCRF_muon2016M_bin16_quad_cT1
1248 prop_binCRF_muon2016M_bin9_sm_lin_quad_cT1
1249 prop_binCRF_muon2017_bin12_quad_cT1
1250 prop_binCRF_muon2017_bin17_quad_cT1
1251 prop_binCRWS_electron2016M_bin20_quad_cT1
1252 prop_binCRWS_electron2016M_bin8_quad_cT1
1253 prop_binCRWS_electron2016M_bin8_sm
1254 prop_binCRWS_electron2018_bin18_quad_cT1
1255 prop_binCRWS_muon2016M_bin18_sm_lin_quad_cT1
1256 prop_binCRWS_muon2018_bin12_sm_lin_quad_cT1
1257 prop_binSR_electron2016M_bin16_sm_lin_quad_cT1
1258 prop_binSR_electron2016M_bin8_quad_cT1
1259 prop_binSR_electron2018_bin11_quad_cT1
1260 prop_binSR_electron2018_bin14_sm_lin_quad_cT1
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.28 −0.26
1261 prop_binSR_muon2016M_bin13_sm_lin_quad_cT1
1262 prop_binSR_muon2016M_bin19_sm_lin_quad_cT1
1263 prop_binCRF_electron2018_bin16_sm_lin_quad_cT1
1264 prop_binCRWS_muon2016M_bin16_quad_cT1
1265 prop_binSR_electron2018_bin14_quad_cT1
1266 prop_binSR_muon2017_bin12_sm_lin_quad_cT1
1267 prop_binCRF_electron2016M_bin12_quad_cT1
1268 prop_binCRF_electron2016M_bin17_sm_lin_quad_cT1
1269 prop_binCRF_electron2016M_bin18_sm_lin_quad_cT1
1270 prop_binCRF_electron2017_bin6_sm_lin_quad_cT1
1271 prop_binCRF_muon2016M_bin13_quad_cT1
1272 prop_binCRF_muon2016M_bin16_sm_lin_quad_cT1
1273 prop_binCRF_muon2016M_bin17_sm_lin_quad_cT1
1274 prop_binCRF_muon2016M_bin19_quad_cT1
1275 prop_binCRF_muon2018_bin9_quad_cT1
1276 prop_binCRWS_electron2016M_bin10_sm
1277 prop_binCRWS_electron2016M_bin16_sm_lin_quad_cT1
1278 prop_binCRWS_electron2016M_bin20_sm_lin_quad_cT1
1279 prop_binCRWS_electron2016M_bin7_quad_cT1
1280 prop_binCRWS_electron2018_bin17_sm_lin_quad_cT1
1281 prop_binCRWS_muon2016M_bin5_sm_lin_quad_cT1
1282 prop_binCRWS_muon2016M_bin8_quad_cT1
1283 prop_binCRWS_muon2017_bin19_quad_cT1
1284 prop_binCRWS_muon2018_bin14_sm_lin_quad_cT1
1285 prop_binCRWS_muon2018_bin6_sm
1286 prop_binSR_electron2017_bin17_sm_lin_quad_cT1
1287 prop_binSR_electron2018_bin16_sm_lin_quad_cT1
1288 prop_binSR_muon2016M_bin17_quad_cT1
1289 prop_binSR_muon2016M_bin19_quad_cT1
1290 prop_binSR_muon2017_bin16_sm_lin_quad_cT1
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1